LEWIS UNIVERSITY LIBRARY

AI EVALUATION TOOLKIT

Complete Reference Guide & Sources Bibliography

33 AI Tools • 94 Activity Rules • 305+ Activities

Research & Teaching • Faculty Resource

This document is a comprehensive reference companion to the interactive Lewis University Library AI Evaluation Tool web application.

CC BY-NC-ND 4.0 • Lewis University Library

Table of Contents

Part 1: AI Tool Catalog

- General-Purpose AI Assistants
  - ChatGPT
  - Claude
  - Google Gemini
  - Microsoft Copilot
- AI-Powered Research & Search
  - Perplexity AI
  - NotebookLM
- Scholarly Literature & Citation Tools
  - Semantic Scholar
  - Elicit
  - Scite
  - ResearchRabbit
  - Scholarcy
  - Rayyan
  - Connected Papers
  - Anara
  - SciSpace
  - ChatPDF
  - Sourcely
  - Inciteful
  - ExplainPaper
  - Petal
- Data Analysis
  - Julius AI
- Writing & Editing
  - Grammarly
- Image Generation
  - Midjourney
  - DALL-E 3
  - Adobe Firefly
  - Stable Diffusion
- Video & Animation
  - Runway
  - Sora
  - Luma AI
  - Synthesia
- Audio & Voice
  - ElevenLabs
  - Otter.ai
  - Suno

Part 2: Activities, Prompts & Use-Case Scenarios

- AI Policy & Academic Integrity
- Accessibility & Universal Design for Learning
- Advanced Discussion Facilitation
- Arts & Media

Authentic Assessment Redesign
Business
Capstone & Semester Projects
Communication, Journalism & Media
Community Engagement & Service Learning
Computer Science & Programming
Conference Presentation Preparation
Course & Module Design
Creative Writing & Composition
Criminal Justice & Legal Studies
Curriculum Mapping & Program Assessment
Data Management & Reproducibility
Data Visualization & Infographics
Education
Environmental & Earth Sciences
Faculty Productivity & Administration
Flipped Classroom & Active Learning
General
Grading, Feedback & Revision
Graduate Student & Thesis Support
Grant Writing & Proposals
Health Sciences & Nursing
Humanities
Interdisciplinary & Cross-Course Applications
Language Learning & Multilingual Support
Library & Information Science
Math, Statistics & Quantitative Reasoning
Multimodal & Creative Assignments
OER & Course Materials
Online & Asynchronous Teaching
Peer Review & Collaborative Learning
Professional Development & Faculty Learning
Publication & Journal Submission
Research Impact & Dissemination
Research: Academic Writing
Research: Literature Review
STEM
Scholarly Communication
Simulations & Role-Play
Social Sciences
Student AI Literacy & Prompt Engineering
Study Skills & Metacognition
Teaching + Research Integration
Teaching: Assignment Assessment
Teaching: Quick Discussion Activities
Teaching: Quick First-Year Activities

Part 3: Sources Bibliography

Foundational Pedagogical Frameworks
AI Pedagogy & Instructional Frameworks
Information Literacy & Source Evaluation
Health Sciences & Evidence-Based Practice
Research Ethics & Scholarly Communication
National & International Policy Guidance
Institutional & Organizational Statements on AI in Education
University Centers for Teaching & Learning
Scholarly & Peer-Reviewed Publications
Faculty Development, Pedagogy & Best Practices
AI Tool Reviews, Comparative Guides & Industry Resources

Part 1: AI Tool Catalog

The following tools are referenced throughout the AI Evaluation Toolkit. Each entry includes the tool name, pricing tier, URL, description, and the teaching/research tasks it is best suited for.

General-Purpose AI Assistants

ChatGPT [Free+]

https://chat.openai.com/

Versatile conversational AI for drafting, brainstorming, and structuring content. Highly responsive to detailed prompts.

Best for: Discussion • Assessment • Course Design • Feedback • Materials • AI Policy

---

Claude [Free+]

https://claude.ai/

Excellent for long-form synthesis, nuanced rewriting, and following complex multi-step instructions. Handles large documents well.

Best for: Literature Review • Research Workflow • Course Design • Feedback • Materials

---

Google Gemini [Free+]

https://gemini.google.com/

Multimodal AI with deep Google/Drive integration. Good for research summaries, multimedia prompts, and slides.

Best for: Course Design • Research • Materials • Discussion

Microsoft Copilot [Free]

https://copilot.microsoft.com/

Integrated into Microsoft 365. Ideal for Word drafts, Teams summaries, and productivity workflows faculty already use.

Best for: Course Design • Feedback • Materials • Admin

AI-Powered Research & Search

Perplexity AI [Free+]

https://www.perplexity.ai/

Search-first AI that cites its sources. Best for quick literature scans, fact-checking, and verifiable research overviews.

Best for: Literature Review • Research • Discussion • AI Policy

---

NotebookLM [Free]

https://notebooklm.google/

Upload YOUR documents; Google's AI summarizes, connects themes, and answers questions grounded in your sources. No hallucination of outside content.

Best for: Literature Review • Research • Materials • Course Design • Assessment

Scholarly Literature & Citation Tools

Semantic Scholar [Free]

https://www.semanticscholar.org/

AI-powered scholarly search with citation graphs and influence scores. Essential for tracking research lineages.

Best for: Literature Review • Research

Elicit [Free+]

https://elicit.org/

Automates literature review by extracting methods, findings, and sample sizes from papers into structured tables.

Best for: Literature Review • Research

Scite [Free+]

https://scite.ai/

Shows exactly how a paper is cited — supporting, contrasting, or simply mentioning. Critical for evaluating evidence claims.

Best for: Literature Review • Research • Assessment

ResearchRabbit [Free]

https://www.researchrabbit.ai/

Visually maps citation networks around seed papers. Perfect for entering a new field or finding influential authors fast.

Best for: Literature Review • Research

Scholarcy [Free+]

https://www.scholarcy.com/

Structured paper summaries, flashcards. Rapid screening for systematic reviews.

Best for: Literature Review • Research

Rayyan [Free+]

https://www.rayyan.ai/

Systematic review management. 90% time savings on screening.

Best for: Literature Review • Research

Connected Papers [Free]

https://www.connectedpapers.com/

Visual citation networks. Find influential works and gaps.

Best for: Literature Review • Research

Anara [Free+]

https://anara.com/

Search PubMed, arXiv, JSTOR. Chat with documents.

Best for: Literature Review • Research • Qualitative

SciSpace [Free+]

https://scispace.com/

Collaborative platform with 270M papers and AI Copilot.

Best for: Literature Review • Research

ChatPDF [Free+]

https://www.chatpdf.com/

Ask questions, get answers from PDF documents.

Best for: Literature Review • Research • Materials

Sourcely [Free+]

https://www.sourcely.net/

Context-aware academic search with automated summarization.

Best for: Literature Review • Research

Inciteful [Free]

https://inciteful.xyz/

Citation network visualization. Discover connected papers, explore research networks, find seminal works.

Best for: Literature Review • Research • Discovery

ExplainPaper [Free]

https://www.explainpaper.com/

AI explains complex research papers. Highlight confusing text, get instant explanations. Makes dense papers accessible.

Best for: Literature Review • Research • Reading

Petal [Free+]

https://www.petal.org/

AI-powered document analysis and research organization. Summarize papers, organize literature, extract insights.

Best for: Literature Review • Research • Organization

Data Analysis

Julius AI [Free+]

https://julius.ai/

Upload a dataset; Julius provides AI-assisted analysis, charts, and interpretation. Always validate statistical outputs independently.

Best for: Research • Assessment • STEM

Writing & Editing

Grammarly [Free+]

https://www.grammarly.com/

AI writing assistant for grammar, clarity, tone, and style. Good scaffolding tool for student writing without full text generation.

Best for: Feedback • Materials • Assessment

Image Generation

Midjourney [Paid]

https://www.midjourney.com/

High-quality AI image generation with strong aesthetic control. Excellent for visual course materials, concept illustrations, and media projects.

Best for: Materials • Arts • Discussion

DALL-E 3 [Paid]

https://openai.com/dall-e-3

Image generation built into ChatGPT. Good for quick concept diagrams, illustrative teaching materials, and accessible alt-text creation.

Best for: Materials • Arts • Course Design

Adobe Firefly [Free+]

https://firefly.adobe.com/

Commercial-safe AI image generation integrated with Adobe Creative Cloud. Professional graphics and marketing materials.

Best for: Materials • Design • Arts • Business

Stable Diffusion [Free]

https://stability.ai/

Open-source image generation. Highly customizable for technical users, research on AI art, custom model training.

Best for: Arts • Research • STEM • Technical

Video & Animation

Runway [Free+]

https://runwayml.com/

AI video generation and editing tools. Useful for short explainer clips, concept visualizations, and creative media assignments.

Best for: Materials • Arts • Research

Sora [Paid]

https://openai.com/sora/

OpenAI's text-to-video model. Hyperreal motion, sound, and cinematic quality. Requires ChatGPT Plus/Pro.

Best for: Creative • Materials • Arts • Research • Simulation

Luma AI [Free+]

https://lumalabs.ai/dream-machine

Dream Machine generates fast AI video (5–10 sec) from text or images. Great for storyboards and quick animations.

Best for: Creative • Materials • Arts • Design

Synthesia [Paid]

https://www.synthesia.io/

AI avatar video creation. Generate lecture videos with realistic presenters in 140+ languages. Accessibility-focused.

Best for: Materials • Online/Async • Accessibility • Training

Audio & Voice

ElevenLabs [Free+]

https://elevenlabs.io/

Realistic AI voice generation for narrated course content, accessibility audio, and language learning materials.

Best for: Materials • Online/Async • Hybrid • Arts

Otter.ai [Free+]

https://otter.ai/

AI transcription and auto-generated meeting notes. Useful for recording office hours, lectures, or research interviews for accessibility.

Best for: Online/Async • Hybrid • Research • Materials

Suno [Free+]

https://suno.com/

AI music and song generation from text prompts. Create original audio for projects, podcasts, presentations.

Best for: Creative • Materials • Arts • Audio

Part 2: Activities, Prompts & Use-Case Scenarios

Activities are organized by category. Each entry includes the activity title, estimated time, Bloom's taxonomy level, a description, the recommended AI tools, and full step-by-step implementation instructions.

AI Policy & Academic Integrity

AI Use Spectrum Activity

Time: 30 min | Bloom's: Evaluate | Tools: claude, chatgpt, perplexity

Students place AI tool uses on a spectrum from 'clearly acceptable' to 'clearly not acceptable' in your course context. Discuss reasoning. Use to co-create class norms.

Implementation Instructions

Create a worksheet listing 10-12 specific scenarios of AI use in your course context, ranging from clearly acceptable (using spell-check) to clearly unacceptable (submitting AI-generated work as original). Include ambiguous middle-ground scenarios that will generate genuine debate (using AI to brainstorm thesis ideas, having AI explain a concept you don't understand, using AI to check grammar on a final draft). Students individually place each scenario on the spectrum, then discuss in small groups where they disagreed and why. Use the class discussion to co-create your course's AI use norms — students who help write the policy are more likely to follow it.

Comparative Policy Analysis

Time: 1–2 hrs | Bloom's: Analyze | Tools: claude, chatgpt, perplexity

Ask ChatGPT to draft three AI policy stances: permissive, moderate, restrictive. Students compare the policies, identify tradeoffs, then draft their own justified position.

Implementation Instructions

Ask ChatGPT to draft three distinct AI policy stances for a college course: permissive (AI encouraged as a tool), moderate (AI allowed with disclosure), and restrictive (AI prohibited except in specific circumstances). Distribute all three to students and have them analyze the tradeoffs of each: What does each policy enable? What does it prevent? What assumptions does it make about learning? Students then draft their own justified position, citing specific elements from the three stances. This builds critical thinking about policy design while preparing students to navigate AI policies they'll encounter throughout their careers.

Syllabi Language Workshop

Time: 1 hr | Bloom's: Create | Tools: claude, chatgpt, perplexity

Use Claude to draft AI-use language for your syllabus in three registers: formal, conversational, and student-facing plain language. Refine for your course context.

Implementation Instructions

Use Claude to draft AI-use language for your syllabus in three different registers: formal/legalistic, conversational/approachable, and student-facing plain language. Compare the three versions: Which one most accurately reflects your actual policy? Which one would students actually read and understand? Which one would hold up in an academic integrity case? Refine your preferred version by combining elements from all three — perhaps the clarity of plain language with the specificity of formal language. Test the final version by asking a student or colleague to read it and tell you back what they think it means.

AI Assessment Scale Implementation

Time: 1-2 weeks | Bloom's: Create | Tools: chatgpt, claude, perplexity

Develop assignment-specific AI usage scale (0-5) showing prohibited, permitted, required uses. Discuss with students, iterate based on feedback.

Implementation Instructions

Develop an assignment-specific AI usage scale from 0 (no AI permitted) to 5 (AI required as a tool), with clear descriptions of what's expected at each level. For example: Level 0 = No AI; Level 1 = AI for brainstorming only; Level 2 = AI for research assistance; Level 3 = AI for drafting with human revision; Level 4 = AI as collaborative partner with documentation; Level 5 = AI required, evaluation focuses on prompt quality and critical engagement. Different assignments within the same course can have different AI levels. Discuss the scale with students at the start of the semester and revisit mid-semester to iterate based on collective experience.

AI Disclosure Workflow

Time: ongoing | Bloom's: Apply | Tools: chatgpt, claude, perplexity

Students submit AI use log with assignments: tools used, prompts, outputs, human modifications. Builds metacognition and transparency.

Implementation Instructions

Create a standardized AI use log that students submit with every assignment. The log includes: which AI tool(s) they used, the prompts they entered, the AI outputs they received, and how they modified or incorporated those outputs into their work. Design the log to be quick to complete (checkboxes plus brief notes) so it doesn't become burdensome. Review logs periodically to identify patterns: Are students over-relying on AI for certain tasks? Are they developing more sophisticated prompting strategies over time? Use aggregated log data to inform your teaching — if most students use AI for the same bottleneck, that's a signal to adjust your instruction.

Departmental Policy Workshop

Time: half-day | Bloom's: Create | Tools: chatgpt, claude, perplexity

Facilitate department discussion on AI policies. Use AI to analyze peer institution policies. Draft consensus guidelines aligned with program values.

Implementation Instructions

Facilitate a half-day workshop where faculty colleagues develop shared AI policies for your department or program. Use Perplexity to gather AI policies from peer institutions in your field as discussion starters. Use Claude to analyze patterns across these policies: What do most agree on? Where is there significant divergence? Guide the department through key decisions: Should policies be course-specific or program-wide? What disclosure is required? How should AI use be handled in capstones vs. introductory courses? The goal isn't a single rigid policy but consensus guidelines that individual faculty can adapt while maintaining program coherence.

Accessibility & Universal Design for Learning

Alternative Text Generation Workflow

Time: ongoing | Bloom's: Create | Tools: claude, elevenlabs, chatgpt

Use AI to draft alt-text for course images. Edit for accuracy and context. Teach students to write effective alt-text for their submissions.

Implementation Instructions

Audit your course materials for images that lack alt-text. For each image, use Claude or ChatGPT to generate an initial alt-text description, specifying: 'Write alt-text for this image that would convey its meaning to someone using a screen reader, in the context of a [your discipline] course.' Review each

AI-generated description for accuracy and context-appropriateness — AI may describe visual elements but miss disciplinary significance. Teach students to write effective alt-text for their own submissions by sharing examples of good and poor alt-text. This ongoing practice improves accessibility compliance across your entire course while building inclusive design habits.

Reading Material Adaptation

Time: per-assignment | Bloom's: Apply | Tools: claude, elevenlabs, chatgpt

AI adapts dense readings to multiple formats: simplified summaries, concept maps, audio versions. Students choose format matching their needs.

Implementation Instructions

Identify your course's most challenging readings — the ones where students consistently struggle. For each, use Claude to generate three alternative access formats: a simplified summary (targeting 2 grade levels below the original), a concept map showing key ideas and relationships, and a structured reading guide with pre-reading questions, key vocabulary, and post-reading reflection prompts. Make all formats available alongside the original text — students self-select the scaffolding they need. Review each adaptation for accuracy, ensuring simplification doesn't distort key arguments. This UDL approach supports diverse learners without reducing rigor for those who engage with the original.

Closed Caption Quality Improvement

Time: ongoing | Bloom's: Apply | Tools: claude, elevenlabs, chatgpt

Auto-generate captions with Otter.ai. AI helps edit for accuracy, punctuation, speaker identification. Essential for accessibility compliance.

Implementation Instructions

Auto-generate captions for your video content using Otter.ai or your LMS's built-in captioning tool. AI captions are a starting point, not a finished product — they consistently struggle with: discipline-specific terminology, proper nouns, accented speech, and mathematical/scientific notation. Use Claude to help edit captions for accuracy: paste the transcript and note your discipline's key terms for correction. Add speaker identification labels if your content includes multiple voices, panels, or guest speakers. Accurate captions are legally required for accessibility and benefit all students — many learners use captions as a study tool even without hearing accommodations.

Alt-Text Generation for Course Materials

Time: 1-2 hrs | Bloom's: Apply | Tools: claude, chatgpt, elevenlabs, otter

Upload image-heavy slides or documents to Claude. AI generates descriptive alt-text for each image. Review for accuracy and discipline-specific context before publishing.

Implementation Instructions

Upload your image-heavy course materials (slides, handouts, web pages) to Claude and ask it to generate descriptive alt-text for each image. Review every alt-text description for accuracy — AI may misidentify graphs, diagrams, or discipline-specific images. For data visualizations, ensure the alt-text conveys the key finding, not just 'a bar chart.' For photographs, verify that descriptions are accurate and culturally sensitive. This one-time investment per course makes your materials accessible to screen reader users and complies with ADA/Section 508 requirements. Many institutions now require alt-text on all course materials — AI makes this achievable at scale.

Multi-Format Content Conversion

Time: 2-3 hrs per module | Bloom's: Create | Tools: claude, chatgpt, elevenlabs, otter

Take a text-heavy lecture and use AI to create: an audio summary (ElevenLabs), a visual concept map, a simplified study guide, and discussion questions. UDL multiple means of representation.

Implementation Instructions

Select a text-heavy lecture or reading and use AI to create multiple representations: (1) Use Claude to generate a plain-language summary at 8th-grade reading level, (2) Ask ChatGPT to create a concept map outline showing relationships between key ideas, (3) Use ElevenLabs to create an audio version of the summary for auditory learners, (4) Generate 5 discussion questions at different Bloom's levels. This provides multiple means of representation (a core UDL principle) without requiring you to create each format from scratch. Students self-select the format that best supports their learning. Review all AI-generated content for accuracy before distributing.

Reading Level Adaptation Ladder

Time: 1-2 hrs | Bloom's: Create | Tools: claude, chatgpt, elevenlabs, otter

AI adapts a complex reading to three levels: original, college introductory, and plain language. Students start at their comfort level and scaffold up. Nobody is excluded from the content.

Implementation Instructions

Take a complex course reading (research article, primary source, technical document) and use Claude to create three versions: the original text, a college-introductory adaptation (reduces jargon, adds context, keeps key arguments), and a plain-language version (8th-grade level, focuses on main ideas). Post all three versions with a note: 'Start with whichever version helps you access the ideas, then work toward engaging with the original.' This scaffolded approach ensures no student is excluded from course content due to reading level while still setting the expectation that all students engage with the original text. Verify AI adaptations for accuracy.

Advanced Discussion Facilitation

Structured Academic Controversy

Time: 2-3 class sessions | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

Students research both sides of controversial issue. AI helps find credible sources. Debate → switch sides → synthesize → position statement.

Implementation Instructions

Select a genuinely controversial issue in your discipline where reasonable people disagree based on evidence interpretation. Assign students to research teams (pairs or trios), randomly assigning each team a position to argue — including positions they may personally disagree with. Students use AI tools (Perplexity for evidence, ChatGPT for argument structure) to prepare their case. In class, run the full protocol: present position → switch sides → synthesize → write individual position statement. The power of this activity is in the position-switching: students must argue against their initial position using the same evidence, building genuine understanding of complexity rather than reinforcing pre-existing beliefs.

Fishbowl Discussion with AI Observer

Time: 1 class session | Bloom's: Analyze | Tools: chatgpt, claude, perplexity

Inner circle discusses while outer circle observes. AI helps outer circle document discussion patterns, participation, evidence use.

Implementation Instructions

Arrange the class in two concentric circles. The inner circle (5-8 students) discusses a text or topic while the outer circle observes and documents. Give outer circle students specific AI-assisted observation tasks: use ChatGPT to categorize types of evidence used, track how often claims are supported vs. asserted, note discussion patterns (who builds on others' ideas vs. introduces new threads). After 15-20 minutes, rotate circles. Debrief using the AI-assisted observations: What patterns emerged? Was the discussion balanced? What kinds of evidence were most persuasive? This meta-level analysis of discussion quality builds both discussion and analytical skills.

Socratic Seminar Preparation

Time: 1-2 weeks | Bloom's: Analyze | Tools: chatgpt, claude, perplexity

Students prepare open-ended questions on text. AI helps refine questions for depth, clarity. Seminar focuses on evidence-based dialogue.

Implementation Instructions

Assign a text for close reading 1-2 weeks before the seminar. Students prepare by writing 3 open-ended discussion questions — questions that don't have a single correct answer and require engagement with the text as evidence. Students then submit their questions to Claude with the prompt: 'How could these questions be improved for depth, specificity, and discussion potential?' Students revise based on AI feedback, documenting what they changed and why. In the seminar, questions drive a student-led evidence-based dialogue. The AI refinement step ensures questions are substantive enough to sustain genuine discussion rather than producing yes/no answers or surface-level responses.

Arts & Media

AI Aesthetics Critique

Time: 1–2 hrs | Bloom's: Evaluate | Tools: midjourney, dalle, elevenlabs, runway, chatgpt

Generate images with Midjourney or DALL-E using a brief from your discipline. Students apply course critique frameworks to the AI output — analyzing composition, intent, and bias.

Implementation Instructions

Choose a visual brief relevant to your discipline (e.g., 'Create a protest poster in the style of 1960s political art' or 'Design a book cover for a dystopian novel'). Generate 4-5 images using Midjourney or DALL-E with the same prompt. Distribute the images and have students apply course critique frameworks: analyze composition, color theory, visual hierarchy, cultural references, and emotional impact. Students must also identify what makes the images read as 'AI art' — what aesthetic patterns does AI default to? This builds critical visual literacy while helping students understand AI's creative capabilities and limitations.

Collaborative AI Storyboard

Time: 2–3 hrs | Bloom's: Create | Tools: midjourney, dalle, elevenlabs, runway, chatgpt

Students use DALL-E 3 for visual panels and ElevenLabs for voice narration to create a short AI-assisted storyboard. Final reflection: where did human creativity exceed the AI's?

Implementation Instructions

Students work in teams to create a short (2-3 minute) AI-assisted storyboard or narrative sequence. Use DALL-E 3 for visual panels and ElevenLabs for voice narration or sound design. Each team must

submit a creative brief before production, documenting their artistic intent and what role AI will play versus human creative direction. The final deliverable includes the storyboard plus a reflection essay: Where did human creativity exceed what the AI could produce? Where was AI genuinely helpful? This balances technical exploration with critical reflection on the creative process.

Remix and Attribution Study

Time: 1 hr | Bloom's: Analyze | Tools: midjourney, dalle, elevenlabs, runway, chatgpt

Generate AI art from prompts inspired by named artists. Examine the ethical and legal dimensions of training data, stylistic influence, and attribution — connecting to IP law.

Implementation Instructions

Have students generate AI art using prompts that reference specific artists, movements, or cultural traditions (e.g., 'in the style of Frida Kahlo' or 'inspired by Japanese woodblock prints'). Students then research and document: What training data likely influenced this output? Whose work is the AI drawing from? What are the ethical implications of using someone's artistic style without permission? Connect to course concepts about intellectual property, fair use, appropriation, and cultural sensitivity. This transforms a technical exercise into a substantive ethical and legal analysis relevant to any creative professional.

Practice Documentation Portfolio

Time: ongoing | Bloom's: Create | Tools: midjourney, claude, runway, chatgpt

Use AI to draft artist statements, process descriptions, contextual analysis. Document creative process alongside final works.

Implementation Instructions

Throughout your creative practice, use AI tools to support documentation: Claude for drafting artist statements and process descriptions, ChatGPT for contextualizing your work within art historical conversations, and image-generation tools for exploring visual concepts before committing to materials. Maintain a comprehensive documentation portfolio that includes: process journals, studio notes, material experiments, and contextual research. AI can help you articulate what you're doing and why, but the creative vision and artistic decisions must be authentically yours. The documentation portfolio serves both as a research record and as a component of practice-based research submissions.

Visual Analysis with AI Support

Time: 1-2 weeks | Bloom's: Analyze | Tools: midjourney, claude, runway, chatgpt

Use Claude to describe formal elements, compositional strategies, historical contexts. Human performs critical aesthetic interpretation.

Implementation Instructions

When analyzing visual works for your research, use Claude to generate initial descriptions: formal elements (line, color, composition), technical approaches, and possible art historical references. Use these AI-generated descriptions as a starting point for your own critical aesthetic interpretation — the AI can identify compositional elements but cannot understand intentionality, cultural meaning, or affective impact. Your contribution is the interpretive layer: What does this visual work mean? How does it function within its cultural context? What theoretical framework best explains its impact? Document the distinction between AI-assisted description and human interpretation in your methodology.

Exhibition Catalog Writing

Time: 2-4 weeks | Bloom's: Create | Tools: midjourney, claude, runway, chatgpt

AI drafts exhibition text at multiple reading levels. Human refines for voice, accuracy, curatorial vision.

Implementation Instructions

Use Claude to draft exhibition text at multiple reading levels: scholarly catalog essay (for specialist audiences), wall text (for general museum visitors), and educational guide (for student groups). Review each version for accuracy, ensuring factual claims about artworks, artists, and movements are correct. Then refine for voice — exhibition text should reflect curatorial vision and institutional tone, which AI cannot authentically replicate. The AI provides a structural scaffold and initial prose that saves significant drafting time, but the curatorial perspective, interpretive framework, and voice editing must be entirely human. Test readability of the general-audience version with non-specialist readers.

AI-Generated Art Critique Workshop

Time: 1 week | Bloom's: Evaluate | Tools: midjourney, runway, claude, elevenlabs

Generate images with Midjourney. Students analyze aesthetic choices, compositional decisions, what makes it read as 'AI art'. Develop critical eye.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. Generate images with Midjourney. Students analyze aesthetic choices, compositional decisions, what makes it read as 'AI art'. Develop critical eye. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Video Art Conceptual Development

Time: 4-6 weeks | Bloom's: Create | Tools: midjourney, runway, claude, elevenlabs

Use Runway for B-roll, transitions, effects in experimental video. Focus on conceptual intent, not technical polish. AI as creative collaborator.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Use Runway for B-roll, transitions, effects in experimental video. Focus on conceptual intent, not technical polish. AI as creative collaborator. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Sound Design for Performance

Time: 2-3 weeks | Bloom's: Create | Tools: midjourney, runway, claude, elevenlabs

Generate ambient soundscapes with AI for theater/dance performance. Students edit, layer, mix for emotional impact. Understand sound as narrative element.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Generate ambient soundscapes with AI for theater/dance performance. Students edit, layer, mix for

emotional impact. Understand sound as narrative element. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Artist Statement Drafting

Time: ongoing | Bloom's: Create | Tools: midjourney, runway, claude, elevenlabs

Claude helps structure artist statements. Students ensure authentic voice describing creative process, influences, intentions. Practice articulating practice.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Claude helps structure artist statements. Students ensure authentic voice describing creative process, influences, intentions. Practice articulating practice. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Remix Culture & Copyright

Time: 1-2 weeks | Bloom's: Evaluate | Tools: midjourney, runway, claude, elevenlabs

Create remixed artwork using AI-generated elements. Research copyright, fair use, attribution. Navigate legal/ethical dimensions of appropriation art.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. Create remixed artwork using AI-generated elements. Research copyright, fair use, attribution. Navigate legal/ethical dimensions of appropriation art. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Authentic Assessment Redesign

Real-World Client Project Design

Time: semester-long | Bloom's: Create | Tools: claude, chatgpt

Students work on actual client problems. AI helps with project scoping, milestone planning. Assessment based on client feedback + process documentation.

Implementation Instructions

Design a capstone or semester-long project where students work on authentic problems for real clients (community organizations, campus departments, small businesses). AI tools can help with project scoping (Claude to break complex problems into manageable phases), milestone planning (ChatGPT to generate Gantt charts and deliverable schedules), and research (Perplexity for background investigation). Assessment should be based on client feedback, process documentation, and final deliverables — not just the end product. The real-world context makes AI use appropriate and professional rather than academic dishonesty, since professionals routinely use AI tools in their work.

Professional Portfolio Development

Time: semester-long | Bloom's: Create | Tools: claude, chatgpt

Students curate work samples with AI-assisted reflections. Portfolio demonstrates learning journey, not just final products.

Implementation Instructions

Design a semester-long portfolio where students curate their best work, with AI-assisted reflective components. For each portfolio piece, students use Claude to help articulate: what they learned, how the piece demonstrates specific competencies, and how it connects to their professional goals. The AI helps with the metacognitive articulation — translating intuitive understanding of their own growth into clear, professional language. Students review and revise AI-assisted reflections to ensure authenticity. The portfolio assessment values the learning journey: growth over time, selection rationale, and self-awareness matter more than polish on any individual piece.

Time-Constrained Performance Task

Time: 1-2 class sessions | Bloom's: Apply | Tools: claude, chatgpt

In-class performance tasks where students apply knowledge under time pressure. AI can help prep, but can't complete task during exam.

Implementation Instructions

This application activity asks students to use AI tools within a structured workflow to practice skills. Begin by reviewing the activity description and adapting it to your specific course content and student level. In-class performance tasks where students apply knowledge under time pressure. AI can help prep, but can't complete task during exam. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. Grade students on the quality of their critical engagement with AI output, not just the final product. The metacognitive process — evaluating, questioning, and refining AI suggestions — is the actual learning outcome.

Business

AI-Generated Case Critique

Time: 1–2 hrs | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity, julius

Ask ChatGPT to generate a business case based on a real company scenario. Students evaluate the AI's strategic analysis against current market data and course frameworks.

Implementation Instructions

Ask ChatGPT to generate a business case study based on a real company or industry scenario relevant to your current unit. Distribute the AI-generated case to students as if it were a standard case assignment. Students analyze the case using course frameworks (Porter's Five Forces, SWOT, BCG Matrix, etc.) but with an additional requirement: verify all factual claims against current market data and flag any inaccuracies or outdated information. Students submit both their strategic analysis and a 'fact-check appendix' documenting what the AI got right and wrong. This builds analytical skills while teaching students that AI-generated business content requires verification.

Market Research Rapid Scan

Time: 1 hr | Bloom's: Analyze | Tools: chatgpt, claude, perplexity, julius

Use Perplexity to conduct a rapid competitive landscape analysis. Students verify AI claims with primary sources and build an annotated bibliography of corrections and confirmations.

Implementation Instructions

Assign students a specific company, product, or industry segment. Have them use Perplexity AI to conduct a rapid competitive landscape analysis in 30 minutes: key competitors, market trends, recent

developments, and strategic positions. Then require students to verify every AI-generated claim with a primary source (company SEC filings, press releases, industry reports) and build an annotated bibliography documenting confirmations and corrections. The final deliverable is a 1-page competitive brief with verified data only. This teaches the workflow of using AI for rapid initial research while maintaining the discipline of primary source verification.

Financial Data Interpretation Lab

Time: 2 hrs | Bloom's: Analyze | Tools: chatgpt, claude, perplexity, julius

Students upload a financial dataset to Julius AI and ask for trend analysis. They write a critique of the AI's interpretation — checking assumptions, noting errors, correcting output.

Implementation Instructions

Provide students with a real financial dataset (public company quarterly reports, market data, or economic indicators). Have them upload the data to Julius AI and request trend analysis, ratio calculations, and visualizations. Students then write a critical interpretation report that addresses: Are the AI's calculations correct? Are the visualizations misleading in any way? What context is the AI missing (industry benchmarks, macroeconomic factors, regulatory changes)? What business decisions would you make based on this data, and what additional information would you need? This builds financial literacy alongside critical evaluation of AI-generated analysis.

Case Study Development from Data

Time: 2-3 weeks | Bloom's: Create | Tools: chatgpt, perplexity, julius, claude

Use Perplexity for background research. Julius for data analysis. Claude structures narrative arc. Human writes final case and teaching notes.

Implementation Instructions

Identify a company or situation with rich publicly available data. Use Perplexity for background research on the company context, industry dynamics, and recent developments. Upload relevant financial or operational data to Julius for analysis: trend identification, ratio calculations, and visualization. Use Claude to help structure the case narrative: situation description, decision point, available data, and teaching note with discussion questions and analysis framework. Write the final case yourself — the AI provides research assistance and organizational scaffolding, but the pedagogical design (what decisions to highlight, what frameworks to apply) requires your expertise as a business educator.

Industry Comparison Matrix

Time: 1 week | Bloom's: Analyze | Tools: chatgpt, perplexity, julius, claude

AI helps extract key metrics across competitor companies. Generate comparison tables. Human interprets strategic implications.

Implementation Instructions

Select 4-6 companies in the same industry and use AI tools to extract key metrics: revenue, market share, growth rates, strategic initiatives, and competitive positioning. Claude can help structure the comparison matrix and identify patterns across companies. Julius can visualize trends and relative performance. Human interpretation is essential: Why are these differences significant? What strategic choices explain different performance trajectories? What external factors (regulation, technology shifts, market conditions) affect the comparison? The AI creates the data infrastructure for analysis; your strategic expertise provides the interpretation that makes the comparison valuable for research or teaching.

Financial Ratio Analysis Automation

Time: 3-5 days | Bloom's: Analyze | Tools: chatgpt, perplexity, julius, claude

Upload financial statements to Julius. Generate standard ratios, trends, peer comparisons. Human contextualizes findings.

Implementation Instructions

Upload financial statements (income statement, balance sheet, cash flow) to Julius AI. Request standard ratio analysis: liquidity ratios, profitability ratios, leverage ratios, and efficiency ratios, along with trend analysis over multiple periods. Julius can generate visualizations showing ratio trends and peer comparisons. Your critical contribution is contextualization: What do these ratios mean in this specific industry? What accounting choices affect the calculations? What non-financial factors explain the trends? How do these ratios compare to industry benchmarks? The AI handles computation; you provide the analytical narrative that transforms numbers into strategic insights.

Financial Statement Ratio Analysis

Time: 1 week | Bloom's: Analyze | Tools: julius, chatgpt, perplexity

Upload 10-K to Julius. Calculate liquidity, profitability, leverage ratios. Compare to industry benchmarks. Interpret financial health for investment decisions.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. Upload 10-K to Julius. Calculate liquidity, profitability, leverage ratios. Compare to industry benchmarks. Interpret financial health for investment decisions. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Stock Valuation Model Building

Time: 2-3 weeks | Bloom's: Create | Tools: julius, chatgpt, perplexity

Build DCF or comparable company valuation model. AI checks formula logic. Students defend assumptions, sensitivity analysis, recommendation to 'buy/sell/hold'.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Build DCF or comparable company valuation model. AI checks formula logic. Students defend assumptions, sensitivity analysis, recommendation to 'buy/sell/hold'. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Market Research Competitor Scan

Time: 1 week | Bloom's: Analyze | Tools: julius, chatgpt, perplexity

Use Perplexity to scan competitor strategies, market positioning, product lines. Synthesize into SWOT analysis. Propose strategic recommendations.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. Use Perplexity to scan competitor strategies, market positioning, product lines.

Synthesize into SWOT analysis. Propose strategic recommendations. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. Emphasize that AI assists with process efficiency but cannot replace the critical thinking, interpretation, and scholarly judgment that constitute genuine research contribution. Require transparent documentation of all AI tool usage in the final submission.

Supply Chain Disruption Scenarios

Time: 1-2 days | Bloom's: Evaluate | Tools: julius, chatgpt, perplexity

AI generates supply chain disruption scenarios (port closures, tariffs, demand shocks). Students analyze impacts, propose mitigation strategies.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI generates supply chain disruption scenarios (port closures, tariffs, demand shocks). Students analyze impacts, propose mitigation strategies. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Business Ethics Case Discussion

Time: 1 class session | Bloom's: Evaluate | Tools: julius, chatgpt, perplexity

AI generates ethically ambiguous business scenarios. Students debate stakeholder impacts, regulatory compliance, corporate social responsibility.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI generates ethically ambiguous business scenarios. Students debate stakeholder impacts, regulatory compliance, corporate social responsibility. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

AI-Assisted Business Plan Critique

Time: 2-3 hrs | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity, julius

Students draft a business plan section. AI plays investor asking tough due diligence questions. Students must defend their assumptions, projections, and competitive positioning.

Implementation Instructions

Students draft a section of their business plan (market analysis, financial projections, or competitive strategy). Submit to Claude with the prompt: 'You are a skeptical angel investor reviewing this business plan section. Ask 5 tough due diligence questions focused on: market size assumptions, revenue projections, competitive threats, team capabilities, and risk factors.' Students must prepare written responses defending their assumptions with evidence. This simulates the investor pitch process and forces students to stress-test their plans before presenting to actual evaluators. The AI asks questions students might not anticipate, building more robust business thinking.

Competitive Intelligence Dashboard

Time: 1 week | Bloom's: Analyze | Tools: chatgpt, claude, perplexity, julius

Students use Perplexity + Julius to build a competitive intelligence dashboard for a real company. Research competitors, market trends, and financial metrics. Present strategic recommendations.

Implementation Instructions

Assign each student or team a publicly traded company. Using Perplexity for market research and Julius for financial data analysis, students build a competitive intelligence dashboard including: market share estimates, competitor SWOT analyses, industry trend analysis, and key financial ratio comparisons. The critical element is synthesis — students must go beyond data collection to identify strategic implications: Where is the company vulnerable? What opportunities are competitors missing? What strategic moves should the company consider? Present findings to the class as a consulting recommendation. AI handles data gathering; students provide strategic judgment.

AI Ethics in Business Decision-Making

Time: 1-2 class sessions | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity, julius

AI generates ethically ambiguous business scenarios involving AI deployment (hiring algorithms, pricing algorithms, surveillance). Students analyze stakeholder impacts using ethical frameworks.

Implementation Instructions

Use ChatGPT to generate 3-4 ethically ambiguous scenarios where a company considers deploying AI: an AI hiring tool that improves efficiency but shows demographic bias, a dynamic pricing algorithm that maximizes revenue but disadvantages low-income customers, workplace monitoring AI that increases productivity but erodes trust. For each scenario, student teams analyze: stakeholder impacts (using stakeholder mapping), ethical dimensions (using utilitarian, deontological, and virtue ethics frameworks), legal considerations, and long-term reputational risks. Teams present recommendations to 'the board' (the class), who votes on the most ethically sound approach.

Capstone & Semester Projects

AI-Integrated Research Portfolio

Time: semester-long | Bloom's: Create | Tools: claude, chatgpt, notebooklm, perplexity, elicit

Students use AI tools across all phases of a research project — with required documentation of every AI interaction and a final reflection on AI's impact on their intellectual process.

Implementation Instructions

Design a semester-long portfolio where students use AI tools across all phases of their research project: literature discovery, methodology refinement, data analysis, and writing. The critical requirement is documentation — students maintain a log of every AI interaction including: the tool used, the prompt given, the output received, and how they used or modified the output. The final portfolio includes their research product plus a substantial reflection (1500-2000 words) analyzing AI's impact on their intellectual process: Where did AI accelerate their work? Where did it lead them astray? How did their use of AI evolve over the semester? This builds transparency habits they'll carry into professional settings.

Original Contribution Audit

Time: end of semester | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm, perplexity, elicit

Before final submission, students audit their capstone project to identify which ideas, arguments, and conclusions are genuinely theirs vs. AI-shaped. Document the boundary.

Implementation Instructions

Before final submission, students conduct a systematic audit of their capstone project. They create a document that categorizes each major element: ideas that are entirely their own, ideas that were AI-

influenced but developed further, ideas that came from AI and were accepted with minor modification, and ideas from other sources properly cited. The audit forces honest reckoning with the boundary between original and AI-assisted thought. It's not about penalizing AI use — it's about developing intellectual honesty and self-awareness about one's own creative process. Include a brief reflection on what constitutes 'original contribution' in an age of AI collaboration.

Communication, Journalism & Media

AI-Generated News Story Analysis

Time: 1 hr | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

AI writes a news story on a current event. Students apply journalism standards: accuracy, sourcing, balance, fairness, and newsworthiness. Identify what AI can't do that journalists must.

Implementation Instructions

Select a current event and have ChatGPT write a news story about it in AP style. Students evaluate the AI's story against professional journalism standards: Is every factual claim verifiable? Are sources identified and credible? Is the story balanced across perspectives? Does it distinguish fact from opinion? Is the lede (opening) newsworthy and accurate? Students rewrite the sections that fail professional standards. This reveals that journalism requires human judgment about newsworthiness, source credibility, and ethical framing that AI cannot replicate — essential understanding for future communication professionals.

Social Media Campaign Critique

Time: 1-2 hrs | Bloom's: Create | Tools: chatgpt, claude, perplexity

AI generates a social media campaign for a cause. Students evaluate messaging strategy, audience targeting, ethical considerations, and potential for misinterpretation or harm.

Implementation Instructions

Ask ChatGPT to design a complete social media campaign for a cause relevant to your course: 5 posts across platforms, suggested hashtags, audience targeting strategy, and key messages. Students evaluate the AI-generated campaign: Does the messaging resonate with the target audience? Are there cultural sensitivities the AI missed? Could any posts be misinterpreted or cause unintended harm? Is the campaign accessible to diverse audiences? Students revise the campaign to address identified issues and present their improved version with strategic justifications for each change.

Media Bias Detection Across AI Tools

Time: 1 hr | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

Ask multiple AI tools the same politically charged question. Students analyze how each tool's response reveals different framings, emphases, and potential biases.

Implementation Instructions

Select a politically charged or socially divisive topic relevant to your course. Submit the same question to ChatGPT, Claude, Perplexity, and Gemini. Students compare the responses: Which perspectives are centered? Which are marginalized? What framing choices does each tool make? Where does each tool hedge or refuse to take positions? Students write a media bias analysis applying course frameworks about framing, agenda-setting, and gatekeeping to AI-generated content. This demonstrates that AI tools are not neutral information sources — they reflect the biases, priorities, and values embedded in their training data and design choices.

Community Engagement & Service Learning

Community Needs Assessment with AI

Time: 1-2 weeks | Bloom's: Analyze | Tools: perplexity, claude, chatgpt

Students use Perplexity to research community demographics, needs, and existing services before a service-learning project. AI provides initial data; community partners verify reality.

Implementation Instructions

Before beginning a service-learning project, students use Perplexity to research: community demographics, prevalent social issues, existing services and organizations, and gaps in current resources. Students compile a preliminary needs assessment report using AI-gathered data. Then, critically, they present this assessment to the community partner for reality-checking: What did the AI get right? What's outdated? What local context is missing? What does the community identify as priorities that wouldn't appear in online data? This teaches students that AI-gathered data is a starting point, not a substitute for genuine community engagement and listening.

Impact Reflection Portfolio

Time: semester-long | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt

Students use AI to help structure reflection entries connecting service experiences to course theories. AI scaffolds the reflection process; genuine experience provides the content.

Implementation Instructions

Throughout the service-learning experience, students maintain a reflection portfolio. For each reflection entry, they first write freely about their experience, then use Claude to help connect their observations to course theories: 'I observed [experience]. Help me identify which course concepts from [topic] might explain or illuminate this observation.' The AI suggests theoretical frameworks; students evaluate which actually fit and develop the connection in their own words. The final portfolio includes both raw reflections and theory-connected analyses. Grade the quality of connections and the depth of critical reflection, not the polish of the writing.

Interview Protocol Development

Time: 2-3 hrs | Bloom's: Create | Tools: claude, notebooklm, perplexity, chatgpt

AI drafts interview questions based on your research framework. Students critique for leading questions, cultural sensitivity, and alignment with research aims. Iteratively refine.

Implementation Instructions

Describe your research topic, theoretical framework, and target population to Claude. Ask it to generate a semi-structured interview protocol with 12-15 questions. Students (or you, for your own research) then critically evaluate each question: Is it leading or biased? Is it culturally sensitive for the target population? Does it align with the research aims? Would a participant understand it as intended? Is the question sequence logical and builds rapport before addressing sensitive topics? Iteratively refine the protocol, documenting every change and its justification. AI generates a solid first draft quickly; human expertise ensures the protocol is ethical, culturally responsive, and methodologically sound.

Focus Group Data Theme Exploration

Time: 2-3 hrs | Bloom's: Analyze | Tools: claude, notebooklm, perplexity, chatgpt

Upload focus group transcripts to NotebookLM. AI suggests initial themes. Students validate, refine, and develop themes through systematic coding. AI accelerates, humans interpret.

Implementation Instructions

Upload anonymized focus group transcripts to NotebookLM. Ask the AI to suggest initial themes and supporting quotes. Use these AI-generated themes as a starting point — not a finished analysis.

Students systematically validate each theme against the raw data: Is there sufficient evidence? Are there disconfirming cases? Is the theme truly emergent or is the AI imposing a framework? Develop a codebook with theme definitions, inclusion/exclusion criteria, and exemplar quotes. Discuss: Where did the AI's initial themes hold up? Where did systematic coding reveal different patterns? This teaches rigorous qualitative analysis while honestly leveraging AI's ability to process large text datasets quickly.

Ethical Research Scenario Analysis

Time: 1 class session | Bloom's: Evaluate | Tools: claude, notebooklm, perplexity, chatgpt

AI generates ethically complex research scenarios (informed consent complications, dual relationships, community harm). Students apply IRB principles and ethical frameworks to navigate dilemmas.

Implementation Instructions

Use ChatGPT to generate 3-4 ethically complex research scenarios relevant to your discipline: informed consent complications with vulnerable populations, dual relationship dilemmas in community-based research, situations where research findings could harm the community studied, or conflicts between methodological rigor and participant wellbeing. Student teams analyze each scenario using the Belmont Report principles (respect for persons, beneficence, justice), professional ethical codes, and IRB guidelines. Teams present their ethical analysis and proposed resolution, then the class discusses whether alternative approaches might better protect participants while maintaining research integrity.

Computer Science & Programming

Code Review as Assessment

Time: 1 hr | Bloom's: Evaluate | Tools: chatgpt, claude, copilot

AI generates a solution to a programming problem with subtle bugs and style issues. Students perform a professional code review: identify bugs, suggest improvements, explain tradeoffs.

Implementation Instructions

Use ChatGPT or Claude to generate a solution to a programming problem that contains 3-5 subtle issues: logical bugs, inefficiencies, poor variable naming, missing edge cases, or security vulnerabilities. Students perform a professional code review as they would in industry: identify all issues, explain why each is problematic, suggest specific improvements, and rate the code's overall quality. This teaches code review skills — a critical professional practice — while revealing common patterns in AI-generated code that students will encounter when using AI coding assistants in their careers.

AI Pair Programming Session

Time: 2 hrs | Bloom's: Create | Tools: chatgpt, claude, copilot

Students use AI as a pair programming partner. They must explain their design decisions to the AI and evaluate its suggestions. Document the human-AI collaboration process.

Implementation Instructions

Students tackle a programming assignment using AI as a pair programming partner. The rules: they must explain their design approach to the AI before asking for code, they must understand every line the AI suggests before accepting it, and they must document every decision point where they chose differently from the AI's suggestion. The final submission includes: working code, a collaboration log documenting human vs. AI contributions, and a reflection on when the AI was helpful vs. when it led them astray. This mirrors how professional developers actually work with AI coding assistants while building the critical evaluation skills to use them effectively.

Algorithm Explanation Peer Teaching

Time: 45 min | Bloom's: Understand | Tools: chatgpt, claude, copilot

Students ask AI to explain an algorithm, then re-explain it to a peer in their own words with their own examples. Peer evaluates whether the explanation was clearer than the AI's.

Implementation Instructions

Assign each student a different algorithm or data structure. They first ask Claude for an explanation, then must re-explain it to a peer in their own words using a novel example (not from the AI or textbook). The peer rates the explanation on clarity, accuracy, and whether it was better than the AI's version. Students who can explain concepts more clearly than the AI demonstrate genuine understanding. Students who struggle reveal that they've memorized the AI's explanation without internalizing the concept. This peer-teaching approach builds deeper understanding while giving you diagnostic information about who actually understands the material.

Conference Presentation Preparation

Conference Talk Script Drafting

Time: 3-5 days | Bloom's: Create | Tools: chatgpt, claude, midjourney, grammarly

Bullet-point key findings. Claude drafts 15-min talk script with transitions. Practice with AI timing feedback. Human refines delivery.

Implementation Instructions

Bullet-point your 5-7 key findings or arguments in the order you want to present them. For each point, note the essential evidence and one transition to the next point. Paste this outline into Claude and ask it to draft a 15-minute talk script with smooth transitions and a compelling opening hook. Practice the AI-drafted script out loud — you'll immediately identify phrases that don't sound like you and passages that are too dense for oral delivery. Revise extensively for your natural speaking voice, cadence, and energy. Time yourself and cut ruthlessly — a focused 12-minute talk with 3 minutes for questions is always better than rushing through 18 minutes of material.

Poster Design Content Organization

Time: 2-3 days | Bloom's: Create | Tools: chatgpt, claude, midjourney, grammarly

AI helps distill research into poster-friendly chunks: 1 research question, 3 key findings, 1 implication. Generate QR codes for supplementary materials.

Implementation Instructions

Identify the single research question, top 3 findings, and one key implication of your work. Use Claude to help distill your research into poster-friendly text chunks: each section should be readable in 30 seconds or less. Generate QR codes linking to your full paper, supplementary data, or contact information. The poster should work as a visual abstract — someone walking by should grasp your key message in under a minute. AI helps with concision and organization, but design decisions (color scheme, visual hierarchy, figure placement) should reflect your institution's brand and the conference norms in your field. Less text is almost always better.

Elevator Pitch Iterations

Time: 1 day | Bloom's: Create | Tools: chatgpt, claude, midjourney, grammarly

Draft 30-second, 2-minute, and 5-minute versions of research pitch. ChatGPT suggests refinements for clarity and impact.

Implementation Instructions

Draft three versions of your research pitch at different lengths: 30 seconds (hallway encounter), 2 minutes (networking event), and 5 minutes (informal presentation). Use ChatGPT to refine each version for clarity, jargon reduction, and narrative hook — the 30-second version especially needs a

compelling opening that makes someone want to hear the 2-minute version. Practice each version out loud and time yourself precisely. Record yourself and listen back — AI can help with word choice, but only you can evaluate whether the pitch sounds natural and enthusiastic when spoken. The pitch should make a non-specialist curious about your work, not summarize your entire methodology.

Course & Module Design

Backward Design with AI

Time: 2–4 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Enter your learning outcomes into Claude. Ask it to suggest assessments and activities aligned to each Bloom's level. Critique with your own disciplinary knowledge.

Implementation Instructions

Start by clearly articulating your 3-5 course or module learning outcomes. Enter them into Claude with the prompt: 'For each outcome, suggest assessments aligned to [specific Bloom's level] and 2-3 learning activities that build toward the assessment.' Review the AI's suggestions critically against your disciplinary knowledge and student population. Use the output as a brainstorming catalyst, not a finished design — modify, combine, or discard suggestions as your expertise dictates. The value is in having a structured starting point that forces alignment between outcomes, activities, and assessments from the very beginning of the design process.

Syllabus Stress Test

Time: 1 hr | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Paste your syllabus into Claude. Ask it to identify: unclear expectations, AI policy gaps, accessibility issues, and misalignment between outcomes and assessments.

Implementation Instructions

Paste your complete syllabus into Claude with the prompt: 'Review this syllabus and identify: (1) unclear expectations, (2) AI policy gaps or ambiguities, (3) accessibility barriers, (4) misalignment between stated outcomes and assessments, and (5) any language that might confuse first-generation college students.' Read the AI's feedback with an open mind — it often catches inconsistencies that are invisible after you've read your own syllabus dozens of times. Prioritize the issues that affect the most students and address them before the semester starts. Share your revised syllabus with a colleague to verify the changes are improvements.

UDL Access Check

Time: 1–2 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Upload course materials to NotebookLM. Ask it to flag content that may present barriers for diverse learners. Use the output as a checklist for UDL revision.

Implementation Instructions

Upload your course materials (syllabus, assignment descriptions, key readings, slide decks) into NotebookLM. Ask the AI to flag content that may present barriers for diverse learners, specifically: dense text without visual breaks, missing alt-text for images, assumptions about prior knowledge, and limited representation in examples. Use the output as a checklist to systematically revise your materials for Universal Design for Learning principles: multiple means of engagement, representation, and action/expression. This isn't about perfection — it's about identifying the highest-impact changes you can make in a limited time.

Creative Writing & Composition

AI as Writing Workshop Partner

Time: per draft | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

Students submit creative writing to AI for craft-level feedback (pacing, voice, structure). They annotate which feedback reflects genuine craft insight vs. generic suggestions.

Implementation Instructions

Students submit a draft of creative writing (short fiction, poetry, creative nonfiction) to Claude with the prompt: 'Provide craft-level feedback on this piece: comment on pacing, voice consistency, imagery effectiveness, structural choices, and emotional resonance.' Students annotate each piece of AI feedback as: genuine craft insight (accepted), generic suggestion (rejected), or interesting but debatable (needs workshop discussion). Bring annotated feedback to the writing workshop for peer discussion. This adds a consistent feedback voice while teaching students to distinguish between surface-level editing and substantive craft development — a key skill for any serious writer.

Genre Convention Analysis

Time: 1 hr | Bloom's: Analyze | Tools: claude, chatgpt, grammarly

AI generates text in three different genres on the same topic. Students analyze what genre conventions the AI follows automatically and which require human artistic judgment to execute well.

Implementation Instructions

Select a single topic or scenario relevant to your course. Ask ChatGPT to write about it in three different genres: academic essay, personal narrative, and investigative journalism (or three genres appropriate to your course). Students analyze the outputs: What conventions does the AI follow automatically for each genre (structure, tone, evidence types, sentence patterns)? Where does the AI fail to execute genre conventions convincingly? What requires human judgment that AI can't replicate? This teaches genre awareness through contrast — students learn conventions by seeing them applied (and misapplied) across forms.

Revision Archaeology Exercise

Time: 1 hr | Bloom's: Analyze | Tools: claude, chatgpt, grammarly

Students submit their draft to AI for revision. They compare the original and revised versions to identify exactly what changed and evaluate whether each change improved the writing.

Implementation Instructions

Students paste their draft into Claude and ask for a revised version focusing on clarity and style. Instead of simply accepting or rejecting the revision, students conduct 'revision archaeology': create a side-by-side comparison documenting every change the AI made. For each change, classify it as: improved (and explain why), neutral (stylistic preference), worsened (and explain why), or incorrect (factual or contextual error). This exercise reveals what students can learn from AI editing (usually sentence-level clarity) and what requires their own judgment (voice, meaning, disciplinary conventions). The classification document is the assignment.

Criminal Justice & Legal Studies

AI Bias in Predictive Policing Analysis

Time: 1-2 hrs | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt

Students research how AI is used in criminal justice (sentencing, policing, parole). Analyze documented cases of algorithmic bias. Propose reforms grounded in course concepts.

Implementation Instructions

Students use Perplexity to research real cases of AI use in criminal justice: predictive policing algorithms, risk assessment tools used in sentencing, and facial recognition in law enforcement. For each case, students analyze: What data trained the algorithm? What biases are embedded? Who is disproportionately affected? What due process concerns arise? Students then propose specific reforms using course concepts from criminal justice theory, constitutional law, or ethics. This connects abstract course concepts to urgent real-world applications while teaching students to critically evaluate AI systems they may encounter professionally.

Case Brief with AI Verification

Time: 1-2 hrs | Bloom's: Analyze | Tools: perplexity, claude, chatgpt

Students draft a case brief independently, then use AI to check for completeness. Compare their brief to AI's version — identify what AI misses about legal reasoning and precedent context.

Implementation Instructions

Students read and brief a case independently using standard format (facts, issue, holding, reasoning, disposition). Then they submit the case to ChatGPT and ask for a case brief. Students compare the two versions: Did the AI identify the same central issue? Did it capture the holding accurately? Did it understand the significance of the reasoning for future cases? Most importantly, what legal context or implications did the AI miss? Students annotate the comparison and submit both versions. This teaches that legal analysis requires contextual understanding of precedent, policy implications, and judicial philosophy that AI frequently oversimplifies.

Curriculum Mapping & Program Assessment

Program-Level Outcome Alignment Audit

Time: 4-6 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Upload all course syllabi in a program to NotebookLM. AI maps where each program outcome is introduced, practiced, and assessed. Identify gaps and redundancies across the curriculum.

Implementation Instructions

Collect syllabi from all courses in a program and upload to NotebookLM. For each program-level student learning outcome (SLO), ask the AI to identify: which courses introduce it, which courses provide practice opportunities, and which courses assess it at mastery level. Generate a curriculum map matrix showing coverage across the program. Look for: outcomes that are only assessed once, outcomes that are never scaffolded (jump from introduction to mastery), and outcomes that are over-assessed (taking time from under-assessed outcomes). This AI-assisted audit that would take weeks manually can be completed in a day, providing evidence for accreditation self-studies.

Accreditation Self-Study Data Mining

Time: 1-2 days | Bloom's: Analyze | Tools: claude, chatgpt, notebooklm

Upload program data (course evaluations, assessment reports, retention data) to Claude. AI extracts trends, flags anomalies, and drafts narrative sections connecting data to standards.

Implementation Instructions

Compile your program's assessment data: course evaluation summaries, student learning outcome assessment results, retention and graduation rates, and faculty qualifications. Upload to Claude with the specific accreditation standards you need to address (AACSB, CAEP, CCNE, ABET, etc.). Ask the AI to identify trends that support or challenge compliance with each standard and draft narrative sections connecting data to standards. Critically review every claim the AI makes against your actual data — AI may overstate positive trends or miss important context. The AI saves enormous time on initial drafting, but the final self-study must reflect your honest institutional self-assessment.

Assessment Rubric Cross-Course Calibration

Time: 2-3 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Upload rubrics from multiple sections of the same course. AI identifies inconsistencies in criteria, language, and standards across sections. Facilitates norming conversations.

Implementation Instructions

Collect assignment rubrics from all sections of a multi-section course. Upload to Claude and ask it to compare: Are the same criteria used across sections? Is the language for performance levels consistent? Are the standards for 'proficient' equivalent? Are there criteria in some sections that are missing from others? Use the AI's comparison report to facilitate a faculty norming session where instructors discuss and resolve inconsistencies. This ensures students receive equitable assessment regardless of which section they're enrolled in — a key equity and accreditation concern.

Annual Report Narrative Drafting

Time: 1-2 days | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Upload department metrics, achievements, and strategic plan. AI drafts annual report sections connecting activities to strategic goals. Faculty review for accuracy and voice.

Implementation Instructions

Compile your department's annual data: enrollment figures, graduation rates, faculty accomplishments, grant awards, new programs, assessment results, and strategic plan goals. Upload to Claude with the prompt: 'Draft annual report sections that connect these departmental activities and achievements to our strategic plan goals. Use a professional but warm institutional tone.' Review the AI draft against your actual data for accuracy, add context the AI couldn't know (e.g., why enrollment dipped due to a specific policy change), and ensure the narrative honestly represents both successes and challenges. The AI handles the tedious task of connecting data points to strategic language.

Data Management & Reproducibility

Data Management Plan Drafting

Time: 1-2 days | Bloom's: Create | Tools: claude, chatgpt, julius

Claude helps structure DMP for grant applications: data types, storage, sharing, preservation. Human ensures institutional compliance.

Implementation Instructions

Use Claude to help structure your Data Management Plan according to your target funder's requirements (NSF, NIH, DOE, etc.). Address each required element systematically: data types and formats, metadata standards, access and sharing policies, storage and backup procedures, preservation plans, and ethical considerations. Have Claude cross-reference your DMP with your

institution's data policies and the funder's specific requirements. Review the AI-drafted plan against your actual research workflow — the DMP must reflect how you will actually manage data, not an idealized version. Share with your institution's research data services for expert review before submission.

Code Documentation Automation

Time: ongoing | Bloom's: Create | Tools: claude, chatgpt, julius

Use Claude to generate inline comments and README files for analysis scripts. Improves reproducibility and code sharing.

Implementation Instructions

As you write analysis code, periodically submit sections to Claude and ask it to generate: inline comments explaining the logic, docstrings for functions, and README files for repositories. AI-generated documentation is a strong starting point but often misses domain-specific context — review every comment for accuracy and add explanations of why certain analytical choices were made (not just what the code does). Maintain a changelog documenting major modifications and their rationale. Well-documented code is dramatically more likely to be reused by other researchers, increasing your work's impact and facilitating reproducibility review.

Supplementary Materials Organization

Time: 2-3 days | Bloom's: Apply | Tools: claude, chatgpt, julius

AI helps structure supplementary materials: appendices, datasets, code. Generate README files explaining contents.

Implementation Instructions

Before submission, compile all supplementary materials: additional analyses, raw data tables, code repositories, extended methods, and alternative model specifications. Use Claude to help structure these materials logically and generate README files explaining each component's purpose and relationship to the main text. Ensure every supplementary item is referenced in the main manuscript and that the README provides clear navigation. AI can help with the organizational structure and documentation prose, but you must verify that every table, figure, and analysis is accurate and properly labeled. Well-organized supplementary materials signal rigorous, transparent scholarship.

Data Management Plan Drafting

Time: 1-2 days | Bloom's: Create | Tools: claude, chatgpt, julius

AI drafts a data management plan based on your funder's requirements. Specify data types, storage, sharing, and preservation plans. Human reviews for accuracy and completeness.

Implementation Instructions

Most grant applications now require a data management plan (DMP). Describe your research project's data to Claude: types of data collected, volume, sensitivity level, and your funder's specific DMP requirements. Ask it to draft a DMP covering: data collection and organization, quality assurance, storage and backup, access and sharing policies, and long-term preservation. Review the AI draft against your funder's specific requirements — different agencies (NSF, NIH, NEH) have different expectations. Verify that the proposed storage, sharing, and preservation approaches are actually available at your institution. The AI handles boilerplate language; you ensure the plan is realistic and compliant.

Codebook Development for Research Data

Time: 1-2 days | Bloom's: Create | Tools: claude, chatgpt, julius

AI drafts a codebook for your research dataset: variable definitions, coding schemes, value labels, and missing data conventions. Human verifies against actual data.

Implementation Instructions

Upload your dataset variable list and any existing documentation to Claude. Ask it to draft a comprehensive codebook including: variable names and descriptions, measurement scales and units, coding schemes for categorical variables, value labels, valid ranges, missing data conventions, and data collection notes. Verify every entry against your actual data — AI may generate plausible but incorrect coding schemes. A well-documented codebook is essential for reproducibility, data sharing, and your future self who will revisit this data in 2 years and won't remember what 'var_37' means. This AI-assisted documentation dramatically reduces a tedious but critical research task.

Statistical Analysis Decision Guide

Time: 1-2 hrs | Bloom's: Apply | Tools: claude, chatgpt, julius

Describe your research question and data characteristics to AI. It suggests appropriate statistical tests with justifications. You verify against methodology literature.

Implementation Instructions

Describe your research question, hypothesis, data characteristics (sample size, variable types, distribution properties, independence assumptions), and study design to Claude. Ask: 'What statistical analyses are most appropriate for this research design? Justify each recommendation and note assumptions that must be verified.' Use the AI's suggestions as a starting point for your analysis plan — not as a final answer. Verify each recommendation against methodology texts in your field, check that assumptions are actually met in your data, and consult a statistician for complex designs. AI provides a useful initial framework that organizes your statistical thinking.

Data Visualization & Infographics

AI-Generated Chart Critique

Time: 45 min | Bloom's: Evaluate | Tools: julius, chatgpt, claude

AI creates data visualizations from a dataset. Students evaluate: Is the chart type appropriate? Is it misleading? Does it tell the story the data actually supports?

Implementation Instructions

Provide students with a dataset relevant to your course. Have them upload it to Julius AI or ChatGPT and request various visualizations (bar chart, pie chart, scatter plot, line graph). Students evaluate each AI-generated visualization: Is this the right chart type for this data? Could it be misleading (truncated axes, cherry-picked ranges, inappropriate scales)? Does the visualization tell the story the raw data actually supports? For each problematic chart, students redesign it with corrections and explain why their version is more honest. This teaches visual data literacy — increasingly important in a world flooded with AI-generated graphics.

Research Poster Data Visualization

Time: 2-3 hrs | Bloom's: Create | Tools: julius, chatgpt, claude

Students use AI to explore multiple ways to visualize their research findings. Critique which visualizations are most accurate and effective for their audience.

Implementation Instructions

Students take data from their research project and use Julius AI to generate 5-6 different visualization options. For each option, they evaluate: Does this accurately represent the data? Is it accessible to my target audience? Does it highlight the key finding or obscure it? Could it be misinterpreted? Students

select the best 2-3 visualizations for their research poster and write a brief justification for each choice. This teaches that data visualization is a rhetorical act — every design choice emphasizes some aspects of data while de-emphasizing others — and that the most visually impressive chart isn't always the most truthful or effective.

Education

Lesson Plan Co-Design

Time: 1–2 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Describe your learning objectives and student context to Claude. Ask for three lesson plan variations using different pedagogical approaches. Critique with your own teaching experience.

Implementation Instructions

Describe your specific learning objectives, student demographics, available class time, and any constraints (e.g., limited technology, large class size) to Claude. Ask it to generate three lesson plan variations using different pedagogical approaches (e.g., direct instruction, inquiry-based, project-based). For each AI-generated plan, critically evaluate: Does the pacing work for your students? Are the activities appropriate for the class size? Do the assessments actually measure the stated objectives? Modify the most promising plan based on your teaching experience and knowledge of your specific students. The AI provides structure and ideas; your pedagogical expertise provides the judgment.

AI Literacy for K–12 Prep

Time: module | Bloom's: Apply | Tools: claude, chatgpt, notebooklm

Future educators design an AI literacy unit for K–12 using ChatGPT to generate age-appropriate explanations, then peer-review for accuracy, appropriateness, and bias.

Implementation Instructions

Future educators design a 3-5 lesson AI literacy unit appropriate for a specific K-12 grade band. Students use ChatGPT to generate age-appropriate explanations of AI concepts (how AI learns, what training data is, why AI makes mistakes) and then peer-review each explanation for accuracy, developmental appropriateness, and potential bias. Students must also design an assessment that measures whether K-12 learners actually understood the concepts — not just whether they can repeat definitions. The final deliverable includes the unit plan, the AI-generated content with annotations, and a reflection on how AI tools could be responsibly integrated into K-12 classrooms.

Assessment Alignment Audit

Time: 2 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Upload your unit plan to NotebookLM. Ask it to audit alignment between stated objectives, instructional activities, and assessments — flag gaps for redesign.

Implementation Instructions

Upload your complete unit plan (learning objectives, instructional activities, assessments, and rubrics) to NotebookLM. Ask the AI to audit the alignment: Does each assessment directly measure at least one stated objective? Are there objectives with no corresponding assessment? Are there activities that don't connect to any objective? Does the cognitive level of assessments match the cognitive level of objectives (e.g., if the objective says 'analyze' but the assessment only requires 'recall')? Use the AI's audit as a structured checklist for revision, addressing the highest-impact misalignments first. Share results with a colleague to validate the AI's findings.

Classroom Observation Coding

Time: 2-4 weeks | Bloom's: Analyze | Tools: notebooklm, claude, chatpdf

Upload observation transcripts to NotebookLM. AI suggests preliminary codes. Human develops codebook, verifies inter-rater reliability.

Implementation Instructions

Upload classroom observation transcripts (with appropriate IRB approval and confidentiality measures) to NotebookLM. Ask the AI to suggest preliminary codes based on recurring patterns: instructional strategies, student interactions, questioning patterns, and engagement indicators. Use these AI-suggested codes as a starting point for developing your own codebook through iterative refinement. Manually verify each code against the original transcripts and establish inter-rater reliability by having a second coder independently apply the codebook to a subset. Document the AI's role in preliminary coding transparently in your methods section, as methodological transparency is essential in qualitative education research.

Curriculum Document Analysis

Time: 1-2 weeks | Bloom's: Analyze | Tools: notebooklm, claude, chatpdf

Upload curricula to Claude. Extract learning objectives, pedagogical approaches, assessment methods. Compare across institutions.

Implementation Instructions

Collect curriculum documents from multiple institutions or programs and upload them to Claude. Ask it to extract and categorize: stated learning objectives, pedagogical approaches mentioned, assessment methods described, alignment with professional standards, and attention to diversity/equity/inclusion. Create a structured comparison matrix showing similarities and differences across documents. Your analytical contribution is interpretation: What patterns emerge? What do differences in curriculum design reveal about institutional values, disciplinary norms, or philosophical approaches to education? The AI accelerates the extraction and categorization work; your expertise provides the critical analysis.

Student Artifact Analysis

Time: 2-3 weeks | Bloom's: Evaluate | Tools: notebooklm, claude, chatpdf

AI helps identify patterns in student work samples. Human codes for learning progressions, misconceptions, skill development.

Implementation Instructions

Collect a representative sample of student work artifacts (with appropriate consent and anonymization). Upload to Claude or NotebookLM and ask the AI to identify patterns: common approaches, recurring misconceptions, skill progression indicators, and quality variation. Use these AI-identified patterns as hypotheses to test through your own systematic coding. Document your coding framework, apply it consistently across all artifacts, and calculate inter-rater reliability with a second coder. The AI's pattern recognition accelerates your initial analysis, but the interpretive work — connecting patterns to learning theories, developmental frameworks, and pedagogical implications — is entirely your scholarly contribution.

Differentiated Instruction Planning

Time: 1-2 weeks | Bloom's: Create | Tools: chatgpt, claude

AI generates multiple versions of lesson for different readiness levels. Pre-service teachers adapt for English learners, gifted, struggling readers.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI generates multiple versions of lesson for different readiness levels. Pre-service teachers adapt for

English learners, gifted, struggling readers. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Classroom Management Scenario Practice

Time: ongoing | Bloom's: Apply | Tools: chatgpt, claude

ChatGPT simulates challenging student behaviors. Pre-service teachers practice de-escalation, restorative practices, positive behavior support strategies.

Implementation Instructions

This application activity asks students to use AI tools within a structured workflow to practice skills. Begin by reviewing the activity description and adapting it to your specific course content and student level. ChatGPT simulates challenging student behaviors. Pre-service teachers practice de-escalation, restorative practices, positive behavior support strategies. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

IEP Goal Writing Workshop

Time: 1 week | Bloom's: Create | Tools: chatgpt, claude

AI drafts IEP goals based on student profile. Students revise for SMART criteria, legal compliance, family-friendly language. Prepare for special education.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI drafts IEP goals based on student profile. Students revise for SMART criteria, legal compliance, family-friendly language. Prepare for special education. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Parent Communication Templates

Time: ongoing | Bloom's: Create | Tools: chatgpt, claude

Draft emails to parents about student progress, behavior concerns, positive news. AI suggests tone adjustments. Practice professional, empathetic communication.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Draft emails to parents about student progress, behavior concerns, positive news. AI suggests tone adjustments. Practice professional, empathetic communication. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Universal Design for Learning Retrofit

Time: 1-2 weeks | Bloom's: Apply | Tools: chatgpt, claude

Take traditional lesson plan. Use UDL principles and AI to add multiple means of representation, expression, engagement. Make learning accessible to all.

Implementation Instructions

This application activity asks students to use AI tools within a structured workflow to practice skills. Begin by reviewing the activity description and adapting it to your specific course content and student level. Take traditional lesson plan. Use UDL principles and AI to add multiple means of representation, expression, engagement. Make learning accessible to all. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Formative Assessment Rapid Prototyping

Time: 1-2 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Describe your learning objectives and student misconceptions to AI. Generate 10+ formative assessment items in varied formats (exit tickets, polls, concept checks). Select and refine the best.

Implementation Instructions

Describe your specific learning objectives, common student misconceptions you've observed, and available class time to Claude. Ask it to generate 10+ formative assessment items in varied formats: exit ticket prompts, multiple-choice concept checks, think-pair-share questions, minute papers, and application scenarios. For each item, evaluate: Does it actually reveal whether students understand or just recall? Will it catch the specific misconceptions you've identified? Is it feasible in the time available? Select and refine the best 5-6 items for your assessment toolkit. AI generates quantity and variety; your pedagogical expertise selects for quality and fit.

Differentiated Assignment Options

Time: 2-3 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm

AI generates multiple ways students could demonstrate the same learning outcome: written, visual, oral, multimedia, experiential. You select options appropriate to your context.

Implementation Instructions

For a key learning outcome in your course, ask Claude: 'Generate 8-10 different ways students could demonstrate mastery of this outcome: [paste outcome]. Include written, visual, oral, multimedia, and experiential options. For each option, suggest assessment criteria.' Review the AI's suggestions against your context: Which options are feasible with your resources? Which genuinely assess the outcome rather than a related but different skill? Which provide appropriate challenge while being accessible to diverse learners? Select 3-4 options to offer students, with a shared rubric that can evaluate different formats fairly. This implements UDL's multiple means of action and expression principle.

Environmental & Earth Sciences

Climate Data Interpretation Lab

Time: 2 hrs | Bloom's: Analyze | Tools: perplexity, julius, claude, chatgpt

Students upload real climate datasets to Julius AI. Evaluate the AI's trend analysis against published climate science. Identify where AI oversimplifies complex Earth systems.

Implementation Instructions

Provide students with real climate datasets (temperature records, CO2 measurements, sea level data — freely available from NOAA, NASA, or IPCC). Students upload to Julius AI and request trend analysis, projections, and correlations. Then they compare the AI's analysis against published climate science: Does the AI account for natural variability? Does it distinguish correlation from causation? Does it acknowledge uncertainty ranges? Students write a report correcting the AI's analysis where needed and demonstrating that climate science requires understanding complex Earth systems, not just statistical trend lines.

Environmental Impact Statement Review

Time: 1 week | Bloom's: Evaluate | Tools: perplexity, julius, claude, chatgpt

AI drafts an environmental impact assessment for a hypothetical development project. Students critique it against regulatory standards and ecological science. Identify omissions and biases.

Implementation Instructions

Describe a hypothetical development project (new factory, housing development, pipeline) to ChatGPT and ask it to draft an environmental impact assessment. Students review the AI's assessment against actual regulatory standards (NEPA, state environmental laws) and ecological science: Does it address all required categories? Does it adequately assess cumulative impacts? Does it consider environmental justice implications? Are the proposed mitigations realistic and sufficient? Students produce a peer-review style critique identifying deficiencies. This teaches environmental assessment skills while revealing AI's tendency to produce plausible-sounding but superficial environmental analysis.

Faculty Productivity & Administration

Letter of Recommendation Drafting

Time: 30-45 min | Bloom's: Create | Tools: claude, chatgpt, copilot, grammarly

Provide Claude with student achievements, course performance, and character notes. AI generates a structured first draft you personalize with specific anecdotes and your authentic voice.

Implementation Instructions

Compile your notes about the student: course grades, notable projects, class participation patterns, character observations, and career goals. Paste these into Claude with the prompt: 'Draft a letter of recommendation for [graduate school/job] emphasizing [specific strengths]. Use a warm, professional academic tone.' The AI will produce a structured first draft covering qualifications, character, and potential. Your critical role is to personalize: replace generic praise with specific anecdotes only you would know, adjust the tone to match your authentic writing voice, and verify all factual claims. This cuts drafting time significantly while ensuring every letter is substantive and personal.

Committee Report Summarization

Time: 20 min | Bloom's: Apply | Tools: claude, chatgpt, copilot, grammarly

Upload lengthy committee meeting notes or minutes to Claude. Generate concise executive summaries with action items, responsible parties, and deadlines extracted automatically.

Implementation Instructions

After a committee meeting, paste your raw notes (or the official minutes) into Claude with the prompt: 'Create a concise executive summary of this meeting. Extract all action items with responsible parties and deadlines. Flag any unresolved issues.' Review the AI summary for accuracy — it may miss political nuance or interpersonal context that matters. Use the formatted summary as your committee's official follow-up document. Over time, this creates a searchable archive of decisions and commitments that prevents the common problem of committee amnesia between meetings.

Email Response Batch Processing

Time: 30 min | Bloom's: Apply | Tools: claude, chatgpt, copilot, grammarly

Draft responses to common student emails (grade inquiries, extension requests, office hour scheduling) using AI templates you customize. Maintain warmth while saving hours weekly.

Implementation Instructions

Identify the 8-10 most common types of student emails you receive (grade inquiries, extension requests, advising questions, recommendation requests, etc.). For each type, draft a template response using Claude that's warm, professional, and covers the key information. Save these as your personal template library. When processing email, paste the student's message and select the closest template as a starting point, then customize with student-specific details. This can cut email processing time by 50% while actually improving response quality — because templates ensure you never forget important policy information in your rush to reply.

Tenure & Promotion Narrative Drafting

Time: 1-2 weeks | Bloom's: Create | Tools: claude, chatgpt, copilot, grammarly

Bullet-point your teaching, research, and service accomplishments. AI helps structure the narrative arc for your file. You revise for voice, accuracy, and strategic emphasis.

Implementation Instructions

Create a comprehensive bullet-point list of your accomplishments organized by category: teaching innovations, research publications, grants, service contributions, and professional development. Paste into Claude with the prompt: 'Help me structure a tenure/promotion personal statement that weaves these accomplishments into a coherent narrative arc. Emphasize impact and trajectory, not just listing achievements.' The AI provides structural scaffolding — connecting themes across categories, suggesting how to frame setbacks as growth, and ensuring completeness. You must heavily revise for your authentic voice, verify all claims, and ensure strategic emphasis aligns with your institution's criteria. The narrative must ultimately be genuinely yours.

Syllabus AI Policy Generator

Time: 45 min | Bloom's: Create | Tools: claude, chatgpt, copilot, grammarly

Input your course type, discipline, and AI philosophy. Claude generates three syllabus AI policy options (permissive, moderate, restrictive) tailored to your context. You select and refine.

Implementation Instructions

Describe your course to Claude: discipline, level, primary assignments, and your personal philosophy about AI use in learning. Ask it to generate three tailored AI policy statements: one permissive (AI as an encouraged learning tool), one moderate (AI allowed with documentation requirements), and one restrictive (AI prohibited except for specified tasks). For each, the AI should explain the pedagogical rationale and suggest enforcement mechanisms. Select the approach that best fits your course, then

refine the language for your institution's academic integrity framework. Test the policy by asking a colleague to identify any ambiguities students might exploit.

Flipped Classroom & Active Learning

Pre-Class AI Comprehension Check

Time: 15 min pre-class | Bloom's: Remember | Tools: chatgpt, claude, notebooklm

Students use ChatGPT to generate 5 quiz questions from the assigned reading, then answer them. Submit both the questions and answers as proof of engagement before class.

Implementation Instructions

Before each class session, students read the assigned material and then prompt ChatGPT: 'Generate 5 quiz questions at different Bloom's levels based on this content: [paste key concepts or summary].' Students answer their own AI-generated questions and submit both the questions and answers before class. This serves multiple purposes: confirms reading completion, generates self-assessment, and creates a bank of student-generated questions you can use for review. In class, select the most interesting student-generated questions for group discussion rather than lecturing on content they've already engaged with.

Think-Pair-Share with AI Third Voice

Time: 20 min | Bloom's: Analyze | Tools: chatgpt, claude, notebooklm

Students think individually, pair to discuss, then add AI as a 'third voice.' Trios compare human consensus to AI perspective. Where do humans and AI diverge?

Implementation Instructions

Pose a discussion question to the class. Students think individually for 2 minutes, then pair to discuss for 5 minutes and reach a tentative position. Next, one partner submits the question to ChatGPT while the other documents their human consensus. Pairs compare their position to the AI's: Where do they agree? Where do they diverge? Is the AI's perspective something they hadn't considered, or is it superficial compared to their discussion? Report out to the class. This evolution of the classic think-pair-share adds a critical AI literacy dimension without sacrificing the peer learning that makes the technique effective.

Jigsaw with AI Expert Panels

Time: 50 min | Bloom's: Analyze | Tools: chatgpt, claude, notebooklm

In jigsaw format, each expert group uses AI to research their subtopic deeply. Home groups synthesize. Tests whether AI-assisted research improves or flattens jigsaw outcomes.

Implementation Instructions

Divide a complex topic into 4-5 subtopics. In expert groups, students use AI tools to research their assigned subtopic deeply: Perplexity for cited overviews, ChatGPT for explanations, NotebookLM for uploaded readings. Expert groups create a 1-page summary and 3 discussion questions. Students then return to home groups where each expert teaches their subtopic. Home groups synthesize across all subtopics to answer an overarching question. Debrief: Did AI-assisted research improve the quality of expert group preparation? Did any group over-rely on AI and arrive with shallow understanding? This reveals whether AI enhances or undermines cooperative learning.

Muddiest Point AI Clarification

Time: 10 min end of class | Bloom's: Understand | Tools: chatgpt, claude, notebooklm

Students submit their 'muddiest point' from today's class. Use AI to draft brief clarification responses. Post curated responses before next session.

Implementation Instructions

At the end of class, students submit their muddiest point — the concept they found most confusing. Collect all responses and paste them into Claude: 'For each of these student confusions about [today's topic], draft a brief, clear clarification that a [level] student would understand.' Review each AI clarification for accuracy and add your own context where needed. Post the curated Q&A before the next session. This transforms the muddiest point from a diagnostic tool into an instructional intervention — students get their specific confusion addressed within 24 hours, and you identify patterns that should shape future instruction.

General

Asynchronous Discussion Synthesis

Time: weekly | Bloom's: Analyze | Tools: chatgpt, claude, otter, notebooklm

After a week of asynchronous discussion posts, AI synthesizes key themes across all posts. Instructor uses synthesis to design the synchronous session around actual student thinking.

Implementation Instructions

At the end of each asynchronous discussion cycle, compile all student posts (anonymized) and paste into Claude: 'Synthesize the key themes, points of agreement, points of disagreement, and unanswered questions from these discussion posts about [topic].' Use the AI synthesis to design your synchronous session around what students actually said — not what you assumed they'd discuss. Open the synchronous session by presenting the synthesis and asking: 'Is this an accurate representation of our discussion?' Then focus synchronous time on resolving disagreements, exploring unanswered questions, and deepening the strongest threads. This bridges the async-sync gap that often plagues hybrid courses.

Equity Parity Check for Modalities

Time: monthly | Bloom's: Evaluate | Tools: chatgpt, claude, otter, notebooklm

Upload assignment descriptions and participation data for in-person and remote students. AI identifies whether both modalities have equitable access to learning opportunities.

Implementation Instructions

Monthly, compile your course's participation data, assignment submissions, and learning activity descriptions for both in-person and remote students. Upload to Claude: 'Analyze whether in-person and remote students in this HyFlex course have equitable access to: participation opportunities, feedback, peer interaction, and assessment accommodations. Identify any disparities.' The AI's analysis may reveal patterns you haven't noticed: remote students getting less informal feedback, in-person students dominating discussion, or assessment formats that disadvantage one modality. Use the findings to adjust your course design for more equitable outcomes across both modalities.

AI Session Recap for Absent Students

Time: per session | Bloom's: Remember | Tools: chatgpt, claude, otter, notebooklm

Use Otter.ai to transcribe class sessions. AI generates a structured recap with key concepts, decisions, and action items. Absent students get equivalent information, not just slides.

Implementation Instructions

Record every class session (Zoom for remote, microphone for in-person) and transcribe using Otter.ai. Feed the transcript into Claude: 'Create a structured session recap covering: key concepts discussed,

examples used, questions raised, decisions made, and action items with deadlines.' Review for accuracy, add context the AI couldn't capture (nonverbal reactions, whiteboard content, spontaneous demonstrations), and post within 24 hours. This ensures students who were absent, needed to attend the other modality, or have accessibility needs receive functionally equivalent information — not just slides, but the actual intellectual content of the session.

Academic Vocabulary Builder

Time: ongoing | Bloom's: Remember | Tools: chatgpt, claude, perplexity

Students encounter unfamiliar academic terms and use AI for context-specific definitions. Build a personal glossary with AI-assisted definitions verified against textbook usage.

Implementation Instructions

First-year students often struggle with academic vocabulary that professors take for granted. Create a running assignment where students identify 3-5 unfamiliar terms from each week's readings and prompt ChatGPT: 'Define [term] as it's used in [discipline]. Give me an example that a first-year college student would understand.' Students verify the AI definition against the textbook usage and note any discrepancies. By semester's end, each student has a personalized glossary of 40-60 discipline-specific terms with definitions in their own words. This bridges the vocabulary gap that disproportionately affects first-generation college students.

Office Hours Preparation Coach

Time: 10 min before office hours | Bloom's: Apply | Tools: chatgpt, claude, perplexity

Students use AI to articulate their questions before visiting office hours. Practice explaining what they don't understand. Arrive prepared for a productive conversation.

Implementation Instructions

Many first-year students avoid office hours because they can't articulate what they don't understand. Before office hours, students prompt ChatGPT: 'I'm confused about [topic] in my [course]. Help me articulate what specifically I don't understand so I can ask my professor a clear question.' The AI helps them narrow vague confusion ('I don't get any of it') into specific, answerable questions ('I understand X but don't see how it connects to Y'). Students bring their AI-refined questions to office hours. This removes a significant barrier to help-seeking and makes office hour conversations dramatically more productive for both student and instructor.

Source Evaluation Bootcamp

Time: 2 class sessions | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

Students evaluate AI-generated, Wikipedia, social media, and peer-reviewed sources on the same topic. Build a source quality hierarchy and learn when each type is appropriate.

Implementation Instructions

Select a topic relevant to your course. Provide students with four sources on that topic: (1) an AI-generated summary from ChatGPT, (2) a Wikipedia article, (3) a social media post or blog, and (4) a peer-reviewed journal article. In session one, students evaluate each source using the CRAAP test (Currency, Relevance, Authority, Accuracy, Purpose). In session two, discuss as a class: When is each source type appropriate? When is it insufficient? How do the sources differ in depth, accuracy, and bias? Students create a personal 'source hierarchy' decision tree they'll use for all future research. This foundational information literacy lesson shapes their entire academic career.

Academic Stress Management Toolkit

Time: 1 class session | Bloom's: Apply | Tools: chatgpt, claude

Students use AI to create personalized study schedules, break down overwhelming assignments into steps, and identify campus resources. AI as organizational assistant, not therapist.

Implementation Instructions

When students feel overwhelmed, teach them to use AI as an organizational tool: 'I have these assignments due [list with dates]. Help me create a realistic daily schedule that includes study blocks, breaks, and sleep.' Or: 'Break this large assignment into manageable daily tasks over [timeframe].' Or: 'What campus resources at [institution] might help with [specific concern]?' Emphasize that AI is an organizational assistant, not a substitute for counseling or professional support. For emotional distress, always direct students to campus counseling services. This teaches a practical, healthy use of AI that reduces academic anxiety through better planning rather than avoidance.

Growth Mindset Reflection Prompts

Time: 5 min per assignment | Bloom's: Evaluate | Tools: chatgpt, claude

After receiving a grade, students use AI-generated reflection prompts to process feedback constructively. Focus on learning from mistakes rather than defending performance.

Implementation Instructions

After returning graded assignments, provide students with AI-generated reflection prompts that encourage growth mindset processing: 'What's one thing you learned from the feedback that you'll apply next time?' 'Which section would you revise first, and what would you change?' 'What study strategy would you try differently for the next assignment?' Students write brief (100-word) responses to 2-3 prompts. Grade for completion, not content. The goal is to build the habit of constructive reflection after receiving feedback — turning the emotional moment of seeing a grade into a learning opportunity. This is especially important for first-year students adjusting to college-level expectations.

Professional Email Communication Lab

Time: 45 min | Bloom's: Create | Tools: chatgpt, claude, perplexity

AI generates professional scenarios (client complaint, colleague disagreement, supervisor update). Students draft emails, get AI feedback on tone and clarity, revise for professional standards.

Implementation Instructions

Create 4-5 professional email scenarios relevant to your discipline: responding to a client complaint, requesting a deadline extension from a supervisor, providing constructive feedback to a colleague, delivering unwelcome news to a stakeholder, or following up after a meeting. Students draft emails for each scenario, then submit to Claude: 'Evaluate this professional email for tone, clarity, completeness, and professional appropriateness. Suggest specific improvements.' Students revise based on AI feedback, then peer-review each other's revised versions. This builds professional communication skills that every graduate needs but few courses explicitly teach.

Industry Trends Rapid Briefing

Time: 1 hr | Bloom's: Analyze | Tools: chatgpt, claude, perplexity

Students use Perplexity to research current trends in their professional field. Synthesize into a 1-page industry briefing. Present to class as 'industry update' to build current awareness.

Implementation Instructions

Each week, assign one student or team to create a 1-page 'industry briefing' on a current trend in your professional field. They use Perplexity to research recent developments, verify key claims with primary sources, and synthesize into a structured brief: What's happening? Why does it matter? How does it connect to what we're learning? What should future professionals in this field know? Present as a 5-

minute 'industry update' at the start of class. By semester's end, every student has practiced professional research, synthesis, and presentation while the class maintains current awareness of their field — a habit essential for lifelong professional competence.

Career-Aligned Portfolio Reflection

Time: end of course | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

Students use AI to identify which course artifacts best demonstrate specific professional competencies. Build a portfolio with AI-assisted career alignment analysis.

Implementation Instructions

At course end, students compile their best work and use Claude: 'I'm building a professional portfolio for a career in [field]. Which of these course artifacts best demonstrate these competencies: [list industry-relevant skills]? How should I frame each piece for a professional audience?' Students use the AI's suggestions to select, curate, and frame their portfolio pieces with career-focused introductions. But they must also add their own reflection: What did they learn from creating each piece? How did it develop their professional capabilities? The AI helps with strategic framing; students provide the authentic professional development narrative.

Cultural Perspective Analysis

Time: 1-2 hrs | Bloom's: Analyze | Tools: perplexity, claude, chatgpt

AI generates explanations of a course concept from different cultural perspectives. Students evaluate which perspectives are represented, which are missing, and what biases the AI reveals.

Implementation Instructions

Select a course concept that varies across cultural contexts (justice, family, health, education, leadership, beauty, success). Ask ChatGPT: 'Explain how [concept] is understood in 5 different cultural traditions.' Students analyze the AI's responses: Which cultural perspectives are well-represented? Which are reduced to stereotypes? Which are entirely absent? What Western-centric assumptions does the AI default to? Students research one underrepresented perspective using primary sources and write a corrective addition. This reveals AI's cultural biases while teaching students to seek perspectives beyond the AI's training data and their own cultural assumptions.

Global Case Study Localization

Time: 1 week | Bloom's: Create | Tools: perplexity, claude, chatgpt

Take a case study set in one context. Students use AI to adapt it for a different cultural, economic, or regulatory context. Analyze how context changes the analysis and solutions.

Implementation Instructions

Select a case study originally set in one national or cultural context (a U.S. healthcare case, a European business case, etc.). Students use Perplexity and Claude to research a different context (a developing nation, a different cultural tradition, a different regulatory framework) and adapt the case: How would the problem manifest differently? What additional factors would matter? What solutions would and wouldn't transfer? Students present their adapted case alongside the original, analyzing how context fundamentally changes both the problem and the appropriate response. This builds the contextual thinking essential for any global professional.

Grading, Feedback & Revision

Dialogic Feedback Protocol

Time: per assignment | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

Students receive AI feedback on their draft, then write a 'response memo' explaining what they accepted, rejected, and why. Builds critical reading of all feedback — not just AI.

Implementation Instructions

After students receive AI feedback on their draft (from ChatGPT, Claude, or Grammarly), require them to write a structured 'response memo' of 300-500 words. The memo must address each major piece of AI feedback: What did they accept and implement (with reasoning)? What did they reject (with justification)? What did they investigate further before deciding? This transforms passive receipt of feedback into active critical engagement. Grade the memo alongside the revised draft — the quality of their reasoning about feedback matters as much as the final product. Over time, students develop the habit of critically evaluating all feedback, not just AI-generated feedback.

Peer Review Calibration

Time: 1 hr | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

Students use Claude to critique a sample paper, then compare their peer review to the AI's. Class discusses: what do human reviewers notice that AI consistently misses?

Implementation Instructions

Select a sample student paper (anonymized or purpose-written) and distribute it to the class. Have students write their own peer review first, using your course's rubric or feedback guidelines. Then submit the same paper to Claude and share the AI's review with the class. In a facilitated discussion, compare the two types of feedback: What did human reviewers consistently notice that AI missed? (Usually: argument strength, disciplinary conventions, creative risk-taking.) What did AI consistently catch? (Usually: structural issues, grammar, clarity.) This calibrates student peer review skills while building awareness of AI's feedback strengths and blind spots.

Revision Trace Assignment

Time: per draft cycle | Bloom's: Create | Tools: claude, chatgpt, grammarly

Students must track every change between draft 1 and draft 2 in a revision log — including noting which changes were AI-suggested vs. self-directed. Transparency by design.

Implementation Instructions

For each draft cycle, require students to maintain a revision log — a simple table documenting every significant change between drafts. Each entry includes: what changed, why it changed, and whether the change was self-directed, peer-suggested, instructor-suggested, or AI-suggested. The revision log is submitted alongside the revised draft. This creates transparency by design: you can see exactly how AI influenced the writing process without prohibiting its use. Over time, patterns emerge — students who over-rely on AI suggestions show narrow revision patterns, while strong revisers integrate multiple feedback sources and make independent editorial decisions.

AI-Augmented Rubric Feedback

Time: per-assignment | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

AI drafts feedback for each rubric criterion. Instructor edits for specificity and tone. Speeds feedback without sacrificing quality.

Implementation Instructions

For each assignment, create rubric-aligned feedback templates using Claude: paste your rubric and ask it to draft specific, constructive feedback for each performance level of each criterion. During grading, select the appropriate feedback level for each criterion and customize with student-specific

details (mentioning specific examples from their work, connecting to their previous submission). This hybrid approach — AI template + human personalization — cuts grading time significantly while maintaining the specificity and warmth that generic rubric comments lack. Review AI-drafted feedback for tone before each grading session and adjust language that feels impersonal.

Common Error Pattern Analysis

Time: after-grading | Bloom's: Analyze | Tools: claude, chatgpt, grammarly

Upload anonymized student work. AI identifies common mistakes, misconceptions. Create targeted mini-lesson addressing patterns.

Implementation Instructions

After grading a batch of assignments, upload anonymized student responses (with identifying information removed) to Claude. Ask it to identify: recurring mistakes, common misconceptions, patterns in where students went wrong, and the 3-5 most frequent errors. Use these patterns to design a targeted mini-lesson for the next class session that addresses the most common issues without calling out individual students. This data-driven approach to formative instruction means you're teaching to actual student needs rather than assumed ones. Share the pattern analysis (without student identification) with students so they can self-identify their own error patterns.

Progressive Feedback Loops

Time: semester-long | Bloom's: Apply | Tools: claude, chatgpt, grammarly

Students submit drafts → AI feedback → revision → peer feedback → revision → instructor feedback. Scaffolded improvement process.

Implementation Instructions

Design a multi-stage submission process: Draft 1 → AI feedback (students submit to ChatGPT or Claude and annotate the response) → Revision 1 → Peer feedback (using your rubric) → Revision 2 → Instructor feedback on final draft. Each stage produces documented evidence of engagement with feedback. Students improve through multiple feedback sources, learning to triangulate suggestions and make independent revision decisions. Grade the process (quality of annotations, peer review thoroughness, revision rationale) as heavily as the final product. This scaffolded approach builds revision habits and feedback literacy that transfer to professional writing contexts.

Rubric-Aligned Feedback Templates

Time: setup: 2 hrs | Bloom's: Create | Tools: claude, chatgpt, grammarly

For each rubric criterion and performance level, AI drafts specific, constructive feedback. You customize per student. Cuts grading time 40% while improving feedback specificity.

Implementation Instructions

Take your assignment rubric and for each criterion at each performance level (e.g., 'Excellent,' 'Proficient,' 'Developing,' 'Beginning'), ask Claude to draft 2-3 specific, constructive feedback paragraphs. Each should identify what the student did well at that level, what's missing, and one concrete suggestion for improvement. During grading, select the appropriate feedback level for each criterion and add student-specific examples from their actual work. Research from Harvard Business School shows this approach cuts grading time by over 25% while students report receiving more helpful feedback because the comments are specific rather than hastily scrawled.

Common Error Pattern Report

Time: 1 hr post-grading | Bloom's: Analyze | Tools: claude, chatgpt, grammarly

After grading a batch, paste anonymized student responses into Claude. AI identifies the top 5 misconceptions and recurring errors. Use this to design targeted mini-lessons.

Implementation Instructions

After grading a set of assignments, compile anonymized excerpts that represent common errors (remove all student names and identifiers). Paste into Claude with the prompt: 'Identify the top 5 recurring errors, misconceptions, or skill gaps in these student responses. For each, suggest a brief instructional intervention.' Use the AI's pattern analysis to design a targeted mini-lesson for the next class session. Share the anonymized patterns (never individual student work) so students can self-identify their own tendencies. This data-driven approach to formative instruction means you're teaching to actual student needs rather than assumed ones.

Peer Review Calibration Exercise

Time: 1 class session | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

AI generates a sample assignment response. Students use your rubric to grade it independently, then compare scores. Calibrate expectations before peer review begins.

Implementation Instructions

Use ChatGPT to generate a sample response to your assignment at an intentionally 'B+' level — good but with identifiable weaknesses. Distribute this sample to students along with your rubric and ask them to independently grade it, providing scores for each criterion with justifications. In class, reveal the range of student scores (which will vary widely) and facilitate a calibration discussion: Why did scores differ? Which rubric criteria were interpreted differently? What does 'proficient' actually look like? This calibration exercise before peer review dramatically improves the quality and consistency of student feedback on each other's work.

Two-Stars-and-a-Wish AI Feedback

Time: per assignment | Bloom's: Evaluate | Tools: claude, chatgpt, grammarly

Students submit work to AI asking for 'two strengths and one area for growth.' They annotate whether they agree with each point before submitting to you. Builds feedback literacy.

Implementation Instructions

Before submitting final work to you, students submit their draft to ChatGPT or Claude with the prompt: 'Identify two specific strengths in this work and one specific area for growth.' Students then annotate the AI's three points: Do they agree with each strength? Is the growth area the same one they would have identified? What would they add or change about the AI's assessment? Submit the original work, AI feedback, and annotations together. This builds feedback literacy — the ability to receive, interpret, and act on feedback — which research shows is one of the most transferable academic skills.

Graduate Student & Thesis Support

Systematic Review Scoping Sprint

Time: half-day | Bloom's: Analyze | Tools: claude, perplexity, elicit, scite, notebooklm, resrabbit

Use Elicit + Semantic Scholar to scope a systematic review: define search terms, run queries, document inclusion/exclusion criteria, and identify AI tool limitations in your process.

Implementation Instructions

Begin by defining your research question and inclusion/exclusion criteria as precisely as possible. Use Elicit to run initial searches and extract structured data from the first 50-100 results: study design, population, key findings, and relevance scores. Simultaneously use Semantic Scholar to identify highly-cited foundational papers you may have missed. Document every search term, database, and AI

tool used — this transparency is required in the methods section of any systematic review. Critically, note the limitations of AI-assisted screening: what types of papers might the AI miss? What biases might the search terms introduce? This self-awareness about methodology is what distinguishes doctoral-level work.

Research Gap Identification

Time: 2–3 hrs | Bloom's: Evaluate | Tools: claude, perplexity, elicit, scite, notebooklm, resrabbit

After citation network mapping in ResearchRabbit, write a gap analysis memo. Claude scaffolds the argument structure; Scite validates that claims are well-supported.

Implementation Instructions

After building your citation network in ResearchRabbit and verifying evidence quality in Scite, write a structured gap analysis memo. Use Claude to help organize your observations: paste your notes about what you've found and ask it to suggest logical frameworks for presenting the gaps (e.g., 'Here are patterns in what's been studied; help me articulate what's missing systematically'). Claude can suggest organizational structures — chronological, methodological, theoretical, or population-based gaps. Write the final memo yourself, using the AI's organizational logic as scaffolding. This memo becomes the foundation for your dissertation's problem statement and justification.

Prospectus Peer Review with AI

Time: 2 hrs | Bloom's: Evaluate | Tools: claude, perplexity, elicit, scite, notebooklm, resrabbit

Exchange prospectus sections. Each student also asks Claude to review theirs. Compare human vs. AI feedback — what did each source notice that the other missed?

Implementation Instructions

Exchange prospectus sections with a fellow graduate student. Each student also submits their own section to Claude with the prompt: 'Review this dissertation prospectus section for clarity, logical coherence, and potential weaknesses in the argument.' Compare the human peer review to the AI review: What did your peer catch that AI missed (likely disciplinary nuance, theoretical positioning)? What did AI catch that your peer missed (likely structural issues, unclear phrasing, logical gaps)? Write a brief reflection on the value of each feedback source. This exercise builds feedback literacy while demonstrating that no single source of feedback — human or AI — is sufficient alone.

Theoretical Framework Integration

Time: 2-4 weeks | Bloom's: Create | Tools: elicit, claude, semscholar, connectedpapers

Map how different theories address your research question. Use Connected Papers to trace theoretical lineages. AI suggests synthesis.

Implementation Instructions

Map how 3-5 different theories address your central research question. Use Connected Papers to trace the intellectual lineage of each theory — who developed it, how it evolved, and which researchers currently work within each framework. Claude can help organize the comparison: Create a matrix showing how each theory conceptualizes your key variables, what mechanisms it proposes, what evidence supports it, and where it falls short. Your original contribution is the synthesis: proposing an integrative framework that combines the strongest elements of multiple theories while addressing their individual limitations. This synthesis becomes a key contribution of your literature review.

Methodological Review Matrix

Time: 1-2 weeks | Bloom's: Analyze | Tools: elicit, claude, semscholar, connectedpapers

Extract methods from 50+ papers using Elicit. Create comparison table: sample sizes, designs, measures, findings. Identify best practices.

Implementation Instructions

Use Elicit to extract methodological details from 50+ papers in your research area: study design, sampling strategy, sample size, measurement instruments, analytical techniques, and key findings. Create a comprehensive comparison matrix that reveals patterns: What methods dominate the field? What populations are overrepresented or underrepresented? What measurement approaches show the strongest validity evidence? What analytical techniques yield the most consistent results? This matrix identifies methodological best practices and gaps — areas where better methods could advance the field. The matrix itself often becomes a publishable contribution, especially in rapidly evolving research areas.

Research Evolution Timeline

Time: 1 week | Bloom's: Analyze | Tools: elicit, claude, semscholar, connectedpapers

Use Semantic Scholar citation data to show how research question evolved over decades. Identify paradigm shifts, key turning points.

Implementation Instructions

Use Semantic Scholar's citation data and publication dates to construct a timeline showing how research on your topic has evolved over decades. Identify key inflection points: landmark papers that shifted the conversation, methodological innovations that opened new questions, and replication crises or paradigm challenges that reshaped understanding. Claude can help organize the chronological narrative, but your expertise is essential for identifying which shifts were truly transformative versus merely fashionable. This timeline provides compelling context for your own research — showing readers where your work fits in the field's intellectual history and why your contribution is timely.

Teaching Philosophy Development

Time: 2-3 weeks | Bloom's: Create | Tools: chatgpt, claude, notebooklm

Grad students draft teaching philosophy. AI suggests examples, frameworks, revision strategies. Peer review with AI feedback integration.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Grad students draft teaching philosophy. AI suggests examples, frameworks, revision strategies. Peer review with AI feedback integration. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Lesson Plan Workshop

Time: module | Bloom's: Create | Tools: chatgpt, claude, notebooklm

Grads design backward-designed lessons. AI provides formative feedback on alignment: objectives → activities → assessment.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Grads design backward-designed lessons. AI provides formative feedback on alignment: objectives →

activities → assessment. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Peer Teaching Observation Protocol

Time: ongoing | Bloom's: Evaluate | Tools: chatgpt, claude, notebooklm

Grads observe each other teaching. AI helps structure observation notes using validated protocols. Debrief focuses on evidence-based feedback.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. Grads observe each other teaching. AI helps structure observation notes using validated protocols. Debrief focuses on evidence-based feedback. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Dissertation Literature Gap Map

Time: 1-2 weeks | Bloom's: Analyze | Tools: claude, elicit, semscholar, scite, perplexity, notebooklm

Graduate students use Elicit + ResearchRabbit to map their field comprehensively. AI identifies under-researched areas. Students validate gaps against advisor expertise.

Implementation Instructions

Graduate students enter 5-7 seed papers central to their dissertation topic into ResearchRabbit and use Elicit to systematically extract methods, findings, and populations from 50+ related papers. The AI helps identify: populations not studied, methods not applied to this question, geographic regions underrepresented, and theoretical frameworks not yet combined. Students create a visual gap map showing established research (dense clusters) and potential gaps (sparse areas). Validate every identified gap with your advisor — some gaps exist for good methodological reasons. The validated gap map becomes the foundation for your dissertation's contribution statement.

Thesis Chapter Draft Review

Time: per chapter | Bloom's: Evaluate | Tools: claude, elicit, semscholar, scite, perplexity, notebooklm

Submit thesis chapter to Claude acting as your committee. AI identifies: argument gaps, unsupported claims, methodological questions, and missing literature. Address before committee review.

Implementation Instructions

Before submitting a chapter to your committee, paste it into Claude with the prompt: 'Act as a demanding but constructive dissertation committee member. Identify: (1) logical gaps in the argument, (2) claims without sufficient evidence, (3) methodological questions a reviewer would raise, (4) missing literature that should be cited, and (5) passages that are unclear or poorly organized.' Address every point the AI raises before your committee sees the chapter. This pre-review catches many issues that would otherwise consume committee meeting time, allowing your actual committee conversations to focus on substantive intellectual questions rather than structural problems.

Research Question Refinement Workshop

Time: 2-3 hrs | Bloom's: Evaluate | Tools: claude, elicit, semscholar, scite, perplexity, notebooklm

Graduate students iteratively refine their research question with AI. Each iteration tests for specificity, feasibility, significance, and originality. Document the evolution.

Implementation Instructions

Graduate students submit their initial research question to Claude and ask it to evaluate on four dimensions: specificity (is it narrow enough to answer?), feasibility (can it be answered with available methods and data?), significance (does it matter to the field?), and originality (what makes it new?). For each dimension, the AI suggests refinements. Students iterate 3-5 times, documenting how their question evolved. The final submission includes the original question, all intermediate versions with rationale for each change, and the refined question with a brief prospectus. This structured refinement process often compresses weeks of advisor meetings into focused self-directed work.

Oral Defense Preparation Simulator

Time: 2-3 sessions | Bloom's: Evaluate | Tools: claude, elicit, semscholar, scite, perplexity, notebooklm

AI plays devil's advocate committee member asking tough defense questions. Students practice articulating their contribution, defending methodology, and addressing limitations.

Implementation Instructions

Prepare for your oral defense by submitting your abstract and key findings to Claude with the prompt: 'Act as a critical but fair dissertation committee member. Ask me 5 challenging questions about my methodology, theoretical framework, findings, limitations, and implications.' Practice answering each question out loud (not in writing — you need oral practice). Then submit your answers back to the AI for follow-up questions. Repeat for 2-3 sessions, covering different aspects of your work each time. Record yourself answering and listen back for clarity, confidence, and precision. This doesn't replace practicing with your actual committee or peers, but it provides unlimited low-stakes practice opportunities.

Grant Writing & Proposals

Significance Section Sprint

Time: 1-2 weeks | Bloom's: Create | Tools: claude, chatgpt, perplexity, grammarly

Bullet-point contribution. ChatGPT drafts in grant language. Claude refines tone. Verify every claim with Perplexity.

Implementation Instructions

Start by bullet-pointing the core argument for your project's significance: What problem exists? Why does it matter? What gap does your research fill? What impact will it have? Paste these bullets into ChatGPT with instructions to draft in the formal register appropriate for your target funding agency. Then use Claude to refine: strengthen the argument arc, ensure every claim is supported, and adjust tone for the specific program officer audience. Use Perplexity to verify every statistical claim or citation. The final version must sound authentically like your scholarly voice — AI helps overcome the blank-page problem, but the argument must be genuinely yours.

Preliminary Data Narrative

Time: 4-6 days | Bloom's: Create | Tools: claude, chatgpt, perplexity, grammarly

Upload pilot visualizations to Claude. AI drafts compelling narrative connecting data to aims. Human revises for accuracy.

Implementation Instructions

Upload your pilot data visualizations and preliminary results to Claude. Ask it to draft a narrative connecting your preliminary data to your specific aims: What do the data show? How do they support the feasibility of your proposed approach? What limitations exist and how will the full study address them? Review the AI-drafted narrative against your actual data — AI may overstate significance or miss important caveats. Revise heavily, ensuring every claim is supported by your actual preliminary results. This section of a grant application must be precisely accurate; AI provides a first-draft scaffold that you then refine for truthfulness and precision.

Budget Justification Workshop

Time: 2-3 days | Bloom's: Create | Tools: claude, chatgpt, perplexity, grammarly

Draft budget items. AI identifies weak justifications. Claude improves clarity without inflating necessity.

Implementation Instructions

List all budget items for your grant application with brief descriptions of their purpose. Submit to Claude and ask it to: (1) identify any items with weak justifications that reviewers might question, (2) suggest stronger language connecting each item to specific project aims, and (3) flag any items that might appear excessive or under-justified. Review AI suggestions critically — it may not understand discipline-specific norms for equipment costs or personnel effort. Ensure every budget item clearly explains why it's necessary (not just what it is) and how it directly supports the proposed research activities.

Broader Impacts Narrative

Time: 1-2 weeks | Bloom's: Create | Tools: claude, chatgpt, perplexity

Draft broader impacts section for NSF-style grants. AI suggests education, outreach, diversity activities. Human aligns with institutional strengths.

Implementation Instructions

Draft the broader impacts section of your NSF-style grant by first listing all ways your research connects to: education (training students, curriculum development), diversity (broadening participation), infrastructure (shared resources, databases), and public benefit (practical applications, policy implications). Ask Claude to help develop each connection into a compelling narrative paragraph that demonstrates concrete plans, not just aspirational statements. Every broader impact should include: what you'll do, how you'll do it, who benefits, and how you'll assess success. Align broader impacts with your institution's existing programs and resources to demonstrate feasibility.

Letter of Support Coordination

Time: 1 week | Bloom's: Apply | Tools: claude, chatgpt, perplexity

AI drafts template letters for collaborators explaining their role, contribution, resources. Personalize for each partner institution.

Implementation Instructions

Draft a template letter of support that includes: (1) a description of the collaboration and each partner's specific role, (2) the resources or expertise they'll contribute, (3) a statement of institutional commitment, and (4) alignment with the partner's mission. Use Claude to customize the template for each collaborator, adjusting language to match their institution type (university, government agency, community organization, industry partner). Send the draft to each collaborator for review and

personalization — they should revise to reflect their authentic voice and institutional commitments. Coordinating letters early avoids the last-minute scramble that often compromises grant quality.

Facilities & Resources Description

Time: 2-3 days | Bloom's: Create | Tools: claude, chatgpt, perplexity

Claude helps document available equipment, core facilities, computational resources. Generate boilerplate adaptable across grants.

Implementation Instructions

Compile a comprehensive inventory of available resources: laboratory equipment, computing infrastructure, core facilities, library resources, office space, and field sites. Use Claude to organize this inventory into the standard grant format, connecting each resource to specific project activities. Create a modular template with sections that can be updated and reused across multiple grant applications. Include both institutional resources (shared equipment, core facilities) and department-level resources (dedicated lab space, specialized software). Describe any resources that need to be acquired (justified in the budget) and how existing resources will be supplemented by the proposed project.

Health Sciences & Nursing

Clinical Reasoning Simulation

Time: 30–45 min | Bloom's: Apply | Tools: chatgpt, perplexity, notebooklm, elicit

Present a patient scenario. Students ask ChatGPT to generate a differential diagnosis, then critique the AI's clinical reasoning against current evidence-based guidelines.

Implementation Instructions

Create or select a patient scenario with presenting symptoms, vital signs, and relevant history. Present it to students and have them individually generate initial assessments. Then have students ask ChatGPT to produce a differential diagnosis for the same case. Students compare their clinical reasoning to the AI's: Did the AI consider the same differentials? Did it weigh evidence appropriately? Did it miss contextual factors (patient age, comorbidities, social determinants)? This exercise reveals both the power and limitations of AI in clinical settings and reinforces that clinical judgment requires human contextual reasoning that AI cannot replicate.

EBP Literature Scan (PICO)

Time: 2 hrs | Bloom's: Evaluate | Tools: chatgpt, perplexity, notebooklm, elicit

Students use Perplexity + Semantic Scholar to find current evidence for a clinical PICO question. Critically appraise the top 3 sources using a structured framework.

Implementation Instructions

Have students formulate a clinical PICO question from a real or realistic patient scenario. Using Perplexity AI and Semantic Scholar together, students conduct a rapid evidence scan to find the 5 most relevant studies. For the top 3 sources, students complete a structured critical appraisal: study design, sample size, relevance to their PICO question, strength of evidence, and applicability to their clinical context. This combines information literacy with clinical reasoning — students learn not just to find evidence, but to evaluate whether it actually answers their specific clinical question.

Patient Education Draft Review

Time: 45 min | Bloom's: Evaluate | Tools: chatgpt, perplexity, notebooklm, elicit

AI drafts a patient education handout on a condition. Students identify inaccuracies, missing warnings, readability issues, and cultural considerations.

Implementation Instructions

Use ChatGPT to generate a patient education handout on a specific condition (e.g., Type 2 Diabetes management, post-surgical care instructions). Distribute the AI-generated handout to students without identifying it as AI-created. Students review it using a structured checklist: medical accuracy, reading level appropriateness (aim for 6th-8th grade), cultural sensitivity, completeness of warnings and side effects, and actionable next steps. Students must identify at least 3 specific issues and propose corrections with evidence from clinical guidelines. This builds critical review skills essential for patient safety.

PRISMA Systematic Review

Time: 4-8 weeks | Bloom's: Analyze | Tools: rayyan, elicit, anara, scite

Use Rayyan for PRISMA-compliant screening. Elicit for data extraction. Document AI involvement transparently.

Implementation Instructions

Use Rayyan for PRISMA-compliant screening of your search results: upload references, apply inclusion/exclusion criteria, and use AI-assisted relevance scoring to accelerate the process. Maintain a PRISMA flow diagram documenting each screening stage. Use Elicit for structured data extraction from included studies, creating tables of study characteristics, outcomes, and quality assessments. Document your AI involvement transparently in the methods section — increasingly, journals require explicit disclosure of AI tools used in systematic reviews. Have a second reviewer independently screen a random 20% sample to establish inter-rater reliability, ensuring the AI-assisted process is rigorous.

Clinical Trial Synthesis

Time: 2-3 weeks | Bloom's: Analyze | Tools: rayyan, elicit, anara, scite

Anara searches ClinicalTrials.gov and PubMed simultaneously. Extract interventions, outcomes, sample sizes.

Implementation Instructions

Use Anara to simultaneously search ClinicalTrials.gov and PubMed for trials relevant to your research question. Extract key data points systematically: intervention type, patient population, primary outcomes, sample sizes, and results. Create a structured evidence table comparing trials across these dimensions. Verify each extraction against the original source — AI can misread statistical results or conflate primary and secondary outcomes. Pay special attention to trial quality: randomization, blinding, follow-up rates, and intention-to-treat analysis. This synthesized evidence table becomes the foundation for your evidence evaluation and any subsequent meta-analysis.

Meta-Analysis Preparation

Time: 1-2 weeks | Bloom's: Analyze | Tools: rayyan, elicit, anara, scite

Elicit extracts effect sizes, CIs, statistics. Verify each extraction. Export to meta-analysis software.

Implementation Instructions

Use Elicit to extract quantitative data from your included studies: effect sizes, confidence intervals, sample sizes, and key covariates. Create a standardized extraction form ensuring you capture the same data points from each study. Verify every extracted number against the original paper — statistical data extraction is where AI is most likely to make errors that could invalidate your meta-analysis. Flag studies where data must be calculated or imputed (e.g., converting medians to means)

and document your conversion methods. Export the verified dataset to your meta-analysis software (RevMan, R, Stata) for pooled analysis.

PICO Question Formulation

Time: 1-2 hours | Bloom's: Create | Tools: elicit, scite, pubmed, anara

AI helps students structure clinical questions using PICO format (Population, Intervention, Comparison, Outcome). Foundation for evidence-based practice.

Implementation Instructions

Present a clinical scenario or patient case that raises a practice question. Guide students through formulating a PICO question: Population (who?), Intervention (what treatment/action?), Comparison (compared to what?), Outcome (what result?). Have students first draft their PICO independently, then use ChatGPT to refine: 'Help me sharpen this PICO question for a literature search: [student's draft].' Compare the original and AI-refined versions: Is the refined version more searchable? More specific? Students learn that a well-formulated PICO question is the foundation of evidence-based practice — and that AI can help with precision without replacing clinical judgment about what questions matter.

Rapid Evidence Appraisal

Time: 1 week | Bloom's: Evaluate | Tools: elicit, scite, pubmed, anara

Use Scite to assess evidence quality for clinical practice guideline. Check if studies are supported or contradicted by later research. Critical appraisal skills.

Implementation Instructions

Assign a clinical practice guideline topic and have students use Scite to assess the evidence underlying the guideline's key recommendations. For each recommendation, students investigate: Are the cited studies mostly supported or contradicted by later research? Has the evidence base shifted since the guideline was published? Are there newer studies that should be considered? Students create an 'evidence freshness report' documenting whether the guideline's recommendations are still well-supported by current evidence. This teaches critical appraisal skills while demonstrating that clinical guidelines require ongoing evaluation — they're not permanent authorities.

Clinical Guideline Gap Analysis

Time: 2-3 weeks | Bloom's: Analyze | Tools: elicit, scite, pubmed, anara

Compare institutional practices to latest evidence-based guidelines using AI literature scans. Identify implementation gaps. Propose quality improvement projects.

Implementation Instructions

Have students compare actual clinical practice at their clinical site (or a case-based simulation) against current evidence-based guidelines for a specific condition. Use AI tools to conduct rapid literature scans identifying the latest guideline updates and supporting evidence. Students document gaps between recommended practice and observed practice, categorizing them as: knowledge gaps (staff unaware of guideline), resource gaps (guideline requires unavailable resources), or implementation gaps (guideline known but not consistently applied). Students propose a quality improvement project addressing the most significant gap — connecting evidence-based practice to real-world implementation challenges.

Patient Education Material Creation

Time: 1 week | Bloom's: Create | Tools: elicit, scite, pubmed, anara

AI generates patient education handouts at 6th-8th grade reading level. Students verify medical accuracy and cultural appropriateness before distribution.

Implementation Instructions

Have students use ChatGPT to generate initial patient education handouts at a 6th-8th grade reading level on a condition relevant to their clinical rotation. Students then verify medical accuracy against current clinical guidelines and assess: reading level appropriateness (using readability tools), cultural sensitivity and inclusivity, completeness of essential information (when to seek emergency care, medication side effects, follow-up instructions), and visual clarity. Students revise the AI-generated content based on their clinical knowledge, creating a final version suitable for actual patient distribution. This builds health literacy skills alongside clinical content knowledge.

Systematic Review Screening

Time: 2-4 weeks | Bloom's: Analyze | Tools: elicit, scite, pubmed, anara

Use Rayyan to screen abstracts for systematic review eligibility. AI accelerates process. Students perform full-text review and quality assessment.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. Use Rayyan to screen abstracts for systematic review eligibility. AI accelerates process. Students perform full-text review and quality assessment. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Drug Interaction Scenario Generator

Time: 30-45 min | Bloom's: Apply | Tools: chatgpt, claude, perplexity, notebooklm

AI generates patient cases with potential drug interactions. Students identify interactions, assess severity, and propose medication management plans. Verify against clinical databases.

Implementation Instructions

Use ChatGPT to generate realistic patient cases involving multiple medications with potential interactions: 'Create a case of a [age] patient taking [medications] presenting with [symptoms].' Students must identify all potential drug interactions, assess their clinical significance using a severity framework, and propose a medication management plan. Crucially, students verify every interaction the AI identifies against clinical databases (Lexicomp, Micromedex, or your institution's drug reference). AI may fabricate interactions or miss real ones — this verification step is both a learning objective and a patient safety lesson.

SBAR Communication Practice

Time: 20 min | Bloom's: Apply | Tools: chatgpt, claude, perplexity, notebooklm

AI plays a physician receiving a nurse's SBAR (Situation, Background, Assessment, Recommendation) report. Students practice clear clinical communication and get immediate feedback.

Implementation Instructions

Configure ChatGPT: 'You are a physician receiving a nurse's SBAR report about a patient. Respond realistically — ask clarifying questions if the report is vague, challenge recommendations that aren't well-supported, and acknowledge clear communication.' Students practice delivering SBAR reports for

clinical scenarios from class. The AI responds in character, asking follow-up questions that reveal gaps in the student's clinical reasoning or communication clarity. Students can practice multiple scenarios in a single session — something impossible with simulated patients alone. Review transcripts for communication quality and clinical accuracy.

Health Equity Case Study Development

Time: 1-2 weeks | Bloom's: Create | Tools: chatgpt, claude, perplexity, notebooklm

Students use AI to research health disparities for a specific population. They develop an original case study incorporating social determinants of health. AI gathers data; students provide clinical judgment.

Implementation Instructions

Students select a specific health disparity affecting a marginalized population. They use Perplexity to research: prevalence data, contributing social determinants of health, systemic barriers to care, and evidence-based interventions. Using this research, students develop an original clinical case study that authentically represents the patient's experience, including social, economic, and cultural factors that affect their health outcomes. The case must include discussion questions that push classmates beyond biomedical solutions to address systemic factors. AI provides research efficiency; students provide the cultural sensitivity, clinical nuance, and ethical awareness that health equity education requires.

Humanities

Human vs. AI Rhetorical Analysis

Time: 1 hr | Bloom's: Analyze | Tools: claude, chatgpt, notebooklm

Students write a brief rhetorical analysis of a text. Claude does the same. Compare: what did the AI miss? What structural elements did it catch that students overlooked?

Implementation Instructions

Select a text your class is currently studying (speech, essay, advertisement, or literary passage). Have students independently write a brief rhetorical analysis (250-400 words) identifying key rhetorical strategies, audience, purpose, and effectiveness. Then submit the same text to Claude with the same analytical prompt. In class, display both a student analysis and the AI analysis side by side (anonymized). Discuss: What did the AI identify that students missed? What nuances did students catch that AI overlooked? This reveals AI's strength in identifying structural patterns but weakness in understanding cultural context, irony, and lived experience.

Primary Source Interrogation

Time: 30 min | Bloom's: Analyze | Tools: claude, chatgpt, notebooklm

Upload primary sources to NotebookLM. Students interrogate the AI's synthesis — identifying gaps in historical context, anachronism, or missing counternarratives.

Implementation Instructions

Upload 3-5 primary source documents related to your current unit into NotebookLM. Have students ask the AI questions about the sources: What themes emerge? What connections exist between documents? What historical context is relevant? Students then interrogate the AI's synthesis: Does it recognize the source's bias? Does it acknowledge gaps in the historical record? Does it impose modern frameworks on historical material? Students write a brief critique of the AI's handling of the sources, identifying at least 2 instances where the AI's synthesis was incomplete, anachronistic, or missing counternarratives.

Stylistic Imitation Exercise

Time: 1 hr | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Ask Claude to imitate the style of an author being studied. Students analyze the imitation for accuracy, then write their own stylistic imitation and compare.

Implementation Instructions

Choose an author your class is studying and prompt Claude: 'Write a 200-word passage in the style of [Author] about [contemporary topic].' Share the AI's imitation with students. Have them analyze the imitation using course concepts: Does it capture the author's syntax? Vocabulary? Thematic preoccupations? Rhetorical moves? Where does it fall flat? Then assign students to write their own stylistic imitation of the same author. Comparing the AI imitation to their own reveals what elements of style are pattern-based (which AI can replicate) versus what requires genuine understanding of an author's worldview and artistic choices.

Archival Document Analysis

Time: ongoing | Bloom's: Analyze | Tools: claude, notebooklm, chatpdf

Upload scanned documents to ChatPDF. Extract key passages, dates, names. Human performs all interpretation.

Implementation Instructions

Scan or photograph archival documents and upload them to ChatPDF. Use the AI to extract basic textual information: dates, names, places, and key passages. This saves significant time on initial document processing, especially for handwritten or damaged documents. However, all interpretation — historical significance, context, bias analysis, and connection to your argument — must be entirely human work. AI can tell you what a document says; only you can determine what it means within your historical framework. Document both the AI-assisted extraction and your interpretive analysis separately to maintain methodological transparency.

Theoretical Framework Synthesis

Time: 2-3 weeks | Bloom's: Create | Tools: claude, notebooklm, chatpdf

Upload theoretical texts to NotebookLM. AI maps connections between theorists. Human writes interpretive synthesis.

Implementation Instructions

Upload key theoretical texts from 3-5 theorists relevant to your research into NotebookLM. Ask the AI to map connections: Where do these theorists agree? Where do they contradict? What concepts appear across multiple frameworks under different names? Use the AI's mapping as a starting point for your own interpretive synthesis — the connections the AI identifies are pattern-based, while your synthesis requires understanding the intellectual traditions, historical contexts, and epistemological commitments behind each theory. Write the integrative framework yourself, using the AI's pattern-finding as scaffolding for your original scholarly contribution.

Close Reading Support

Time: ongoing | Bloom's: Analyze | Tools: claude, notebooklm, chatpdf

Claude generates initial readings of complex texts. Challenge AI interpretation. Use disagreements to sharpen analysis.

Implementation Instructions

When encountering particularly dense or challenging texts in your research, paste passages into Claude and ask for an initial reading: summary, key arguments, rhetorical strategies, and potential interpretations. Use the AI's reading as a sparring partner — where do you agree? Where does the AI miss crucial context or nuance? Where does its interpretation seem too literal or too surface-level? This productive disagreement with AI's reading sharpens your own analytical eye. Document instances

where your interpretation diverges from the AI's and explain why — these moments of divergence often contain your most original insights.

AI Poetry Style Mimicry Critique

Time: 1 class session | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm

Prompt AI to mimic Emily Dickinson, John Donne, or Anne Carson. Students critique why AI lacks the weirdness and surprise of real poetry. AI is too predictable.

Implementation Instructions

Choose a distinctive poet your class is studying and prompt Claude or ChatGPT: 'Write a 12-line poem in the style of [Poet] about [contemporary topic].' Share the result without identifying it as AI-generated and ask students to evaluate it as if reviewing a workshop submission. Then reveal it's AI-generated and discuss: Why does the AI version feel predictable? What makes the real poet's work surprising, strange, or compelling in ways AI can't replicate? What elements of style are pattern-based (syntax, vocabulary) versus rooted in lived experience and artistic vision? This exercise sharpens close reading skills while demonstrating AI's creative limitations.

Archaic Text Glossary Generation

Time: ongoing | Bloom's: Understand | Tools: claude, chatgpt, notebooklm

Students input older literary texts into AI for instant definitions of archaic terms. Like a dynamic glossary. Focus time on ideas, not vocabulary lookup.

Implementation Instructions

For courses involving historical or archaic texts (Shakespeare, Chaucer, classical philosophy, religious texts), have students paste challenging passages into ChatGPT and ask for definitions of unfamiliar terms in context. Students build a running glossary throughout the semester, with AI-generated definitions verified against scholarly dictionaries and annotated with their own contextual notes. The AI functions as an instant reference tool that removes vocabulary barriers, allowing students to focus class time on interpretation and analysis rather than basic comprehension. Periodically quiz students on key terms to ensure the AI-assisted glossary is building actual vocabulary knowledge.

Character Image Generation Analysis

Time: 2-3 hours | Bloom's: Create | Tools: claude, chatgpt, notebooklm

Generate images of literary characters with Midjourney. Analyze what AI gets right/wrong about textual descriptions. Discuss interpretation vs visualization.

Implementation Instructions

Select a literary character with detailed textual descriptions. Have students create detailed prompts for Midjourney or DALL-E based solely on textual evidence from the assigned reading. Generate images and analyze: What did the AI get right about the textual description? What did it get wrong or interpret differently? What assumptions did the AI make that weren't in the text (race, setting, clothing, mood)? This exercise teaches close reading through visual translation — students must cite specific textual evidence for their prompts. Discuss how visualization is always interpretation, and how AI's visual choices reveal embedded assumptions in training data.

Comparative Literary Movement Summaries

Time: 1-2 days | Bloom's: Analyze | Tools: claude, chatgpt, notebooklm

AI explains Romanticism, Modernism, Postmodernism. Students verify against scholarly sources. Identify what AI oversimplifies or misses about historical context.

Implementation Instructions

Ask ChatGPT to generate 500-word summaries of 3-4 literary movements your course covers (Romanticism, Modernism, Postmodernism, etc.). Distribute these AI summaries to students as study materials — but with a critical assignment attached. Students must verify each summary against scholarly sources, identifying: oversimplifications, missing cultural context, inaccurate periodization, and excluded voices (movements described primarily through canonical white male authors). Students annotate the AI summaries with corrections and additions. This teaches that AI-generated reference material is a starting point, not an authority — and that literary history is contested, not settled.

Student-as-Text-Generator Exercise

Time: 1 class session | Bloom's: Apply | Tools: claude, chatgpt, notebooklm

Students manually mimic AI by 'predicting' next words in their own essays. Embodied understanding of AI constraints. Reveals why human writing matters.

Implementation Instructions

Give students a paragraph from a well-known text and ask them to manually predict the next 3-5 sentences — just as a language model would — by guessing what comes next based on patterns they recognize. Then compare their predictions to the actual text and to ChatGPT's predictions. Discuss: Where did human predictions and AI predictions overlap? Where did the actual author surprise both? This embodied experience of text generation builds intuitive understanding of how language models work (pattern prediction) and why human writing differs (intentional surprise, lived experience, artistic risk). It's a powerful way to discuss AI's creative limitations through direct experience.

Primary Source Contextualization

Time: 1-2 weeks | Bloom's: Analyze | Tools: chatpdf, claude, perplexity, notebooklm

Upload historical documents to ChatPDF. AI extracts dates, names, events. Students perform all historical interpretation and contextualization.

Implementation Instructions

Upload historical documents (letters, speeches, government records, newspaper articles) to ChatPDF. Use the AI to extract basic factual information: dates, names, locations, key events mentioned, and institutional affiliations. This extraction work is time-consuming when done manually, especially for large archival collections. Students then perform all historical interpretation: placing documents in their political, social, and cultural context; analyzing authorial bias and purpose; connecting to broader historical narratives; and identifying what the documents reveal about power dynamics. The AI handles the 'what' (extraction); students provide the 'so what' (interpretation and significance).

Historiographical Debate Mapping

Time: 2-3 days | Bloom's: Analyze | Tools: chatpdf, claude, perplexity, notebooklm

Use Claude to map how historians debate an event differently (e.g., causes of WWI). Students identify paradigm shifts and methodological differences.

Implementation Instructions

Select a historical event or period where significant historiographical debate exists (e.g., causes of the French Revolution, origins of the Cold War, interpretation of Reconstruction). Use Claude to map the major interpretive schools: Who argued what? When? What evidence did they use? How did later historians challenge earlier interpretations? Students create a visual timeline showing how the historical interpretation evolved, identifying paradigm shifts and methodological innovations. Then

students take a position: Which interpretation is best supported by current evidence? Why? This teaches that history is an active argument, not a fixed narrative.

Philosophy Concept Clarification

Time: ongoing | Bloom's: Understand | Tools: chatpdf, claude, perplexity, notebooklm

Students prompt AI to explain dense philosophical concepts (Kant's categorical imperative). Use as starting point, then engage with primary texts for nuance.

Implementation Instructions

For dense philosophical texts, have students first attempt to understand key concepts through their own reading. When they encounter particularly difficult passages (Kant's categorical imperative, Heidegger's Dasein, Foucault's biopower), they can prompt AI for plain-language explanations as a scaffolding tool. The critical pedagogical step: students must then return to the primary text and evaluate whether the AI's simplification accurately captures the philosopher's meaning or whether important nuance was lost. Keep a class log of concepts where AI explanation was helpful versus concepts where it oversimplified dangerously. This builds philosophical reading stamina while providing appropriate support.

Historical Counterfactual Analysis

Time: 1 week | Bloom's: Evaluate | Tools: chatpdf, claude, perplexity, notebooklm

Prompt AI with counterfactual scenarios ('What if Rome never fell?'). Students critique AI's determinism and lack of historical contingency.

Implementation Instructions

Prompt ChatGPT or Claude with historical counterfactual scenarios relevant to your course: 'What if the printing press had been invented 200 years earlier?' or 'What if the Confederacy had won the Battle of Gettysburg?' Distribute the AI's analysis to students for critical evaluation. Students must identify: Where does the AI's reasoning rely on deterministic logic rather than historical contingency? What factors does it ignore? What are the limits of counterfactual reasoning in historical analysis? This exercise teaches critical thinking about causation and contingency in history while revealing AI's tendency toward linear, oversimplified causal narratives.

Digital Archive Navigation

Time: module | Bloom's: Apply | Tools: chatpdf, claude, perplexity, notebooklm

Use AI to scan digitized archives for relevant documents. Students curate findings, verify authenticity, contextualize within historiography.

Implementation Instructions

Introduce students to AI-assisted digital archive tools by assigning a research task that requires navigating a digitized archive (Library of Congress, National Archives, institutional collections). Use AI to help scan large volumes of digitized documents for relevant keywords, dates, or themes. Students must then curate their findings: select the most relevant documents, verify authenticity and provenance, contextualize within the broader historical record, and build a coherent narrative from primary sources. The AI accelerates the discovery phase; students perform the interpretive and curatorial work that constitutes genuine historical research.

Interdisciplinary & Cross-Course Applications

Cross-Field Literature Mining

Time: 1-2 weeks | Bloom's: Analyze | Tools: semscholar, consensus, claude

Use Semantic Scholar to search across disciplinary boundaries. Identify methods from adjacent fields. Claude translates terminology.

Implementation Instructions

Use Semantic Scholar's broad search capabilities to find research on your topic from fields outside your home discipline. Search for methodological approaches, theoretical frameworks, and empirical findings that address similar questions but from different disciplinary perspectives. Claude can help translate discipline-specific terminology: paste an abstract from an unfamiliar field and ask it to explain the jargon in terms relevant to your discipline. Track cross-disciplinary connections in a structured document, noting potential methodological transfers, theoretical complementarities, and collaborative opportunities.

Methodological Transfer Analysis

Time: 3-5 days | Bloom's: Analyze | Tools: semscholar, consensus, claude

Find papers using methods common in other fields. Track methodology diffusion. Adapt for your context.

Implementation Instructions

Identify a methodology that is standard in one field but rarely used in yours. Find 5-10 papers applying this method in its home discipline using Semantic Scholar. Document: How is the method typically applied? What data requirements does it have? What assumptions does it make? Then assess transferability: How would you adapt it for your research context? What modifications would be needed? What limitations would emerge in a new disciplinary context? This systematic analysis of methodological transfer can become the foundation for a methodological innovation paper in your field.

Consensus Across Disciplines

Time: 2-3 days | Bloom's: Evaluate | Tools: semscholar, consensus, claude

Consensus shows how different fields address similar questions. Identify paradigm differences. Propose integrative frameworks.

Implementation Instructions

Use Consensus or Semantic Scholar to explore how different academic disciplines address the same fundamental question (e.g., 'What drives cooperation?' is studied in biology, economics, psychology, political science, and philosophy). Map the different answers each field provides, noting where disciplinary paradigms create different conclusions from similar evidence. Identify opportunities for integrative frameworks that bridge disciplinary divides. This exercise often reveals that what seems like a settled question in your field is actively debated in adjacent fields — creating opportunities for novel contributions that draw on cross-disciplinary insights.

Cross-Disciplinary Concept Mapping

Time: 1-2 hrs | Bloom's: Analyze | Tools: perplexity, claude, chatgpt, notebooklm

Students explore how a concept from your course appears in 3 other disciplines using Perplexity. Build a concept map showing connections, tensions, and different framings.

Implementation Instructions

Select a key concept from your course that has cross-disciplinary applications (e.g., 'systems thinking' appears in biology, business, engineering, and social work; 'narrative' appears in literature, psychology, law, and medicine). Students use Perplexity to research how 3 other disciplines define and apply the concept. They create a concept map showing: how definitions differ across fields, where

approaches conflict, and what insights emerge from cross-disciplinary comparison. This builds the integrative thinking that characterizes the most innovative professionals — the ability to see connections across silos that specialists miss.

AI-Mediated Expert Panel Simulation

Time: 1 hr | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt, notebooklm

Configure AI to represent experts from different disciplines addressing the same problem. Students moderate the 'panel discussion' and synthesize insights across perspectives.

Implementation Instructions

Choose a complex problem relevant to your course (climate change, healthcare access, AI ethics, urban planning). Configure ChatGPT to respond as experts from 4-5 different disciplines: 'Respond as an economist analyzing this problem,' then 'Respond as a public health expert,' then 'Respond as an ethicist,' etc. Students 'moderate' the panel by reading each perspective and writing a synthesis that: identifies where experts agree, where they conflict, what each perspective reveals that others miss, and what a comprehensive solution would look like. This teaches integrative thinking without requiring actual experts from multiple departments.

Wicked Problem Analysis Framework

Time: 2 weeks | Bloom's: Create | Tools: perplexity, claude, chatgpt, notebooklm

Teams tackle a 'wicked problem' using AI to research multiple disciplinary perspectives. Final deliverable: a policy brief that integrates evidence from at least 3 fields.

Implementation Instructions

Assign student teams a 'wicked problem' — a complex challenge with no clear solution that spans multiple disciplines (homelessness, food security, misinformation, antibiotic resistance). Teams use Perplexity and Claude to research the problem from at least 3 disciplinary perspectives, documenting how each field frames the problem differently. The final deliverable is a policy brief that integrates evidence from all perspectives into actionable recommendations. Teams must address: trade-offs between approaches, implementation barriers, and whose voices are typically excluded from the conversation. This mirrors the kind of complex problem-solving graduates will face in every profession.

Language Learning & Multilingual Support

AI Conversation Partner for Language Practice

Time: 15 min daily | Bloom's: Apply | Tools: chatgpt, claude, elevenlabs, grammarly

Students practice target language conversation with ChatGPT. AI adjusts difficulty level. Students log vocabulary learned and grammatical patterns practiced.

Implementation Instructions

Students engage ChatGPT in conversation in their target language with the prompt: 'Have a conversation with me in [language] at [beginner/intermediate/advanced] level. If I make grammar mistakes, gently correct me and explain the rule.' Students practice for 15 minutes daily and maintain a learning log documenting: new vocabulary encountered, grammar corrections received, and phrases they found difficult. Periodically, students submit their conversation transcripts and logs. This provides unlimited low-stakes practice opportunities that complement classroom instruction, especially valuable for students who lack native speaker contacts outside class.

Translation Quality Evaluation

Time: 45 min | Bloom's: Evaluate | Tools: chatgpt, claude, elevenlabs, grammarly

AI translates a complex text. Students evaluate the translation for accuracy, nuance, cultural sensitivity, and idiomatic correctness. Reveals what machines miss about language.

Implementation Instructions

Select a complex text rich in idioms, cultural references, and nuanced meaning (a poem, a political speech, a literary passage). Have ChatGPT translate it into/from the target language. Students evaluate the translation across multiple dimensions: grammatical accuracy, preservation of meaning, handling of idioms and metaphors, cultural sensitivity, and register/tone appropriateness. For each issue found, students provide their own improved translation with justification. This exercise powerfully demonstrates that translation is interpretation, not mere word replacement, and builds deeper understanding of both languages' cultural dimensions.

Multilingual Content Accessibility

Time: 1-2 hrs | Bloom's: Create | Tools: chatgpt, claude, elevenlabs, grammarly

Faculty use AI to create study guides, assignment instructions, or key definitions in multiple languages for multilingual classrooms. Students verify translations for accuracy.

Implementation Instructions

For courses with multilingual students, use Claude to translate key course materials — study guides, assignment instructions, critical definitions, and frequently asked questions — into the primary languages represented in your classroom. Have bilingual students verify the translations for accuracy and naturalness. Post both English and translated versions as supplementary resources. This doesn't replace English-language learning expectations but removes unnecessary comprehension barriers. Research shows that providing key concepts in students' home languages actually accelerates their academic English acquisition by reducing cognitive load on non-linguistic content.

Library & Information Science

AI vs. Database Source Comparison

Time: 1 hr | Bloom's: Evaluate | Tools: perplexity, semscholar, notebooklm, chatgpt

Students use an AI tool and a library database to answer the same research question. They compare source quality, citation accuracy, and bias — building critical evaluation skills.

Implementation Instructions

Assign students the same research question to investigate using two different tools: an AI tool (Perplexity or ChatGPT) and a library database (JSTOR, PubMed, ProQuest, or your institution's discovery layer). Students create a comparison matrix documenting: number of sources found, source types (peer-reviewed, popular, gray literature), citation accuracy, recency, and relevance to the research question. Discuss as a class: When is AI search more efficient? When is database search more reliable? When should you use both? This builds genuine information literacy by helping students understand the strengths and limitations of each tool rather than simply prohibiting one.

Prompt Literacy Workshop

Time: 45 min | Bloom's: Apply | Tools: perplexity, semscholar, notebooklm, chatgpt

Introduce prompt engineering as an information literacy skill. Students iteratively refine prompts to improve AI output quality, documenting what changed and why it worked.

Implementation Instructions

Frame prompt engineering as an information literacy skill — the ability to formulate effective queries is valuable whether the target is a search engine, database, or AI tool. Give students a research scenario and have them write an initial prompt. Test it in ChatGPT or Perplexity and evaluate the output. Then

iteratively refine: add constraints, specify audience, request citations, adjust scope. Document each iteration: what changed in the prompt, what changed in the output, and why the refinement worked. Students create a 'prompt engineering cheat sheet' based on their discoveries. This teaches transferable query formulation skills that apply across all information systems.

Misinformation Detection Lab

Time: 1–2 hrs | Bloom's: Evaluate | Tools: perplexity, semscholar, notebooklm, chatgpt

Use AI-generated text as sample 'sources.' Students apply source evaluation frameworks (CRAAP, SIFT) to detect AI-generated content, hallucinated citations, and fabricated claims.

Implementation Instructions

Generate several AI-produced 'sources' on a topic: a fake research abstract with hallucinated citations, a plausible but fabricated news summary, and a real source for comparison. Without telling students which is which, have them apply source evaluation frameworks (CRAAP test, SIFT method, or lateral reading) to all three. Students document their evaluation process and identify which sources they trust and why. Reveal the AI-generated sources and discuss: What clues did they miss? What made the AI content convincing? How can they adapt their evaluation practices for an era where AI-generated misinformation is increasingly sophisticated?

Information Behavior Analysis

Time: 2-3 weeks | Bloom's: Analyze | Tools: perplexity, claude, semscholar, scite

Analyze user search logs, help desk transcripts. AI identifies patterns. Human interprets information seeking behaviors.

Implementation Instructions

Collect user search logs, help desk transcripts, or information-seeking behavior data (with appropriate IRB approval). Upload to Claude or NotebookLM for preliminary pattern analysis: common search strategies, frequent query types, recurring barriers, and help-seeking patterns. Use AI-identified patterns as hypotheses for systematic analysis. Code the data using established information behavior frameworks (Kuhlthau's ISP, Wilson's model, etc.) and compare your expert coding to the AI's preliminary analysis. Document discrepancies — these often reveal the most interesting findings about how users actually seek information versus how models predict they should.

Collection Gap Analysis

Time: 1-2 weeks | Bloom's: Evaluate | Tools: perplexity, claude, semscholar, scite

Use Semantic Scholar to identify high-impact papers in field. Compare against library holdings. Generate acquisition recommendations.

Implementation Instructions

Use Semantic Scholar to identify high-impact papers in your library's key subject areas over the past 5 years: most-cited articles, influential reviews, and emerging research fronts. Compare this list against your library's holdings to identify gaps: important papers not in your collection, journals not subscribed to, and research areas underrepresented in your resources. Generate a prioritized acquisition recommendation list with impact metrics and usage predictions. This AI-assisted analysis provides data-driven evidence for collection development decisions, supplementing traditional selector judgment with systematic gap identification.

Metadata Quality Assessment

Time: ongoing | Bloom's: Evaluate | Tools: perplexity, claude, semscholar, scite

AI helps audit metadata for consistency, completeness, adherence to standards. Generate improvement recommendations.

Implementation Instructions

Use Claude to audit a sample of your library's metadata records for: consistency in field formatting, completeness of required elements, adherence to cataloging standards (MARC, Dublin Core, etc.), and controlled vocabulary compliance. AI can quickly identify patterns of inconsistency that would take hours to find manually — incorrect date formats, inconsistent name authorities, missing subject headings. Generate a structured quality report with specific examples and recommended corrections. This AI-assisted quality assessment doesn't replace expert cataloger judgment but significantly accelerates the identification of systematic quality issues across large collections.

AI vs Database Search Comparison

Time: 1 class session | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt, scite

Students search same topic in Perplexity and library databases. Compare results: depth, source quality, citations, comprehensiveness. When to use each?

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. Students search same topic in Perplexity and library databases. Compare results: depth, source quality, citations, comprehensiveness. When to use each? Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Lateral Reading with AI Assistance

Time: 1-2 hours | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt, scite

AI helps students quickly verify source credibility by checking citations, author credentials, publication venue. Practice fact-checking workflows.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI helps students quickly verify source credibility by checking citations, author credentials, publication venue. Practice fact-checking workflows. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Search Query Optimization Workshop

Time: 1 class session | Bloom's: Apply | Tools: perplexity, claude, chatgpt, scite

Test different search queries in AI and databases. Analyze how phrasing affects results. Develop query formulation and Boolean logic skills.

Implementation Instructions

This application activity asks students to use AI tools within a structured workflow to practice skills. Begin by reviewing the activity description and adapting it to your specific course content and student level. Test different search queries in AI and databases. Analyze how phrasing affects results. Develop

query formulation and Boolean logic skills. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Citation Accuracy Verification

Time: ongoing | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt, scite

AI generates citations. Students verify format correctness against style guides. Check that cited sources actually support claims. Combat hallucinations.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI generates citations. Students verify format correctness against style guides. Check that cited sources actually support claims. Combat hallucinations. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Misinformation Detection Protocol

Time: 1-2 weeks | Bloom's: Evaluate | Tools: perplexity, claude, chatgpt, scite

Students develop checklist for identifying misinformation. Test against AI-generated content, social media, news sources. Build critical evaluation skills.

Implementation Instructions

This evaluation activity asks students to critically assess AI-generated output against disciplinary standards. Begin by reviewing the activity description and adapting it to your specific course content and student level. Students develop checklist for identifying misinformation. Test against AI-generated content, social media, news sources. Build critical evaluation skills. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Math, Statistics & Quantitative Reasoning

AI Solution Error Analysis

Time: 45 min | Bloom's: Evaluate | Tools: chatgpt, claude, julius

AI solves a math/stats problem. Students verify each step, identify errors, and explain where and why the AI went wrong. Builds deeper procedural understanding.

Implementation Instructions

Select a multi-step math or statistics problem appropriate for your course level. Have ChatGPT solve it step-by-step (AI frequently makes subtle mathematical errors — this is a feature, not a bug, for this exercise). Students verify each step independently, marking where the AI's work is correct and where it goes wrong. For each error, students must explain why it's wrong and provide the correct solution. This inverts the typical homework dynamic: instead of solving from scratch, students develop the higher-order skill of error detection and correction — a skill that transfers directly to debugging their own work.

Statistical Interpretation Audit

Time: 1 hr | Bloom's: Evaluate | Tools: chatgpt, claude, julius

Upload a dataset to Julius AI. Students audit the statistical analysis: Are the right tests used? Are assumptions met? Are conclusions supported? Fix what the AI gets wrong.

Implementation Instructions

Provide students with a real dataset that has some deliberately messy characteristics (missing data, non-normal distributions, potential outliers). Have them upload it to Julius AI and request a full statistical analysis. Students then audit the AI's choices: Did it check assumptions before running parametric tests? Did it handle missing data appropriately? Are the effect sizes meaningful or just statistically significant? Did it account for multiple comparisons? Students write an audit report correcting the AI's analysis and explaining better methodological choices. This teaches statistical reasoning, not just statistical computation.

Word Problem Generator & Critic

Time: 30 min | Bloom's: Create | Tools: chatgpt, claude, julius

Students prompt AI to generate word problems at increasing difficulty. They solve them, then identify poorly written problems (ambiguous, unsolvable, unrealistic) and improve them.

Implementation Instructions

Students prompt ChatGPT: 'Generate 5 word problems about [current topic] at increasing difficulty levels.' They solve each problem, then critically evaluate the problem design: Is each problem solvable? Are any ambiguous? Do the real-world contexts make mathematical sense (e.g., negative people, fractional items)? For problems with flaws, students rewrite them to be clear, solvable, and realistic. This develops both mathematical proficiency and the metacognitive skill of recognizing problem quality — students who can spot a poorly designed problem understand the mathematics more deeply than students who can only solve well-designed ones.

Multimodal & Creative Assignments

Podcast Episode as Literature Review

Time: 3-4 weeks | Bloom's: Create | Tools: midjourney, runway, elevenlabs, claude

Students create podcast reviewing course concepts. Use ElevenLabs for intro/outro. Must cite sources, explain complex ideas accessibly.

Implementation Instructions

Students create a 10-15 minute podcast episode that reviews and synthesizes 3-5 course readings or research sources on a specific topic. Use ElevenLabs for professional intro/outro music or narration segments. Students must cite sources verbally, explain complex ideas accessibly (as if their audience is an educated non-specialist), and provide a written reference list. The podcast format forces students to understand material deeply enough to explain it conversationally — you can't hide behind academic jargon in a podcast. Grade on: accuracy of content, quality of synthesis, accessibility of explanation, and proper citation.

Video Essay with AI-Generated B-Roll

Time: 4-6 weeks | Bloom's: Create | Tools: midjourney, runway, elevenlabs, claude

Students make video essay analyzing course topic. Runway/Midjourney for visuals. Focus on argumentation, not production value.

Implementation Instructions

Students create a 5-8 minute video essay analyzing a course topic, using AI-generated visuals (Runway for motion graphics, Midjourney for still images) as B-roll to illustrate their arguments. The focus is on argumentative quality, not production value — a well-argued video with simple visuals

earns more than a slick production with weak analysis. Require a written script submitted alongside the video, so you can assess the argument independent of the visual delivery. Students reflect on where AI visuals enhanced their argument versus where they were just decorative. This teaches visual rhetoric alongside content knowledge.

Interactive Digital Story

Time: semester-long | Bloom's: Create | Tools: midjourney, runway, elevenlabs, claude

Students build choice-based narrative exploring ethical dilemmas, historical scenarios, or scientific concepts. AI assists with branching logic.

Implementation Instructions

Students build a choice-based interactive narrative exploring ethical dilemmas, historical scenarios, or scientific decision trees relevant to your course. AI assists with branching logic (ChatGPT can help map decision trees) and content generation for alternative paths. Each choice point must present genuinely different options with distinct consequences grounded in course content. The final deliverable includes the interactive story plus a design document explaining the learning objectives for each branch. This project format works across disciplines: nursing ethics scenarios, historical counterfactuals, business strategy decisions, or scientific methodology choices.

OER & Course Materials

Plain-Language Resource Adaptation

Time: 2–3 hrs | Bloom's: Create | Tools: claude, chatgpt, dalle, notebooklm, elevenlabs

Upload a complex reading to NotebookLM. Generate a plain-language summary, study guide, and discussion questions. Review for accuracy before distributing to students.

Implementation Instructions

Upload a complex course reading (research article, textbook chapter, or technical document) into NotebookLM. Ask the AI to generate three outputs: a plain-language summary (targeting 10th-grade reading level), a study guide with key terms and concepts, and five discussion questions that require critical engagement with the material. Review each output against the original for accuracy — AI summaries can oversimplify nuance or miss key caveats. Edit the outputs to correct any errors and add disciplinary context the AI missed. Distribute alongside (not instead of) the original reading to scaffold student access to complex material.

Visual Explainer Creation

Time: 1–2 hrs | Bloom's: Create | Tools: claude, chatgpt, dalle, notebooklm, elevenlabs

Use DALL-E 3 or Midjourney to generate concept diagrams for course materials. Pair each visual with ChatGPT-drafted alt-text — building accessible, illustrated resources.

Implementation Instructions

Identify 3-5 concepts in your course that students consistently struggle to understand from text alone. Use DALL-E 3 or Midjourney to generate visual representations: concept diagrams, process flowcharts, visual metaphors, or annotated illustrations. For each generated image, use ChatGPT to draft descriptive alt-text that makes the visual accessible to screen reader users. Review both the image and alt-text for accuracy before including in your materials. This creates accessible, illustrated resources without requiring graphic design expertise — and you can iterate on prompts until the visuals accurately represent the concepts.

Narrated Module with AI Voice

Time: 2–3 hrs | Bloom's: Remember | Tools: claude, chatgpt, dalle, notebooklm, elevenlabs

Draft a module script in Claude. Generate narration audio with ElevenLabs. Pair with slides for an accessible, narrated online module — no recording studio needed.

Implementation Instructions

Write a clear, conversational script for your online module content using Claude to help with structure and flow (paste your bullet points and ask for a polished script). Upload the script to ElevenLabs and select a voice that matches your desired tone — professional but warm works well for educational content. Generate the audio narration and pair it with your existing slides or visual materials. Add the narrated module to your LMS as an alternative to reading-only content. This gives students UDL-compliant access to content in multiple formats without requiring you to record yourself — particularly useful for modules you want to update frequently.

Online & Asynchronous Teaching

Weekly AI Controversy Anchor

Time: ongoing | Bloom's: Evaluate | Tools: chatgpt, claude, otter, notebooklm

Post a weekly AI-generated claim or scenario. Students respond with a source-backed critique before engaging peers. Scales asynchronously across any time zone.

Implementation Instructions

Each week, post an AI-generated claim or provocative statement related to the course content in your discussion forum. The claim should be plausible but debatable — something that sounds authoritative but requires investigation. Students must respond within 48 hours with a source-backed critique (minimum 2 cited sources) before they can engage with peers' responses. This creates a structured asynchronous discussion rhythm that scales across time zones and doesn't require synchronous attendance. Over the semester, students develop a habit of critically evaluating claims rather than accepting them at face value.

AI Feedback Loop for Drafts

Time: per assignment | Bloom's: Evaluate | Tools: chatgpt, claude, otter, notebooklm

Students submit drafts to ChatGPT, screenshot and annotate the AI response, then submit: original draft + AI feedback + revision rationale. Three artifacts, one grade.

Implementation Instructions

Structure the assignment submission process in three stages: (1) Students submit their first draft to ChatGPT or Claude and screenshot the complete AI feedback. (2) Students annotate each piece of AI feedback — accepting, rejecting, or questioning it with brief justifications. (3) Students revise their draft based on their own critical judgment of the AI feedback. The final submission package includes all three artifacts: original draft, annotated AI feedback, and revised draft. You grade the quality of their engagement with feedback (annotation quality) as heavily as the final product. This works beautifully in asynchronous settings because all interaction is documented.

Lecture Audio Accessibility

Time: setup: 1 hr | Bloom's: Remember | Tools: chatgpt, claude, otter, notebooklm

Use Otter.ai to auto-transcribe recorded lectures. Students use transcripts for review, note-taking practice, and accessibility accommodations.

Implementation Instructions

Record your lectures using your existing setup (Zoom, LMS, phone). Upload recordings to Otter.ai for automatic transcription. Review the transcripts for accuracy — especially technical terms, names, and discipline-specific vocabulary that AI transcription commonly mishandles. Post both the recording and

the corrected transcript as course materials. Students can use transcripts for review, note-taking comparison, keyword searching, and accessibility accommodations. This is a one-time setup that dramatically improves accessibility compliance and gives all students — not just those with accommodations — flexible access to lecture content.

AI-Narrated Micro-Lectures

Time: 1-2 hrs per module | Bloom's: Create | Tools: chatgpt, claude, otter, elevenlabs, notebooklm

Write concise lecture scripts. Use ElevenLabs to generate narrated audio. Post as supplementary mini-lectures (5-8 min) for asynchronous learners who benefit from audio content.

Implementation Instructions

For each module, write a concise 600-800 word script covering the key concepts students need before engaging with readings or assignments. Use ElevenLabs to generate narrated audio (or record yourself for a more personal touch). Post as 5-8 minute supplementary mini-lectures alongside your existing materials. This serves multiple purposes: provides an audio learning option for students who absorb content better through listening, creates study material for commuting students, and supplements dense readings with an accessible overview. AI narration saves time if you have many modules; your own voice builds connection if you have capacity.

Discussion Post Quality Scaffolding

Time: setup: 1 hr | Bloom's: Evaluate | Tools: chatgpt, claude, otter, elevenlabs, notebooklm

Use AI to generate exemplar discussion posts at A, B, and C quality levels. Post as models with annotations explaining why each earned its grade. Students calibrate before posting.

Implementation Instructions

For each discussion forum, use Claude to generate three sample posts at different quality levels: an A-level post (original insight, course concepts applied, cites evidence, engages the prompt deeply), a B-level post (adequate but generic, surface-level engagement), and a C-level post (minimal effort, doesn't engage course material). Annotate each with specific comments explaining the grade. Post all three as models before students write their own posts. This transparent approach gives students concrete understanding of expectations — especially important in asynchronous settings where they can't read the room or ask clarifying questions in real time.

Weekly Video Check-In with AI Summary

Time: 15 min weekly | Bloom's: Apply | Tools: chatgpt, claude, otter, elevenlabs, notebooklm

Record brief weekly video check-ins. Use AI to transcribe and generate searchable summaries. Students can search for specific topics across all check-ins when studying.

Implementation Instructions

Record a brief (5-7 minute) weekly video check-in addressing: what we covered this week, common questions from student work, preview of next week, and any course logistics. Upload to Otter.ai for transcription. Post both the video and the searchable transcript. By semester's end, students have a searchable archive of all your weekly updates — when studying for the final, they can search transcripts for specific topics. This builds instructor presence in online courses (the single biggest factor in online student satisfaction) while the AI transcription adds a study tool dimension that benefits all students.

Peer Review & Collaborative Learning

Co-Author Contribution Matrix

Time: 1 day | Bloom's: Apply | Tools: claude, notebooklm, chatgpt

Document each author's contributions using CRediT taxonomy. AI ensures nothing is missed. Transparent authorship from project start.

Implementation Instructions

At the project's outset, create a document using the CRediT (Contributor Roles Taxonomy) framework listing all 14 contributor roles (Conceptualization, Methodology, Software, Validation, etc.). Have each team member indicate their anticipated contributions. Use Claude to ensure no role is overlooked and to suggest role descriptions that match your project's specific activities. Revisit the matrix at mid-project and before submission, updating contributions as they evolve. This transparent documentation from day one prevents authorship disputes, ensures fair credit, and satisfies journal requirements for contributor statements. Share the matrix with all co-authors for ongoing agreement.

Meeting Notes Synthesis

Time: ongoing | Bloom's: Apply | Tools: claude, notebooklm, chatgpt

Upload meeting transcripts to NotebookLM. Generate action items, decisions, unresolved questions. Track progress across meetings.

Implementation Instructions

Upload meeting recordings or transcripts to NotebookLM after each team research meeting. Ask the AI to extract: decisions made, action items assigned (with responsible person), unresolved questions, and timeline commitments. Share the AI-generated summary with all team members for correction and confirmation. Maintain a running document that tracks action items across meetings: what was assigned, what was completed, what needs follow-up. This documentation is invaluable for large research teams where miscommunication is common, and it creates an audit trail that helps with project management and eventual contribution documentation.

Multi-Author Manuscript Integration

Time: 1-2 weeks | Bloom's: Create | Tools: claude, notebooklm, chatgpt

Claude helps merge sections written by different authors. Ensure consistent voice, style, formatting. Flag contradictions for team discussion.

Implementation Instructions

When multiple authors write different sections, the resulting manuscript often has inconsistent voice, style, terminology, and formatting. Upload all sections to Claude and ask it to: identify terminology inconsistencies, flag contradictions between sections, suggest transitions between author-written sections, and recommend a unified voice and style. Use the AI's analysis as a checklist for your editorial pass — don't accept AI's rewording wholesale, but use its identification of inconsistencies to guide your own harmonization work. The lead author should make final editorial decisions, ensuring the manuscript reads as a unified document while respecting each contributor's scholarly voice.

AI-Calibrated Peer Review Training

Time: 1 class session | Bloom's: Evaluate | Tools: claude, chatgpt

Before peer review, students practice on an AI-generated sample using your rubric. Compare scores, discuss calibration, then review actual peer work with shared standards.

Implementation Instructions

Generate an AI sample response to your assignment at approximately a B/B+ level — good enough to be useful but with clear areas for improvement. Distribute the sample and your rubric to all students. Each student independently scores the sample on every rubric criterion with brief justifications. In

class, reveal the score distribution (which will be surprisingly wide) and discuss: Why did scores differ? Which criteria were most ambiguous? What does 'proficient' actually look like? Establish consensus interpretations for each criterion. Only then do students proceed to review actual peer work. This calibration step dramatically improves peer review quality and fairness.

Group Project AI Mediator

Time: ongoing | Bloom's: Apply | Tools: claude, chatgpt

When group project conflicts arise, teams submit their disagreement to AI for structured mediation. AI suggests compromise frameworks. Teams decide — AI facilitates, humans choose.

Implementation Instructions

When group project teams encounter disagreements (about direction, workload distribution, quality standards, or design choices), have them submit the disagreement to Claude in a structured format: 'Team members A and B disagree about [issue]. A's position: [summary]. B's position: [summary]. Our project goals are: [goals].' The AI suggests 2-3 compromise frameworks that address both positions. The team discusses the AI's suggestions and reaches their own decision, documenting their reasoning. This teaches conflict resolution skills, provides a neutral third perspective, and prevents disagreements from derailing projects — while keeping final decision-making power with the humans.

Professional Development & Faculty Learning

Teaching Philosophy Statement Revision

Time: 2-3 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm, perplexity

Upload your teaching philosophy to Claude. AI identifies cliches, vague claims, and unsupported assertions. Revise with specific evidence and authentic voice.

Implementation Instructions

Paste your current teaching philosophy statement into Claude with the prompt: 'Identify cliches, unsupported claims, vague language, and areas where this teaching philosophy lacks specific evidence or examples. Suggest where concrete teaching examples would strengthen the statement.' Many teaching philosophy statements fall into generic language that could describe any instructor. Use the AI's critique to replace every vague claim ('I believe in student-centered learning') with a specific example ('In my [course], students choose their own research topics because...'). The revision should sound unmistakably like you — the AI identifies problems, but only you can provide the authentic examples that make a teaching philosophy compelling.

Scholarship of Teaching & Learning Design

Time: 1-2 weeks | Bloom's: Create | Tools: claude, chatgpt, notebooklm, perplexity

Faculty design a SoTL study about their own teaching innovation. AI helps with literature review, methodology selection, and IRB language. Human expertise drives the research question.

Implementation Instructions

If you've implemented a teaching innovation and want to study its effectiveness, use AI to accelerate the SoTL research design process. Use Perplexity to survey existing SoTL literature on similar innovations. Use Claude to help select appropriate research methodology (action research, quasi-experimental, mixed methods) based on your context and questions. Draft IRB application language with AI assistance — but ensure all human subjects protections reflect your actual study design, not generic templates. The research question must come from your genuine curiosity about your teaching practice; AI helps with the infrastructure of research design, not the intellectual core.

Course Evaluation Response Plan

Time: 1-2 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm, perplexity

Upload anonymized course evaluation comments to Claude. AI identifies themes, separates actionable feedback from noise, and suggests concrete changes. You prioritize and implement.

Implementation Instructions

After receiving course evaluations, compile the written comments (anonymized) and paste into Claude with the prompt: 'Organize these student comments into themes: praise (continue doing), constructive criticism (consider changing), and outliers (single complaints unlikely to reflect broad experience). For each theme of constructive criticism, suggest one concrete, implementable change.' Use this analysis to create a focused improvement plan for next semester — you can't address everything, so prioritize the 2-3 changes that would benefit the most students. Document your response plan as evidence of reflective practice for your teaching portfolio.

Publication & Journal Submission

Journal Target Selection Matrix

Time: 1-2 days | Bloom's: Evaluate | Tools: claude, grammarly, chatgpt, semscholar

List 5 target journals. Use AI to analyze your abstract against each journal's scope, recent publications, and acceptance criteria. Rank by fit.

Implementation Instructions

List 5-8 potential target journals for your manuscript. For each journal, use Semantic Scholar and Perplexity to analyze: recent publication patterns, citation metrics, typical article types, scope statements, and editorial board composition. Create a structured comparison matrix scoring each journal on: fit with your topic, typical review timeline, impact factor, open access options, and recent articles similar to yours. Use Claude to draft a brief rationale for your top 3 choices, explaining why your manuscript aligns with each journal's scope and audience. This systematic approach prevents the common mistake of submitting to journals that aren't the right fit.

Cover Letter Drafting

Time: 2-3 hours | Bloom's: Create | Tools: claude, grammarly, chatgpt, semscholar

Provide paper abstract and journal info. Claude drafts cover letter highlighting significance and fit. Human ensures authentic voice and accuracy.

Implementation Instructions

Provide Claude with: your paper's abstract, the target journal's scope and aims, and 2-3 recent articles from that journal that relate to your topic. Ask Claude to draft a cover letter that: (1) introduces the paper's core contribution, (2) explains why it fits this specific journal's scope and readership, (3) highlights the significance and novelty, and (4) suggests potential reviewers if the journal allows it. Review and heavily revise the AI's draft to ensure it reflects your authentic scholarly voice and doesn't overclaim your results. The cover letter is often the editor's first impression — it must be compelling, honest, and precisely tailored.

Graphical Abstract Creation

Time: 3-5 hours | Bloom's: Create | Tools: claude, grammarly, chatgpt, semscholar

Use Midjourney or DALL-E to generate visual elements for graphical abstract. Combine with text to create compelling one-page summary.

Implementation Instructions

Identify the single most important finding or framework in your paper. Sketch a rough layout: What visual elements convey the key message? Use Midjourney or DALL-E to generate visual components (icons, diagrams, background elements) and combine them with text in a presentation or design tool.

The graphical abstract should be understandable without reading the paper — it tells the story in one image. Review against the journal's graphical abstract guidelines for size, format, and content requirements. Have a colleague outside your sub-field look at the graphical abstract and tell you what they think the paper is about — if they can't, revise.

Reviewer Comment Categorization

Time: 1 day | Bloom's: Analyze | Tools: claude, chatgpt, grammarly

Upload all reviewer comments to Claude. AI categorizes by: major vs minor, methodology vs interpretation vs writing. Prioritize major issues first.

Implementation Instructions

Upload all reviewer comments from all reviewers into Claude. Ask it to: (1) categorize each comment as major or minor, (2) group comments by theme (methodology, interpretation, writing, theory, etc.), (3) identify where multiple reviewers raised the same concern, and (4) flag any contradictions between reviewers. This organized view is dramatically more useful than reading reviewer comments sequentially. Prioritize your revision by addressing: first, issues raised by multiple reviewers (consensus concerns); second, major methodological issues; third, major interpretive concerns; and finally, minor issues. This systematic prioritization ensures your revision addresses the most critical concerns first.

Point-by-Point Response Drafting

Time: 3-5 days | Bloom's: Create | Tools: claude, chatgpt, grammarly

For each comment, draft response showing: (1) what you changed, (2) where in manuscript, (3) rationale. Claude helps with diplomatic language for disagreements.

Implementation Instructions

For each reviewer comment, draft a structured response with three elements: (1) what you changed in the manuscript, (2) exactly where the change appears (page, paragraph, line), and (3) your rationale for the change (or respectful explanation of why you didn't change). Use Claude to help with diplomatic language, especially for comments you disagree with — ask it to suggest phrasing that is firm but respectful. The response letter should demonstrate that you took every comment seriously, even those you ultimately didn't act on. A well-crafted response letter significantly increases the probability of acceptance on revision.

Track Changes Documentation

Time: 1-2 days | Bloom's: Apply | Tools: claude, chatgpt, grammarly

Use AI to generate summary of all changes made for cover letter. Highlight major revisions that strengthen the paper.

Implementation Instructions

After completing all revisions, use Claude to generate a summary of changes for your resubmission cover letter. Organize changes by category: major revisions (new analyses, restructured arguments, additional data), moderate revisions (strengthened justifications, expanded literature review), and minor corrections (grammar, formatting, citation updates). Highlight how the major revisions strengthen the paper and address the core concerns of reviewers. This summary gives the editor a roadmap for evaluating your revision and demonstrates thorough engagement with the review process. A clear, organized summary of changes signals professionalism and respect for reviewers' time.

Research Impact & Dissemination

Plain Language Summary Creation

Time: 1-2 days | Bloom's: Create | Tools: chatgpt, claude, perplexity, grammarly

AI adapts technical research for general audience. Multiple reading levels (8th grade, college, expert). Human ensures accuracy.

Implementation Instructions

After your paper is accepted, use Claude to adapt your abstract and key findings into plain language summaries at three reading levels: 8th grade (general public), college undergraduate (educated non-specialist), and discipline-adjacent expert. For each version, verify that simplification hasn't introduced inaccuracies — it's easy for AI to oversimplify nuance or drop important caveats when reducing complexity. Test the 8th-grade version with a non-academic friend or family member: Can they explain your findings back to you? If not, revise for clarity. These summaries become press release drafts, social media content, and public engagement materials.

Research Brief for Policymakers

Time: 2-3 days | Bloom's: Create | Tools: chatgpt, claude, perplexity, grammarly

Distill research into 2-page policy brief. Claude formats: problem, evidence, recommendations, next steps. Human verifies policy implications.

Implementation Instructions

Distill your research findings into a 2-page policy brief following the standard format: problem statement (1 paragraph), key evidence (2-3 paragraphs), policy recommendations (bulleted), and next steps. Use Claude to help with structure and concision — academic writers tend to include too much methodological detail for policy audiences. Every recommendation must be directly supported by your evidence and must include implementation feasibility considerations. Review against sample policy briefs from relevant organizations (Brookings, RAND, your professional association). Have a policy-oriented colleague review the brief for practical relevance and political awareness.

Social Media Thread Strategy

Time: 1 day | Bloom's: Create | Tools: chatgpt, claude, perplexity, grammarly

Turn publication into Twitter/X thread, LinkedIn post, blog post. AI drafts multiple formats. Human selects tone and platform.

Implementation Instructions

After publication, plan a dissemination strategy across platforms. Use ChatGPT to draft: a Twitter/X thread (5-7 posts summarizing key findings with a hook), a LinkedIn post (professional framing emphasizing implications for practice), and a blog post (800-1000 words for a general audience). For each format, verify that simplification hasn't introduced inaccuracies and that your tone is appropriate for the platform. Schedule posts strategically — share the paper when your academic community is most active. Tag relevant accounts, use discipline-specific hashtags, and include a direct link to the paper. Engagement with your research starts with making it accessible.

Research: Academic Writing

Abstract Scaffold with AI

Time: 1 hr | Bloom's: Create | Tools: claude, chatgpt, notebooklm, perplexity

Upload research notes to NotebookLM. Ask it to outline the key contribution, methods, and findings. Use Claude to tighten the prose — then verify every claim.

Implementation Instructions

Upload your research notes, data summaries, and key findings into NotebookLM. Ask it to identify the core contribution, methodological approach, and primary findings from your materials. Use this output as a structural scaffold — not a draft — to organize your thinking about the abstract's narrative arc. Then use Claude to tighten the prose: paste your rough abstract and ask for suggestions on clarity, concision, and flow while maintaining your voice. Critically, verify every factual claim in the AI-assisted output against your actual data. The abstract must be yours; AI just helps you organize and polish.

Grant Narrative Sprint

Time: 2–4 hrs | Bloom's: Create | Tools: claude, chatgpt, notebooklm, perplexity

Bullet-point your Significance section. Ask ChatGPT to draft it in grant-register prose. Use Claude to align tone and sharpen argument. Peer-review the AI-assisted version.

Implementation Instructions

Write your Significance section as bullet points first — focus on the problem, why it matters, what gap your research fills, and what impact it will have. Paste these bullets into ChatGPT with the prompt: 'Convert these into formal grant narrative prose suitable for [NSF/NIH/etc.].' Then paste the draft into Claude and ask it to: strengthen the argument arc, ensure logical flow, and adjust tone for the specific funding agency. Have a colleague review the AI-assisted version with a critical eye. The final text should sound like you, not like AI — use these tools to overcome writer's block, not to replace your scholarly voice.

Gap Analysis Memo

Time: 2–3 hrs | Bloom's: Evaluate | Tools: claude, chatgpt, notebooklm, perplexity

After mapping literature in ResearchRabbit + Scite, use Claude to help structure a gap analysis. Write the final memo yourself — AI scaffolds the argument arc.

Implementation Instructions

After mapping your literature using ResearchRabbit for citation networks and Scite for evidence verification, you'll have a rich understanding of what's been studied and what hasn't. Use Claude to help structure a gap analysis by pasting your notes and asking: 'Help me organize these observations into a coherent argument about what the field is missing.' Claude can suggest logical frameworks for presenting gaps (chronological, methodological, theoretical, population-based). Write the final memo yourself, using the AI's organizational suggestions as a starting point. This memo becomes the justification for your entire research project.

Argument Structure Scaffolding

Time: 2-4 weeks | Bloom's: Create | Tools: claude, grammarly, chatgpt, notebooklm

Bullet-point your argument. Use Claude to outline structure. Iteratively refine organization. Write final prose yourself.

Implementation Instructions

Before writing, create a detailed bullet-point outline of your argument: thesis, supporting claims, evidence for each claim, counterarguments, and your responses to counterarguments. Paste this outline into Claude and ask: 'Suggest improvements to the logical structure of this argument — are there gaps, circular reasoning, or unsupported jumps?' Use Claude's structural feedback to reorganize

and strengthen your outline. Then write the actual prose yourself — the AI helps with architecture, not construction. Plan for 2-4 iterative cycles over several weeks, refining the argument structure before committing to final prose. This prevents the common mistake of starting to write before the argument is sound.

Methods Section Precision

Time: 3-5 days | Bloom's: Create | Tools: claude, grammarly, chatgpt, notebooklm

Draft methods. Ask Claude to identify ambiguities, missing details, replicability gaps. Revise for clarity.

Implementation Instructions

Draft your methods section with as much detail as you can, then submit it to Claude with the prompt: 'Read this methods section as a reviewer trying to replicate the study. Flag any ambiguities, missing details, or procedural steps that would prevent exact replication.' Pay special attention to AI's identification of: unstated assumptions, missing sample selection criteria, vague measurement descriptions, and inadequate statistical justification. Revise each flagged item for maximum clarity and specificity. Have a colleague in a different sub-field read the revised version — if they can follow the procedures, you've achieved methodological clarity.

Multi-Tool Revision Sprint

Time: 1 week | Bloom's: Evaluate | Tools: claude, grammarly, chatgpt, notebooklm

Submit sections to Claude, ChatGPT, Grammarly. Compare feedback. Prioritize revisions flagged by multiple tools.

Implementation Instructions

Submit different sections of your manuscript to multiple AI tools: Claude for argument structure and logic, ChatGPT for clarity and readability, and Grammarly for grammar, style, and tone. Create a tracking document that records feedback from each tool for each section. Prioritize revisions that are flagged by multiple tools — these are the most likely genuine issues. For conflicting suggestions, use your disciplinary judgment. This cross-tool validation approach catches more issues than relying on a single AI tool while helping you distinguish between genuine problems and tool-specific idiosyncrasies.

Research: Literature Review

Citation Network Map

Time: 2–3 hrs | Bloom's: Analyze | Tools: perplexity, semscholar, elicit, scite, resrabbit, notebooklm

Start with 2–3 seed papers in ResearchRabbit. Map the citation network to identify key nodes, influential authors, and emerging sub-fields in your research area.

Implementation Instructions

Begin with 2-3 foundational papers in your research area and enter them into ResearchRabbit. Explore the visual citation network to identify papers that are frequently cited (hub papers), recent papers citing your seeds (emerging work), and clusters of related research. Spend 2-3 hours following the most promising branches, saving papers to a collection as you go. Use the network visualization to identify influential authors you may not have encountered and potential collaborators working on adjacent questions. Document your discovery path — this becomes the backbone of your literature review's organizational structure.

Evidence Extraction Pipeline

Time: 4–6 hrs | Bloom's: Analyze | Tools: perplexity, semscholar, elicit, scite, resrabbit, notebooklm

Use Elicit to extract methods, sample sizes, and findings from 20+ papers into a structured table. Verify each row against the original before writing.

Implementation Instructions

Select 20-30 papers from your literature search and upload them to Elicit. Use Elicit's extraction feature to pull structured data: study design, sample size, key methods, primary findings, and limitations into a sortable table. Before trusting any extracted data point, verify it against the original paper — AI extraction is good but not perfect, especially with complex statistical results. This structured table becomes your evidence synthesis matrix, making it dramatically easier to identify patterns, contradictions, and gaps across the literature. Export the verified table as your working document for writing the review.

Claim Verification via Scite

Time: 1–2 hrs | Bloom's: Evaluate | Tools: perplexity, semscholar, elicit, scite, resrabbit, notebooklm

Run central claims from your literature through Scite. Identify whether they're supported, contrasted, or just mentioned across the field — dramatically strengthens your argument.

Implementation Instructions

Identify 5-10 central claims that your literature review depends on — the foundational assertions your argument is built on. Enter each claim into Scite to see how the supporting literature actually treats it: is it supported by subsequent research, contrasted by conflicting findings, or merely mentioned without evaluation? Pay special attention to claims where Scite shows significant contrasting evidence — these are where your literature review can demonstrate critical engagement rather than uncritical acceptance. Document the verification results and use them to strengthen your argumentation by acknowledging complexity and counter-evidence.

Systematic Review Sprint

Time: 1-2 weeks | Bloom's: Analyze | Tools: elicit, rayyan, semscholar, scite, resrabbit

Use Elicit for data extraction, Rayyan for screening (90% time savings), Scite for citation verification. Document AI limitations in methods.

Implementation Instructions

Begin by clearly defining your PICO/research question and registering your protocol (PROSPERO or similar). Use Elicit for systematic data extraction across your identified papers, creating structured tables of study characteristics, methods, and findings. Employ Rayyan for screening — its AI-assisted relevance scoring can reduce screening time by up to 90%, but always have a second reviewer verify a random sample. Run your key claims through Scite to verify citation contexts (supporting vs. contrasting evidence). Document every AI tool used, its role, and its limitations in your methods section — transparency about AI-assisted systematic review processes is increasingly expected by journals.

Citation Network Mapping

Time: 3-5 days | Bloom's: Analyze | Tools: elicit, rayyan, semscholar, scite, resrabbit

Start with 3 seed papers in ResearchRabbit. Map forward/backward citations. Use Semantic Scholar to identify influential nodes.

Implementation Instructions

Enter 3 seed papers representing the core of your research area into ResearchRabbit. Systematically explore three directions: backward citations (foundational work your seeds build on), forward citations (newer work building on your seeds), and lateral connections (papers cited alongside your seeds but that you haven't encountered). Use Semantic Scholar's influence scores to identify the most impactful nodes in the network. Spend 3-5 days following promising branches and saving papers to organized

collections. The resulting network map becomes your literature review's organizational skeleton — showing not just what you read, but how the field's conversation is structured.

Evidence Quality Verification

Time: 2-3 days | Bloom's: Evaluate | Tools: elicit, rayyan, semscholar, scite, resrabbit

Run key claims through Scite. Document supporting vs contrasting citations. Visualize consensus evolution with Connected Papers.

Implementation Instructions

Select the 10-15 most critical claims your research depends on — the assertions that, if wrong, would undermine your entire argument. Run each through Scite to see the citation context: is the claim supported, contrasted, or merely mentioned by subsequent research? For claims with significant contrasting evidence, dig deeper into the original papers to understand the discrepancy. Use Connected Papers to visualize how consensus has evolved over time. Document your verification results in a structured table: claim, original source, Scite evidence profile, your assessment. This table becomes a powerful appendix that demonstrates rigorous scholarship.

STEM

AI-Assisted Data Analysis Lab

Time: 2 hrs | Bloom's: Analyze | Tools: julius, chatgpt, perplexity, claude

Students upload a real dataset to Julius AI. They must identify which analyses are appropriate, run them, then critique the AI's methodology and validate conclusions.

Implementation Instructions

Prepare a real dataset relevant to your STEM course (or use a publicly available research dataset). Have students upload it to Julius AI and first predict what analyses would be appropriate before asking the AI to run them. Students must document: what analysis they requested, why it was appropriate for the data type, what the AI produced, and whether the methodology was sound. Require students to validate at least one AI-generated statistical result by hand calculation or using a trusted statistical package. The deliverable is a lab report that critically evaluates the AI's analytical choices, not just the results.

Model Assumptions Audit

Time: 45 min | Bloom's: Evaluate | Tools: julius, chatgpt, perplexity, claude

ChatGPT generates a simplified model of a STEM concept. Students systematically identify assumptions, limitations, edge cases, and real-world applicability gaps.

Implementation Instructions

Have ChatGPT generate a simplified model or explanation of a STEM concept your students are currently studying (e.g., ideal gas behavior, population growth, circuit analysis). Present the AI's model to students and assign them to systematically audit it: What assumptions does the model make? Under what conditions do these assumptions break down? What real-world factors are ignored? Students document at least 3 specific limitations and provide examples of edge cases where the model would fail. This teaches critical evaluation of models — a fundamental scientific skill — while revealing AI's tendency to present simplified versions without caveats.

Code Debugging with AI

Time: 1 hr | Bloom's: Analyze | Tools: julius, chatgpt, perplexity, claude

Students attempt a coding problem first. Then AI provides a solution. Students compare approaches, explain differences, and document what the AI got wrong.

Implementation Instructions

Assign a coding problem at an appropriate difficulty level for your course. Students must attempt a solution independently first, documenting their approach and any errors they encounter. Then have them submit the same problem to ChatGPT or Claude and receive the AI's solution. Students write a comparison analysis: How did their approach differ from the AI's? Was the AI's solution more efficient, more readable, or more correct? Did the AI make any subtle errors? This teaches students to learn from AI code without becoming dependent on it, while building debugging and code review skills.

Dataset Exploration & Visualization

Time: 1-2 weeks | Bloom's: Analyze | Tools: julius, claude, chatpdf

Upload dataset to Julius. Generate exploratory visualizations. Identify patterns. Always validate statistical claims.

Implementation Instructions

Upload your research dataset to Julius AI and request an initial exploratory analysis: summary statistics, distribution plots, correlation matrices, and missing data patterns. Use the AI-generated visualizations to identify unexpected patterns, potential outliers, and variables worth investigating further. Always validate statistical claims by running key analyses in a trusted statistical package (R, SPSS, Stata) — use Julius for rapid exploration, not final analysis. Document your exploration process and the questions each visualization raised, as this exploratory phase often generates the most interesting research questions and methodological decisions.

Computational Model Documentation

Time: ongoing | Bloom's: Create | Tools: julius, claude, chatpdf

Claude drafts clear model documentation. AI suggests organization. Human ensures technical accuracy.

Implementation Instructions

As you develop your computational model, use Claude to draft clear documentation at each stage: model assumptions, input parameters, mathematical formulations, implementation decisions, and validation approaches. AI excels at organizing technical information into readable formats but may misunderstand domain-specific nuances — always verify technical accuracy. Create documentation at two levels: a technical reference for fellow specialists and a plain-language overview for interdisciplinary audiences. Maintain this documentation throughout the project, updating as the model evolves. Well-documented computational models are significantly more likely to be adopted and cited by other researchers.

Methods Paper Drafting

Time: 2-4 weeks | Bloom's: Create | Tools: julius, claude, chatpdf

Document novel method. AI ensures clarity for readers outside sub-field. ChatPDF compares to similar papers.

Implementation Instructions

Document your novel method in structured sections: motivation, theoretical foundation, implementation details, validation results, and limitations. Use Claude to help ensure clarity for readers outside your immediate sub-field — paste each section and ask: 'Would a researcher in [adjacent field] understand this?' Use ChatPDF to compare your methods paper to similar published methods papers in your field:

How do successful papers structure their validation sections? What level of detail do they provide? This comparative analysis helps calibrate your paper's scope and detail level to field norms.

Pre-Lab AI Simulation

Time: before lab | Bloom's: Apply | Tools: julius, claude, chatgpt

Students use AI to simulate lab procedures, predict outcomes. Compare predictions to actual results. Discuss discrepancies and experimental error.

Implementation Instructions

Before a laboratory session, have students use ChatGPT or Claude to simulate the planned experiment: describe the procedure, predict expected outcomes based on theoretical principles, identify potential sources of error, and outline safety considerations. Students bring their AI-generated predictions to lab and compare against actual results. Discrepancies between prediction and reality become the focus of the lab report discussion: Why did the experiment diverge from AI's prediction? What factors did the simulation miss? This pre-lab exercise ensures students understand the procedure conceptually before handling equipment and materials, reducing both confusion and safety risks.

Error Propagation Calculator

Time: during lab | Bloom's: Analyze | Tools: julius, claude, chatgpt

Julius AI helps calculate error propagation in multi-step experiments. Students verify calculations manually. Understand uncertainty quantification.

Implementation Instructions

During multi-step laboratory experiments, have students use Julius AI to calculate error propagation: How does measurement uncertainty in early steps compound through the analysis? Students input their measurements with uncertainty ranges and AI calculates the propagated error for final results. Critically, students must verify at least one calculation manually to understand the mathematical process, not just the AI's output. Discuss: Which measurement contributed the most uncertainty? How could the experimental design be improved to reduce overall error? This builds quantitative reasoning skills while teaching students that all measurements have uncertainty — a foundational concept in scientific thinking.

Lab Report Methods Clarity Check

Time: post-lab | Bloom's: Create | Tools: julius, claude, chatgpt

Submit methods section to Claude. AI flags ambiguities preventing replication. Students revise for specificity and procedural completeness.

Implementation Instructions

After students write their lab report methods section, have them submit it to Claude with the prompt: 'Could you replicate this experiment from this methods section alone? Flag any ambiguities, missing steps, or unclear descriptions.' Students then revise based on the AI's identification of gaps in procedural clarity. This teaches the scientific communication principle that methods sections must be precise enough for independent replication — a standard that students consistently underestimate. The AI serves as a 'naive reader' who takes everything literally, catching ambiguities that students (who know what they meant) tend to overlook.

Experimental Design Peer Review

Time: 1-2 weeks | Bloom's: Evaluate | Tools: julius, claude, chatgpt

AI reviews proposed experimental designs for confounds, sample size, controls. Students address AI and human feedback before conducting experiment.

Implementation Instructions

Before students conduct their proposed experiments, have them submit their experimental design to both Claude (for structural review) and a peer (for disciplinary review). The AI evaluates: sample size adequacy, control conditions, potential confounding variables, and measurement validity. The peer evaluates: theoretical justification, practical feasibility, and disciplinary norms. Students address both AI and human feedback before beginning data collection. This dual-review process catches design flaws before they become data problems — far more efficient than discovering methodological issues after the experiment is complete.

Data Visualization Best Practices

Time: ongoing | Bloom's: Create | Tools: julius, claude, chatgpt

Generate multiple visualizations of same data in Julius. Compare for clarity, misleading features, appropriate chart types. Learn data honesty.

Implementation Instructions

Provide students with a single dataset and have them generate multiple visualizations in Julius AI: bar charts, scatter plots, line graphs, box plots, and pie charts. For each visualization, students evaluate: Does this chart type accurately represent the data? Could any visualization be misleading? What story does each visualization tell, and is that story faithful to the data? Introduce concepts of data honesty: truncated axes, misleading scales, cherry-picked time ranges, and correlation vs. causation in visual representation. Students submit a 'visualization ethics report' documenting which charts they would and wouldn't use professionally and why.

Code Review for Logic Errors

Time: ongoing | Bloom's: Analyze | Tools: chatgpt, claude, julius

Submit code to ChatGPT for logic error detection. AI suggests fixes. Students must understand and explain fixes before implementing. No blind copying.

Implementation Instructions

Assign a coding problem and have students write their solution independently. Then have them submit their code to ChatGPT or Claude for review, specifically asking: 'Identify logic errors, edge cases not handled, and potential efficiency improvements in this code.' Students must understand every suggestion before implementing it — require a written explanation of each fix in their own words. If the AI suggests a fix the student can't explain, they must flag it and research the concept independently. This teaches code review as a professional practice while preventing blind copy-paste reliance on AI-generated solutions.

Algorithm Complexity Analysis

Time: 1-2 hours | Bloom's: Evaluate | Tools: chatgpt, claude, julius

Ask AI to analyze Big O complexity of student algorithms. Verify AI's analysis. Optimize code based on insights. Focus on computational efficiency.

Implementation Instructions

Have students write a solution to an algorithmic problem, then ask ChatGPT or Claude to analyze its Big O time and space complexity. Students verify the AI's analysis: Does the runtime analysis correctly identify the dominant operations? Is the space complexity assessment accurate? Students then optimize their code based on the complexity analysis and compare performance. For advanced students, have the AI generate both an optimal and a naive solution — students explain why one is

more efficient and under what conditions the difference matters. This builds computational thinking skills essential for any quantitative discipline.

Engineering Documentation Generation

Time: per project | Bloom's: Create | Tools: chatgpt, claude, julius

Use Claude to draft technical documentation for engineering projects. Students edit for accuracy and completeness. Learn professional documentation standards.

Implementation Instructions

For each engineering project, require students to maintain technical documentation using AI-assisted drafting. Submit project descriptions to Claude and ask for documentation in standard engineering format: problem statement, design requirements, approach, implementation details, testing results, and limitations. Students edit the AI-generated documentation for technical accuracy, adding domain-specific details and correcting any misunderstandings. The goal is fluency in professional documentation standards — AI provides the structure and format conventions, students provide the technical substance. Grade documentation quality alongside project quality to emphasize that engineering communication is as important as engineering execution.

Unit Test Case Generation

Time: ongoing | Bloom's: Create | Tools: chatgpt, claude, julius

AI generates initial unit tests for code. Students add edge cases AI missed. Emphasizes thorough testing and defensive programming.

Implementation Instructions

After students write a function or module, have them submit it to ChatGPT and ask: 'Generate comprehensive unit tests for this code, including edge cases.' Students then evaluate the AI-generated tests: Are they testing the right behaviors? What edge cases did the AI miss? What test cases would be important in a production environment? Students add at least 3 original test cases covering scenarios the AI didn't consider. This teaches defensive programming and thorough testing habits while demonstrating that AI-generated test suites are a starting point, not a complete quality assurance strategy.

Thermodynamics Concept Clarification

Time: ongoing | Bloom's: Understand | Tools: chatgpt, claude, julius

AI explains abstract thermodynamics concepts (entropy, reversibility). Students apply to problem-solving. Human validates numerical work; AI for conceptual scaffolding.

Implementation Instructions

For abstract thermodynamics concepts that students struggle with (entropy, enthalpy, Gibbs free energy, reversibility), have students first attempt problems independently, then use AI for conceptual clarification when stuck. The critical rule: AI is for understanding concepts, not for solving problems. Students must show their own work on calculations — they can ask ChatGPT to explain why entropy increases in a certain process, but they must calculate the entropy change themselves. Keep a class log of concepts where AI explanation was most helpful, building a resource of 'AI-assisted concept guides' that benefits future students.

Pre-Lab Protocol Comprehension Check

Time: 15 min pre-lab | Bloom's: Understand | Tools: chatgpt, claude, julius, perplexity

Students paste lab protocol into AI and ask for a simplified step-by-step summary. They annotate where the AI oversimplified critical safety or procedural details.

Implementation Instructions

Before lab sessions, students paste the full lab protocol into ChatGPT and ask: 'Summarize this lab protocol in simple step-by-step instructions a first-year student could follow.' Students compare the AI's simplified version to the original, annotating where the AI: omitted safety precautions, oversimplified critical steps, combined distinct procedures, or used imprecise measurement language. Submit annotated comparison as a pre-lab ticket. This confirms protocol comprehension and builds the critical habit of recognizing that simplified instructions can be dangerously incomplete in laboratory settings.

Field Observation Data Logger

Time: per field session | Bloom's: Apply | Tools: chatgpt, claude, julius, perplexity

Students dictate field observations to AI-transcription tools. AI helps organize raw observations into structured field notes. Students add interpretive layers the AI cannot provide.

Implementation Instructions

During fieldwork, students use their phones to dictate raw observations (weather, species, measurements, site conditions) and transcribe using Otter.ai or similar. After the session, paste transcripts into Claude: 'Organize these raw field observations into structured field notes with categories for: environmental conditions, species/specimens observed, quantitative measurements, and qualitative observations.' Students then add the critical interpretive layer: patterns noticed, hypotheses generated, connections to course concepts, and questions for follow-up. AI handles the tedious transcription and organization; students provide the scientific interpretation that makes field notes valuable.

Lab Report Methods Section Precision

Time: 1 hr | Bloom's: Create | Tools: chatgpt, claude, julius, perplexity

Students draft their lab report methods section. AI reviews for replicability: could another scientist follow these steps exactly? Identifies ambiguities and missing details.

Implementation Instructions

Students draft their lab report methods section with as much detail as they can. Submit to Claude: 'Read this methods section as a scientist trying to replicate the experiment. Flag any ambiguities, missing details, or steps that would prevent exact replication.' Students revise every flagged item for maximum precision. This teaches the core skill of scientific writing: the methods section must enable replication. AI catches the vague language students don't notice ('some,' 'approximately,' 'a few minutes') and pushes toward the precision science demands ('2.5 mL ± 0.1 mL,' 'incubated at 37°C for exactly 24 hours').

Experimental Design Challenge

Time: 1-2 hrs | Bloom's: Create | Tools: chatgpt, claude, julius, perplexity

Students design an experiment to test a hypothesis. AI critiques the design for confounding variables, sample size adequacy, and methodological flaws. Revise before conducting.

Implementation Instructions

Students propose an experimental design to test a hypothesis relevant to your course. Submit to Claude: 'Critique this experimental design for: confounding variables not controlled, sample size adequacy, potential sources of bias, measurement validity, and statistical analysis appropriateness.' Students address every critique before conducting the experiment (or submitting the final design for courses without wet labs). This pre-experiment review catches design flaws that would otherwise only

become apparent after data collection — saving time, resources, and frustration while teaching students to think critically about methodology before they invest in execution.

Scholarly Communication

Hostile Peer Review Simulation

Time: 2-3 days | Bloom's: Evaluate | Tools: claude, scite, semscholar, grammarly

Submit manuscript to Claude acting as critical reviewer. Address every critique before submission. Strengthens argumentation.

Implementation Instructions

Before submitting your manuscript, paste it into Claude with the prompt: 'Act as a critical peer reviewer who is looking for every possible weakness, logical gap, methodological flaw, and unsupported claim. Be thorough and demanding.' Read the AI's critique carefully and address every point — even the ones you initially disagree with, as reviewers will likely raise similar concerns. For each critique, either revise the manuscript to address the issue or prepare a response explaining why no change is needed. This pre-submission exercise can significantly reduce the number of revisions required after actual peer review and strengthens your manuscript before it reaches real reviewers.

Conference Abstract Optimization

Time: 1-2 days | Bloom's: Create | Tools: claude, scite, semscholar, grammarly

Draft abstract. AI ensures exact guideline compliance. Test multiple framings. Human selects best version.

Implementation Instructions

Draft your conference abstract according to the specific submission guidelines (word count, required sections, format). Submit to Claude with the guidelines and ask it to verify compliance: word count, required elements present, proper formatting. Then ask Claude to suggest three alternative framings of your opening sentence — the hook that determines whether reviewers keep reading. Test each framing with a colleague to see which generates the most interest. Ensure the final abstract accurately represents your research without overclaiming, and that every factual statement can be supported by your actual data or analysis.

Response to Reviewers Drafting

Time: 3-5 days | Bloom's: Create | Tools: claude, scite, semscholar, grammarly

Claude organizes comments by theme. Draft point-by-point responses. AI suggests diplomatic phrasing for disagreements.

Implementation Instructions

Upload all reviewer comments into Claude and ask it to: (1) organize comments by theme rather than by reviewer, (2) identify which comments are major vs. minor, and (3) flag contradictions between reviewers. For each themed group, draft a point-by-point response showing what you changed, where in the manuscript the change appears, and your rationale. For comments you disagree with, ask Claude to help draft diplomatic but firm language explaining your reasoning. The final response letter must demonstrate respectful engagement with every comment — even those you ultimately don't act on — while clearly documenting all changes made.

Simulations & Role-Play

AI Virtual Patient Interviews

Time: 1-2 hours | Bloom's: Apply | Tools: chatgpt, claude

ChatGPT acts as virtual patient with specific case. Students conduct history-taking. AI provides dynamic responses. Debrief on communication and clinical reasoning.

Implementation Instructions

Configure ChatGPT with a detailed patient scenario including: demographic information, presenting complaint, medical history, current medications, psychosocial factors, and communication style. Students conduct a simulated patient interview through the chat interface, practicing: open-ended questions, active listening responses, history-taking sequence, and therapeutic communication. After the interview, students document their clinical findings and compare against the original case information to identify: what they missed, what they discovered effectively, and where their communication could improve. Debrief focuses on communication skills as much as clinical content.

Deteriorating Patient Scenario

Time: 1 class session | Bloom's: Analyze | Tools: chatgpt, claude

AI describes progressively worsening patient symptoms. Students identify deterioration patterns, prioritize interventions. Practice rapid assessment skills.

Implementation Instructions

Create a progressive patient scenario where symptoms worsen over time. Present the initial patient state (stable vitals, mild symptoms) and have students assess and plan. Then reveal new information indicating deterioration (changing vitals, new symptoms, lab results) in stages. At each stage, students must: identify the change, assess its significance, prioritize interventions, and communicate using SBAR format. AI provides the dynamic patient responses, allowing more complex scenarios than static case studies. Debrief focuses on: recognition of deterioration patterns, appropriate escalation decisions, teamwork communication, and the clinical reasoning process.

Medication Dosage Verification

Time: ongoing | Bloom's: Apply | Tools: chatgpt, claude

Students calculate drug dosages. AI provides instant feedback on calculations. Emphasize double-checking and safe medication administration practices.

Implementation Instructions

Provide patient scenarios requiring drug dosage calculations (weight-based dosing, drip rate calculations, unit conversions, pediatric dosing). Students calculate dosages manually first, showing all work. Then they verify their calculations using ChatGPT or a pharmaceutical calculator. Discrepancies require students to identify their error and correct it. Emphasize the 'five rights' of medication administration and discuss: Why is double-checking critical in clinical practice? What are the consequences of calculation errors? Why can't clinicians blindly trust AI or calculator results? This builds both mathematical confidence and the safety mindset essential for healthcare professionals.

Cultural Competency Scenarios

Time: 2-3 hours | Bloom's: Apply | Tools: chatgpt, claude

AI-generated patients from diverse cultural backgrounds. Students navigate language barriers, cultural health beliefs, family dynamics. Develop cultural humility.

Implementation Instructions

Configure ChatGPT with patient scenarios representing diverse cultural backgrounds: different health beliefs, communication norms, family decision-making structures, language barriers, and religious considerations affecting care. Students navigate the encounter, practicing: cultural assessment

techniques, use of interpreter services, respectful inquiry about health beliefs, and family-inclusive care planning. After each scenario, debrief: What assumptions did students make? Where did cultural misunderstanding occur? How could the care plan be more culturally responsive? This builds cultural humility — the recognition that competence is an ongoing process, not a destination.

Nursing Diagnosis Generation Practice

Time: ongoing | Bloom's: Apply | Tools: chatgpt, claude

Given patient data, students generate nursing diagnoses. AI suggests alternatives. Compare NANDA classifications. Build diagnostic reasoning skills.

Implementation Instructions

Present patient assessment data (vital signs, lab values, symptoms, functional status, psychosocial assessment) and have students generate nursing diagnoses independently using NANDA-I taxonomy. Then submit the same patient data to ChatGPT and compare: Did the AI identify the same priority diagnoses? Did it miss relevant nursing diagnoses? Did it suggest diagnoses unsupported by the data? Students evaluate the clinical reasoning behind each diagnosis — both their own and the AI's. This comparison exercise builds diagnostic reasoning skills while demonstrating that nursing diagnosis is an interpretive clinical judgment, not a pattern-matching exercise.

AI-Powered Socratic Dialogue

Time: 30 min | Bloom's: Analyze | Tools: chatgpt, claude

Students engage ChatGPT in Socratic dialogue: AI asks probing questions, students defend their reasoning. Document the conversation and reflect on where their argument broke down.

Implementation Instructions

Assign students a position related to course content. They prompt ChatGPT: 'Engage me in a Socratic dialogue about [position]. Ask me probing questions to test my reasoning. Challenge my assumptions and request evidence.' Students carry on the dialogue for 10-15 exchanges, genuinely attempting to defend their position. They submit the full transcript with annotations highlighting: where they were strongest, where their reasoning broke down, which AI questions they couldn't answer, and what they need to study further. This builds argumentation skills through practice rather than lecture, and the annotations develop metacognitive awareness.

Stakeholder Negotiation Simulation

Time: 1-2 hrs | Bloom's: Evaluate | Tools: chatgpt, claude

AI plays different stakeholders in a policy debate. Student teams negotiate with 'AI stakeholders,' then present compromise proposals. Cross-disciplinary applicability.

Implementation Instructions

Design a scenario with multiple stakeholders who have competing interests (e.g., environmental regulation affecting industry, community health vs. economic development, budget allocation). Assign student teams to represent one stakeholder, and have them negotiate with ChatGPT playing the other stakeholders. Instruct AI to stay in character and push back realistically. Teams must document their negotiation strategy, compromises offered, and final position. Present compromise proposals to the class for evaluation. This works across disciplines — business, political science, health policy, environmental science, education — anywhere stakeholder conflict is relevant.

Historical Figure Interview

Time: 45 min | Bloom's: Analyze | Tools: chatgpt, claude

Students 'interview' a historical figure through AI. They prepare research-based questions, critique the AI's responses for anachronisms, and identify what AI gets wrong about historical context.

Implementation Instructions

Students select a historical figure relevant to your course and prepare 10 research-based interview questions. They prompt ChatGPT: 'Respond as [historical figure] in an interview setting. Stay in character based on what is historically documented about this person's views and experiences.' Students conduct the interview and then critically analyze every response: Is this historically accurate? Does the AI introduce anachronistic ideas? Does it oversimplify the figure's actual complexity? Students write a 'fact-check report' comparing AI responses to primary source evidence. This teaches research skills while revealing AI's tendency to flatten historical complexity.

Client Consultation Role-Play

Time: 30-45 min | Bloom's: Apply | Tools: chatgpt, claude

AI simulates a client/patient/student with specific needs. Students practice professional communication, intake questions, and problem-solving. Applicable to health, business, education, social work.

Implementation Instructions

Configure ChatGPT to role-play a realistic client scenario relevant to your discipline: a patient presenting symptoms (health/nursing), a client with a business problem (business), a parent concerned about their child (education), or a community member seeking services (social work). Students practice their professional intake process: asking appropriate questions, building rapport, identifying key issues, and proposing next steps. The AI responds realistically, including providing incomplete information or expressing frustration — mimicking real professional interactions. Students record or transcript the session and self-evaluate against professional standards.

Social Sciences

AI Bias Audit in Survey Design

Time: 1–2 hrs | Bloom's: Evaluate | Tools: perplexity, elicit, julius, claude, chatgpt

Ask ChatGPT to generate a survey on a sensitive social topic. Students audit for question wording bias, leading questions, demographic assumptions, and missing perspectives.

Implementation Instructions

Ask ChatGPT to generate a complete survey (10-15 questions) on a sensitive social topic relevant to your course (racial attitudes, political beliefs, health behaviors). Distribute the AI-generated survey to students and have them audit each question for: leading language, double-barreled questions, assumption-laden response options, demographic blind spots, and cultural insensitivity. Students must identify at least 5 specific problems and propose improved versions with methodological justification. This reveals AI's tendency to embed dominant cultural assumptions in research instruments and teaches critical survey design skills simultaneously.

Policy Brief with AI Scaffold

Time: 1–2 weeks | Bloom's: Create | Tools: perplexity, elicit, julius, claude, chatgpt

Students use Perplexity for evidence, Elicit for literature extraction, and Claude to help structure the argument arc. Final brief must be fully student-authored with AI cited as a tool.

Implementation Instructions

Students select a policy issue relevant to the course and build a policy brief using multiple AI tools: Perplexity for current evidence gathering, Elicit for systematic literature extraction, and Claude for argument structure scaffolding. The brief must follow a standard format: problem statement, background, policy options, recommendations, and references. The critical requirement is that the final text must be entirely student-authored — AI is cited as a research and organizational tool in the methods section, not used for final prose generation. This teaches the professional workflow of using AI as a research assistant while maintaining authorial responsibility.

Qualitative Coding Comparison

Time: 2–3 hrs | Bloom's: Analyze | Tools: perplexity, elicit, julius, claude, chatgpt

Students manually code a set of interview excerpts. Then use ChatGPT to code the same excerpts. Compare codebooks — discuss consistency, interpretation, and emergent themes.

Implementation Instructions

Select a set of 5-8 interview excerpts (real or simulated) relevant to your course content. Have students manually code the excerpts using open coding, developing their own codebook with definitions and examples. Then submit the same excerpts to ChatGPT with the prompt: 'Code these interview excerpts thematically and provide a codebook.' Compare the human and AI codebooks: How do the themes differ? Where does AI over-categorize or miss subtlety? Where do human coders show interpretive bias? This exercise builds essential qualitative research skills while demonstrating that coding is an interpretive act, not a mechanical one.

Interview Transcript Analysis

Time: 2-3 weeks | Bloom's: Analyze | Tools: notebooklm, julius, claude, anara

Upload transcripts to NotebookLM. AI suggests preliminary themes. Manually verify and refine. Document reliability.

Implementation Instructions

Upload your interview transcripts to NotebookLM (ensure participant confidentiality — remove identifying information first). Ask the AI to suggest preliminary thematic codes based on recurring patterns in the data. Use these AI-suggested codes as a starting point, not a final codebook — manually review each suggested theme against the original transcripts to verify it represents actual patterns rather than AI artifacts. Develop your own codebook iteratively, adding codes the AI missed and refining the ones it suggested. Document inter-rater reliability between your manual coding and the AI-assisted preliminary codes as a methodological transparency measure.

Survey Mixed Methods Analysis

Time: 1-2 weeks | Bloom's: Analyze | Tools: notebooklm, julius, claude, anara

Julius for quantitative analysis. NotebookLM for open-ended responses. Human integrates findings.

Implementation Instructions

For the quantitative component, upload survey data to Julius for statistical analysis: descriptive statistics, cross-tabulations, and hypothesis testing. For open-ended responses, upload to NotebookLM for preliminary thematic coding. The critical methodological step is integration — how do qualitative themes illuminate quantitative patterns? AI can process each data type separately, but the interpretive synthesis that connects them must be human work. Document your mixed methods

integration approach explicitly: What convergences emerged? What divergences require explanation? How do the two data types together tell a story that neither tells alone?

Policy Document Comparison

Time: 1-2 weeks | Bloom's: Analyze | Tools: notebooklm, julius, claude, anara

Upload policy documents to Claude. Extract key provisions, compare jurisdictions. Human performs critical analysis.

Implementation Instructions

Upload policy documents from multiple jurisdictions or organizations to Claude. Ask it to extract: stated goals, key provisions, enforcement mechanisms, stakeholder definitions, and implementation timelines. Create a structured comparison matrix showing similarities and differences across documents. AI excels at this extraction and comparison work, significantly reducing the time needed for document analysis. However, your critical analysis — why these differences exist, what power dynamics they reflect, what unintended consequences they may produce — is entirely your scholarly contribution. The AI provides the organized data; you provide the interpretation.

Thematic Saturation Tracking

Time: ongoing | Bloom's: Analyze | Tools: notebooklm, claude, chatpdf

After each interview, AI suggests emerging themes. Track when new themes stop appearing (saturation). Document sample size justification.

Implementation Instructions

After each interview or data collection session, upload the transcript to Claude and ask it to identify emerging themes, comparing to themes identified in previous sessions. Track when new themes stop appearing — this is an indicator of thematic saturation, though the final determination requires researcher judgment. Maintain a running document showing: session number, new themes identified, themes that deepened or evolved, and your assessment of saturation progress. This systematic tracking provides evidence for your sample size justification in the methods section — a common weakness in qualitative research. Document the AI's role transparently as a preliminary coding tool.

Member Checking Protocol

Time: 2-3 weeks | Bloom's: Evaluate | Tools: notebooklm, claude, chatpdf

AI generates participant-friendly summaries of findings. Send to participants for validation. Incorporate feedback into analysis.

Implementation Instructions

After completing your initial analysis, use Claude to generate participant-friendly summaries of your findings: plain language descriptions of each theme, how you interpreted the data, and preliminary conclusions. Send these summaries to participants and ask: Does this accurately reflect your experience? What did we miss? What would you change? Incorporate participant feedback into your analysis, documenting how it affected your interpretations. This AI-assisted member checking makes the process more accessible to participants (academic language can be alienating) while maintaining rigor. The AI generates the accessible draft; you ensure it accurately represents your analysis.

Reflexivity Journal Analysis

Time: ongoing | Bloom's: Analyze | Tools: notebooklm, claude, chatpdf

Keep researcher reflexivity journal. AI helps identify potential biases, assumptions affecting interpretation. Strengthens qualitative rigor.

Implementation Instructions

Maintain a regular researcher reflexivity journal documenting your assumptions, emotional reactions to data, analytical decisions, and potential biases. Periodically upload journal entries to Claude with the prompt: 'Identify potential biases, assumptions, or blind spots in my research process based on these journal entries.' Use the AI's analysis as a mirror — it may surface patterns you haven't noticed in your own thinking. This doesn't replace genuine self-reflection but adds an additional layer of analytical distance. Document how reflexive practice influenced your analytical decisions, as this transparency strengthens the credibility of qualitative findings.

Interview Guide Development

Time: 1-2 days | Bloom's: Create | Tools: notebooklm, claude, otter

Draft semi-structured interview questions. AI suggests follow-up probes. Human ensures culturally sensitive, non-leading phrasing aligned with research aims.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. Draft semi-structured interview questions. AI suggests follow-up probes. Human ensures culturally sensitive, non-leading phrasing aligned with research aims. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Memo Writing with AI Prompts

Time: ongoing | Bloom's: Analyze | Tools: notebooklm, claude, otter

After each interview, use AI prompts to structure analytic memos. What surprised you? What patterns emerge? What needs follow-up? Build reflexive practice.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. After each interview, use AI prompts to structure analytic memos. What surprised you? What patterns emerge? What needs follow-up? Build reflexive practice. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Focus Group Transcript Analysis

Time: 2-3 weeks | Bloom's: Analyze | Tools: notebooklm, claude, otter

Upload focus group transcripts to NotebookLM. AI identifies speaker patterns, agreement/disagreement. Human codes for group dynamics and emergent themes.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. Upload focus group transcripts to NotebookLM. AI identifies speaker patterns,

agreement/disagreement. Human codes for group dynamics and emergent themes. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Grounded Theory Constant Comparison

Time: ongoing | Bloom's: Analyze | Tools: notebooklm, claude, otter

AI helps track evolving codes across interviews. Suggests when categories are saturated. Human maintains theoretical sensitivity and makes conceptual leaps.

Implementation Instructions

This analytical activity asks students to examine AI outputs and compare them against domain knowledge. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI helps track evolving codes across interviews. Suggests when categories are saturated. Human maintains theoretical sensitivity and makes conceptual leaps. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Participant Validation Materials

Time: 1 week | Bloom's: Create | Tools: notebooklm, claude, otter

AI generates participant-friendly summaries of preliminary findings for member checking. Students send to participants, incorporate validation feedback.

Implementation Instructions

This creative assignment asks students to produce original work using AI as a collaborative tool. Begin by reviewing the activity description and adapting it to your specific course content and student level. AI generates participant-friendly summaries of preliminary findings for member checking. Students send to participants, incorporate validation feedback. Provide students with clear expectations about what AI tools to use, how to document their process, and what the final deliverable should include. After completing the activity, have students reflect briefly on how AI contributed to their learning process and where human judgment was essential. This builds the metacognitive awareness and AI literacy that are increasingly important across all disciplines.

Student AI Literacy & Prompt Engineering

Prompt Engineering Tournament

Time: 45 min | Bloom's: Apply | Tools: chatgpt, claude, perplexity, copilot

Teams compete to craft the best AI prompt for a course-related task. Judge outputs on accuracy, depth, and usefulness. Winners share their prompt strategies.

Implementation Instructions

Choose a course-related task that benefits from precise prompting (e.g., 'Get AI to explain [complex concept] accurately for a first-year student'). Teams of 3-4 have 20 minutes to iteratively refine their prompts, documenting each iteration. Judges (instructor + class vote) evaluate the AI outputs on accuracy, depth, usefulness, and clarity. Winning teams present their prompting strategies: What constraints did they add? How did they specify audience? Did they use examples or role-playing? This gamified approach teaches prompt engineering as a transferable information literacy skill while building collaborative problem-solving abilities.

AI Hallucination Scavenger Hunt

Time: 30 min | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity, copilot

Students ask AI factual questions in your field and race to find hallucinated claims. First team to verify 5 errors with real sources wins. Builds critical verification habits.

Implementation Instructions

Give each team a set of 10 factual questions in your discipline, ranging from common knowledge to obscure details. Teams submit all questions to ChatGPT and then race to identify hallucinated or inaccurate claims in the responses. For each error found, they must provide the correct answer with a verified source (peer-reviewed article, authoritative website, or textbook). First team to verify 5 genuine errors wins. Debrief by discussing patterns: What types of questions produced more hallucinations? Were errors more common in recent events, niche topics, or statistical claims? This builds lasting verification habits through competitive, experiential learning.

AI Output Comparison Lab

Time: 1 hr | Bloom's: Analyze | Tools: chatgpt, claude, perplexity, copilot

Same prompt submitted to ChatGPT, Claude, Perplexity, and Gemini. Students compare outputs for accuracy, tone, citation quality, and bias. Understand tool strengths and weaknesses.

Implementation Instructions

Select a complex, discipline-specific question and have students submit the exact same prompt to 3-4 different AI tools (ChatGPT, Claude, Perplexity, Gemini). Students create a comparison matrix evaluating each response across dimensions: factual accuracy, citation presence and quality, depth of analysis, potential biases, and tone. Discuss as a class: Why did the tools produce different responses? Which tool was best for which aspects? When would you choose one over another? This prevents students from defaulting to a single tool and builds nuanced understanding of AI as a diverse ecosystem, not a monolith.

Role-Play Prompting Workshop

Time: 30 min | Bloom's: Apply | Tools: chatgpt, claude, perplexity, copilot

Students craft prompts assigning AI specific roles (skeptical reviewer, patient educator, devil's advocate). Compare how role assignment changes output quality and perspective.

Implementation Instructions

Introduce the concept of role-based prompting: assigning AI a specific persona changes its output significantly. Students experiment with the same question using different role prompts: 'You are a skeptical peer reviewer,' 'You are a patient tutor for beginners,' 'You are a devil's advocate,' 'You are a [discipline] expert.' Compare outputs: How did the role change the depth, tone, and perspective of the response? Which role produced the most useful output for the student's actual need? This teaches students that AI interaction is a skill — the quality of the output depends heavily on the quality of the input.

AI Transparency Documentation Practice

Time: ongoing | Bloom's: Apply | Tools: chatgpt, claude, perplexity, copilot

Students maintain an AI use log for every assignment: tool used, prompt given, output received, how it was modified. Builds professional documentation habits.

Implementation Instructions

Require students to maintain a standardized AI use log throughout the course. For each AI interaction related to coursework, they document: (1) the tool used, (2) the exact prompt, (3) a summary of the output, (4) how they used or modified the output, and (5) what they contributed that was uniquely their own. This log is submitted with every assignment. Grade it not on whether they used AI, but on the

quality and honesty of their documentation. This prepares students for professional environments where AI use documentation is increasingly required and builds habits of intellectual transparency.

Study Skills & Metacognition

AI-Generated Practice Exam

Time: 30 min | Bloom's: Remember | Tools: chatgpt, claude, notebooklm

Students prompt AI to generate practice questions from course material. They answer, check, and identify which topics need more study. Teaches self-assessment and exam preparation.

Implementation Instructions

Students prompt ChatGPT: 'Generate 15 practice questions about [specific topics] at [course level]. Include a mix of multiple choice, short answer, and application questions. Provide an answer key.' Students take the practice exam under timed conditions, then check their answers. For each wrong answer, they identify the specific knowledge gap and find the correct information in course materials. The metacognitive value is in the self-assessment: students learn to identify their own weak areas rather than studying everything equally. This leverages the well-documented 'testing effect' — practice retrieval strengthens memory more than re-reading.

Concept Map Construction with AI Check

Time: 30-45 min | Bloom's: Analyze | Tools: chatgpt, claude, notebooklm

Students build a concept map of a unit by hand. Then ask AI to generate one. Compare: what connections did students see that AI missed? What did AI connect that students overlooked?

Implementation Instructions

After completing a unit, students create a concept map by hand showing relationships between key ideas. Then they ask ChatGPT: 'Create a concept map showing relationships between these concepts from [unit topic]: [list key terms].' Students compare the two maps: What connections did they identify that the AI missed? What connections did the AI make that they hadn't considered? Are the AI's connections accurate? This comparison reveals the depth of student understanding — students who identify connections the AI missed demonstrate genuine conceptual integration, while AI-only connections they missed highlight areas for further study.

Elaborative Interrogation Practice

Time: 20 min | Bloom's: Understand | Tools: chatgpt, claude, notebooklm

Students read a key concept, then ask AI 'why?' and 'how?' questions to deepen understanding. Document the dialogue and identify the point where AI stops adding value.

Implementation Instructions

Students select a key concept from the current unit and engage ChatGPT in an elaborative interrogation dialogue: ask 'Why does this happen?' then follow up with 'How does that mechanism work?' then 'What would happen if [condition changed]?' Students continue asking increasingly specific questions until the AI either starts repeating itself or provides unsatisfying answers. They document the full dialogue and identify: the point where AI explanations were most helpful, the point where AI stopped adding value, and the questions they still need to explore through course readings or instructor consultation. This teaches active study strategies far more effective than passive re-reading.

Teaching + Research Integration

Research-to-Teaching Pipeline

Time: ongoing | Bloom's: Apply | Tools: notebooklm, elicit, claude, chatgpt

NotebookLM summarizes your publications for undergraduate lectures. Maintain research accuracy, improve accessibility.

Implementation Instructions

Upload your published papers or working manuscripts to NotebookLM. Ask it to generate summaries at three levels: graduate seminar, undergraduate upper-level, and introductory course. Review each summary for accuracy — AI may oversimplify key nuances or miss important caveats that affect teaching. Create course materials from the verified summaries: lecture outlines, discussion questions, and case studies drawn from your own research. This pipeline ensures your courses reflect cutting-edge scholarship (yours!) while making complex research accessible to students at different levels. Update annually as your research evolves.

Student Co-Authorship Scaffolding

Time: semester-long | Bloom's: Create | Tools: notebooklm, elicit, claude, chatgpt

Elicit trains students in systematic review methods. Document AI vs manual work. Students gain skills, contribute to scholarship.

Implementation Instructions

Design a semester-long project where students contribute to your ongoing research under structured mentorship. Use Elicit to train students in systematic review methods — the AI tool provides immediate feedback on search strategies and data extraction, reducing the mentoring burden while teaching rigorous methods. Clearly document each student's contribution using the CRediT taxonomy from the beginning of the project. Students gain genuine research skills and potential co-authorship credit; you gain research assistance that is transparent, documented, and high-quality. Ensure IRB compliance and student consent for any research involving human subjects.

Literature Review as Teaching Tool

Time: module | Bloom's: Analyze | Tools: notebooklm, elicit, claude, chatgpt

Students use ResearchRabbit and Semantic Scholar to map research area. Their work feeds your lit review. Mutual benefit.

Implementation Instructions

Design a course assignment where students use ResearchRabbit and Semantic Scholar to map a specific area of your research field. Structure the assignment so that student work directly feeds into your own literature review: each student maps a different sub-area, and you synthesize their maps into a comprehensive field overview. Students gain genuine research skills by doing real research work, and their effort contributes to scholarship rather than existing only as a graded exercise. Provide clear guidelines about citation network mapping methodology so student work meets the rigor standards required for your literature review.

Teaching: Assignment Assessment

AI-Augmented Rubric Workshop

Time: 2–4 hrs | Bloom's: Create | Tools: chatgpt, claude, notebooklm

Ask ChatGPT to draft a rubric from your learning outcomes. Students critique and revise it — building metacognitive understanding of quality criteria before the assignment.

Implementation Instructions

Start by entering your assignment's learning outcomes into ChatGPT and ask it to generate a detailed rubric with 4-5 criteria and performance levels. Print or share the AI-generated rubric with students before they begin the assignment. In class, have students work in pairs to critique each criterion: Is it measurable? Fair? Aligned with what they're actually learning? Students revise the rubric collaboratively, submitting a final annotated version with justifications for every change. Use the student-revised rubric for actual grading — this gives students ownership of the quality standards and deepens their metacognitive understanding of what 'good work' looks like.

Annotated AI Feedback Portfolio

Time: module-long | Bloom's: Evaluate | Tools: chatgpt, claude, notebooklm

Students submit drafts to an AI tool for feedback, then annotate the response — accepting, rejecting, or questioning each suggestion — and submit both with a rationale memo.

Implementation Instructions

Assign students to submit a draft of their work to an AI tool (ChatGPT or Claude) and screenshot the full AI feedback response. Students then create an annotation document where they categorize each piece of AI feedback as: Accepted (with explanation of why), Rejected (with reasoning), or Needs Investigation (requires further thought or research). They revise their draft based on their own judgment, not the AI's wholesale suggestions. The final submission includes: original draft, AI feedback screenshot, annotated response, revised draft, and a 1-page rationale memo. Grade the quality of their critical engagement with AI feedback, not just the final product.

Authentic Assessment Redesign

Time: 1 week | Bloom's: Evaluate | Tools: chatgpt, claude, notebooklm

Use NotebookLM to load pedagogical readings on authentic assessment. Redesign one traditional exam prompt into an AI-proof, real-world task.

Implementation Instructions

Upload 2-3 key readings on authentic assessment design into NotebookLM and explore how the sources define 'authentic' versus 'traditional' assessment. Select one existing exam question or assignment from your course that could be easily completed by AI. Use NotebookLM's synthesis to brainstorm redesign principles, then draft a new version that requires real-world application, personal experience, or process documentation that AI cannot replicate. Share your before-and-after with a colleague for feedback. The goal is not to be 'AI-proof' but to design tasks where AI is a tool, not a shortcut — where the human thinking is the point.

Teaching: Quick Discussion Activities

AI Devil's Advocate

Time: 20 min | Bloom's: Analyze | Tools: chatgpt, claude, perplexity

Assign ChatGPT to argue the opposite position on a course topic. Students critique its argument using course readings — surface-level AI is easy to beat with real evidence.

Implementation Instructions

Choose a debatable topic from your current unit and paste it into ChatGPT with the prompt: 'Argue the opposite of [position].' Distribute the AI's response to small groups of 3-4 students. Each group has 15 minutes to identify weak reasoning, unsupported claims, and logical fallacies using their course readings as evidence. Groups then present their strongest rebuttal to the class. Close with a reflection: Was the AI's argument persuasive on the surface? What made it easy or hard to counter? This builds critical evaluation skills while demonstrating AI's tendency toward plausible but shallow argumentation.

Question Cascade

Time: 15 min | Bloom's: Remember | Tools: chatgpt, claude, perplexity

Students each submit one genuine question. Paste all into Claude; it clusters and ranks them by theme. Class explores the most-voted cluster first.

Implementation Instructions

At the start of class, give students 3 minutes to write one genuine question about the current topic on a slip of paper or digital form. Collect and paste all questions into Claude with the prompt: 'Cluster these questions by theme and rank the clusters by frequency.' Display the clustered results and let students vote on which cluster to explore first. Facilitate a 10-minute discussion on the top-voted cluster, using student questions as the driving force. This transforms passive lecture formats into student-driven inquiry and helps you quickly identify gaps in understanding.

Fact-Check in Real Time

Time: 25 min | Bloom's: Evaluate | Tools: chatgpt, claude, perplexity

Students use Perplexity to fact-check a bold claim from the lecture. Each group presents their findings, citations included.

Implementation Instructions

Select a bold or provocative claim from the lecture material — ideally one that sounds true but requires verification. Divide students into groups and have each use Perplexity AI to investigate the claim's accuracy within 15 minutes. Each group must find at least two cited sources that either support or refute the claim. Groups present their findings, including source quality evaluation. Debrief by discussing how AI search tools handle contested claims differently than traditional databases. This activity builds information literacy muscles students will use throughout their academic careers.

Teaching: Quick First-Year Activities

Misconception Warm-Up

Time: 15 min | Bloom's: Evaluate | Tools: copilot, chatgpt, perplexity

Pose a common misconception in your field. Students use Perplexity to investigate and report back with a cited response — immediate information literacy practice.

Implementation Instructions

Identify a common misconception in your field that first-year students frequently hold (e.g., 'We only use 10% of our brains' or 'The Great Wall of China is visible from space'). Present it to the class as a true/false warm-up question. After initial guesses, give students 10 minutes to use Perplexity AI to

investigate, requiring at least one cited source. Students share their findings and discuss how the misconception persists despite available evidence. This builds immediate information literacy skills while addressing content knowledge gaps early in the course.

Explain It Three Ways

Time: 20 min | Bloom's: Remember | Tools: copilot, chatgpt, perplexity

Students ask ChatGPT to explain a core term at beginner, intermediate, and expert level. Class votes on which was most accurate and discusses why.

Implementation Instructions

Choose a core concept or vocabulary term from the week's material. Have students individually prompt ChatGPT: 'Explain [concept] at a beginner level, an intermediate level, and an expert level.' Students read all three versions and vote on which level was most accurate and helpful for their understanding. Facilitate a brief class discussion about what changed between levels — what was oversimplified at the beginner level? What jargon appeared at the expert level? This helps first-year students calibrate their own comprehension and builds vocabulary awareness without overwhelming them.

AI Note-Check

Time: 20 min | Bloom's: Remember | Tools: copilot, chatgpt, perplexity

Students paste their lecture notes into ChatGPT and ask for a structured summary. Compare the AI's version to their own — annotate gaps and errors.

Implementation Instructions

After a lecture segment, have students paste their handwritten or typed notes into ChatGPT with the prompt: 'Create a structured summary of these notes and identify any gaps or inaccuracies.' Students then compare the AI's organized summary to their original notes, using a highlighter or annotation tool to mark what the AI added, what it missed, and what it got wrong. This is not about replacing note-taking — it's about building metacognitive awareness of how well students captured key ideas. Collect a brief reflection: 'What did the AI catch that you missed? What did you have that the AI didn't?'

Part 3: Sources Bibliography

The following sources informed the design, tool selection, activity creation, pedagogical frameworks, and implementation logic of the AI Evaluation Toolkit. Sources are organized thematically and formatted per APA conventions.

Foundational Pedagogical Frameworks

Anderson, L. W., & Krathwohl, D. R. (Eds.). (2001). A taxonomy for learning, teaching, and assessing: A revision of Bloom's taxonomy of educational objectives (Complete ed.). Longman.

CAST. (2018). Universal Design for Learning guidelines version 2.2. https://udlguidelines.cast.org/

Wiggins, G., & McTighe, J. (2005). Understanding by design (Expanded 2nd ed.). Association for Supervision and Curriculum Development.

AI Pedagogy & Instructional Frameworks

Mollick, E. R., & Mollick, L. (2023a). Assigning AI: Seven approaches for students, with prompts. Wharton School, University of Pennsylvania. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4475995

Mollick, E. R., & Mollick, L. (2023b). Using AI to implement effective teaching strategies in classrooms: Five strategies, including prompts. Wharton School, University of Pennsylvania. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4391243

Speicher, S. (2023, December 22). Unlocking human-AI potential: 10 best practices for AI assignments in higher ed. Association of College and University Educators (ACUE). https://acue.org/resources/blog/unlocking-human-ai-potential-10-best-practices-for-ai-assignments-in-higher-ed/

Information Literacy & Source Evaluation

Association of College & Research Libraries. (2016). Framework for information literacy for higher education. American Library Association. https://www.ala.org/acrl/standards/ilframework

Blakeslee, S. (2004). The CRAAP test. LOEX Quarterly, 31(3), 6–7.

Caulfield, M. (2019). SIFT (The four moves). Hapgood. https://hapgood.us/2019/06/19/sift-the-four-moves/

Health Sciences & Evidence-Based Practice

Herdman, T. H., & Kamitsuru, S. (Eds.). (2018). NANDA International nursing diagnoses: Definitions and classification 2018–2020 (11th ed.). Thieme.

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The PRISMA 2021 statement: An updated guideline for reporting systematic reviews. BMJ, 372, n71. https://doi.org/10.1136/bmj.n71

Sackett, D. L., Rosenberg, W. M. C., Gray, J. A. M., Haynes, R. B., & Richardson, W. S. (1996). Evidence based medicine: What it is and what it isn't. BMJ, 312(7023), 71–72. https://doi.org/10.1136/bmj.312.7023.71

Research Ethics & Scholarly Communication

Brand, A., Allen, L., Altman, M., Hlava, M., & Scott, J. (2015). Beyond authorship: Attribution, contribution, collaboration, and credit. Learned Publishing, 28(2), 151–155. https://doi.org/10.1087/20150211

Seckel, E., Stephens, B. Y., & Rodriguez, F. (2024). Ten simple rules to leverage large language models for getting grants. PLOS Computational Biology, 20(3), Article e1011863. https://doi.org/10.1371/journal.pcbi.1011863

National & International Policy Guidance

Miao, F., & Holmes, W. (2023). Guidance for generative AI in education and research. United Nations Educational, Scientific and Cultural Organization (UNESCO). https://unesdoc.unesco.org/ark:/48223/pf0000386693

U.S. Department of Education, Office of Educational Technology. (2023). Artificial intelligence and the future of teaching and learning: Insights and recommendations. https://tech.ed.gov/ai-future-of-teaching-and-learning/

Institutional & Organizational Statements on AI in Education

AWAC Working Group on AI in WAC. (2025). AWAC statement on AI and writing across the curriculum (Version 2.0). Association for Writing Across the Curriculum. https://wacassociation.org/ai-statement/

MLA-CCCC Joint Task Force on Writing and AI. (2023). Working paper 1: Overview of the issues, statement of principles, and recommendations. Modern Language Association; Conference on College Composition and Communication. https://aiandwriting.hcommons.org/working-paper-1/

MLA-CCCC Joint Task Force on Writing and AI. (2024). Working paper 2: Generative AI and policy development: Guidance from the MLA-CCCC task force. Modern Language Association; Conference on College Composition and Communication. https://aiandwriting.hcommons.org/working-papers/

University Centers for Teaching & Learning

Columbia University Center for Teaching and Learning. (2026). Faculty use cases. https://ai.ctl.columbia.edu/explore/examples/

Harvard University Derek Bok Center for Teaching and Learning. (2026). Examples and ideas for using AI for your teaching. https://bokcenter.harvard.edu/examples-and-ideas-for-using-AI-for-your-teaching

Stanford Medicine. (2025, July 23). 10 simple rules for using AI in grant writing. Stanford University. https://medicine.stanford.edu/news/current-news/standard-news/10-rules-for-ai-in-grant-writing.html

Syracuse University Center for Teaching and Learning Excellence. (2026). Generative AI. https://teachingexcellence.syr.edu/resources/generative-ai/

University of Central Florida. (2026). Teaching repository of AI-infused learning (TRAIIL). https://stars.library.ucf.edu/traiil/

University of Florida. (2026). AI across the curriculum. https://ai.ufl.edu/teaching-with-ai/ai-across-the-curriculum/

University of Florida. (2026). Expanding the AI curriculum. https://ai.ufl.edu/teaching-with-ai/expanding-the-ai-curriculum/

University of Florida Institutional Assessment. (2024, June 11). UF QEP AI across the curriculum: Assessment guide. https://assessment.aa.ufl.edu/media/assessmentaaufledu/UF_QEP_AI_Curriculum_AssessmentGuide.pdf

University of Maryland, Baltimore Writing Center. (2026, February 5). Writing and AI. https://www.umaryland.edu/writing/writing-resources/writing-and-ai/

University of North Carolina Wilmington. (2025, November). AI for academic administration: Practical applications in graduate education.

University of Pittsburgh University Center for Teaching and Learning. (2026). Teaching with generative AI. https://teaching.pitt.edu/resources/teaching-with-generative-ai/

University of Texas at Austin Center for Teaching and Learning. (2026). Generative AI in teaching and learning: Assignments. https://ctl.utexas.edu/generative-ai-teaching-and-learning-assignments

University of Utah Center for Teaching Excellence. (2026, February 3). AI tools for education. https://cte.utah.edu/instructor-education/ai-for-teaching.php

Washington University in St. Louis Center for Teaching and Learning. (2026). Generative AI teaching activities: Online repository. https://ctl.wustl.edu/resources/generative-ai-teaching-activities-online-repository/

Scholarly & Peer-Reviewed Publications

Nerantzi, C., Abegglen, S., Karatsiori, M., & Martínez-Arboleda, A. (Eds.). (2023). 101 creative ideas to use AI in education: A crowdsourced collection. #creativeHE. https://doi.org/10.5281/zenodo.8072950

OpenAI. (2026, January). AI as a scientific collaborator. https://cdn.openai.com/pdf/f4b4a5da-b2de-418d-9fcd-6b293e9dc157/oai_ai-as-a-scientific-collaborator_jan-2026.pdf

Faculty Development, Pedagogy & Best Practices

Berklee Online. (2026). AI in music: Composition, production, and analysis [Online course]. https://online.berklee.edu/courses/ai-in-music-composition-production-and-analysis

EDUCAUSE. (2024). 2024 EDUCAUSE AI landscape study. https://www.educause.edu/research-and-publications

Gardner, T. (2024, June 17). MLA-CCCC Joint Task Force on Writing and AI working paper. Teaching WAC with AI, Virginia Tech. https://teachingwacwithai.tracigardner.com/ai-concept/bias-in-ai/mla-cccc-joint-task-force-on-writing-and-ai-working-paper/

QuadC. (2025, August 11). AI for faculty: How to save time on administrative work. https://www.quadc.io/blog/ai-for-faculty-how-to-save-time-on-administrative-work

SchoolAI. (2025). AI for college instructors: A 2025 guide. https://schoolai.com/blog/ai-for-college-instructors

AI Tool Reviews, Comparative Guides & Industry Resources

Capacity. (2025). 15 must-try AI tools for education in 2025. https://capacity.com/blog/ai-tools-for-education/

DeepWriter AI. (2026). Best AI deep research tools in 2026. https://deepwriter.com/blog/best-ai-deep-research-tools-in-2026/

Giugliano, A. (2025, November 26). Best AI tools to improve academic writing 2026. Thesify. https://www.thesify.ai/blog/best-ai-tools-improve-academic-writing-2026

Glean Technologies. (2026, February 12). How AI is enhancing administrative efficiency in higher education. https://www.glean.com/blog/enterprise-search-evaluation-2026

Interactico. (2026). 16 best AI tools for teachers: Smarter planning and teaching. https://www.interactico.com/blog/ai-tools-for-teachers

John, J. (2026). I tested the 6 best AI tools for research in 2026, and here's my honest take. Jotform Blog. https://www.jotform.com/ai/best-ai-tools-for-research/

Labguru. (2026). Challenges of using AI in the lab. https://www.labguru.com/blog/challenges-of-using-ai-in-the-lab

Lumivero. (2026, January 23). Best AI tools for academic research in 2026. https://lumivero.com/resources/blog/ai-tools-for-academic-research/

Paperguide. (2025, July 4). 5 best AI tools for academic writing in 2026 (R. Reddy, Author). https://paperguide.ai/blog/ai-tools-for-academic-writing/

ResearchRabbit. (2025, October 7). Best AI tools for literature review in 2025: Stage by stage. https://www.researchrabbit.ai/articles/best-ai-tools-for-literature-review

ScholarsReview Team. (2026, February 16). The best AI peer review tool of 2026: Why 60% of researchers are making the switch. https://scholarsreview.com/blogs/best-ai-peer-review-tool-2026-for-researchers

Sindhuja, A. (2025, October 10). Top AI tools every researcher should know in 2025. Editage Insights. https://www.editage.com/insights/ai-tools-for-researchers

Sonix. (2025). 5 best AI tools for lab managers in 2025. https://sonix.ai/ai/best-ai-for-lab-managers/